EXAM PACK

EEE459

Compound Semiconductor and Hetero-Junction Devices



Exam Pack

Session 2018–19 *Solutions and formula sheet of the 4th Year 1st Semester Final Examination, 2022, and Term Tests 1 and 2*

IN THIS PACK

1 PAST PAPER SOLUTIONS
Question Paper Solutions 2018–19 pages 3--79 77 pages

2 FORMULA SHEET
Formula Sheet pages 80--94 15 pages

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Prepared by Nur Mohammad

November 2026

IN THIS PACK

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1 PAST PAPER SOLUTIONS
Question Paper Solutions 2018–19

pages 3--79

Question Paper 5
1(a) Three Crystal Structures 12
1(b) Metal–Semiconductor Junction with an Interfacial Layer and Interface States 14
1(c) Gold on n-Type Germanium Schottky Diode 16
1(d) Vapour-Phase Epitaxy 19
2(a) Band Diagrams of pN and nN Heterojunctions 22
2(b) Depletion Width of a Heterojunction 24
2(c) Excess Minority Electrons in the Base of a BJT 26
2(d) n-Ge / P-GaAs Heterojunction: Doping and Depletion Width 29
3(a) Specific Contact Resistance 32
3(b) Multiple Negative Differential Resistance in a Resonant Tunnelling Diode 34
3(c) Types of Ohmic Contact 36
3(d) Four Metal–GaAs Junctions 38
4(a) Structure and Doping Profile of a Double-Heterojunction Bipolar Transistor 41
4(b) Advantages of the HBT and Its Injection Ratio 43
4(c) Charge, Field, Potential and C–V of an Ideal MOS Capacitor 45
4(d) Flat-Band Voltage and Threshold Voltage 49
5(a) Emitter Injection Efficiency of a BJT 52
5(b) Early Effect and Current Crowding 55
5(c) The Gummel–Poon Model 58
5(d) Base Width and Emitter Doping for a Gain of 150 60
6(a) Advantages of the HEMT 63
6(b) Operation of the HEMT 64
6(c) Ebers–Moll Equivalent Circuit 66
6(d) Maximum Cut-Off Frequency of a MESFET and the Channel Doping 68
6(e) Channel Thickness of a GaAs MESFET for a Threshold of 0.1 V 70
7(a) Ohmic Contacts and Their Types 72
7(b) Gold on n-Type Germanium (Term Test) 73
8(a) MOS Energy-Band Diagrams Under Bias 76
8(b) Threshold Voltage with a Gold Gate 78

2 FORMULA SHEET
Formula Sheet

pages 80--94

1 Crystals and growth 83
2 Metal–semiconductor contacts 83
3 Heterojunctions 84
4 Bipolar transistors 86
5 MOS capacitor 87
6 HEMT and MESFET 89
7 Key formulas 90
8 Symbol index 91

PAST PAPER SOLUTIONS

EEE459

Compound Semiconductor and Hetero-Junction Devices

Question Paper Solutions

Session 2018–19

4th Year 1st Semester Final Examination, 2022, with Term Tests 1 and 2

QUESTION MAP

Q1	Q2	Q3	Q4	Q5	Q6
25 marks	25 marks	25 marks	25 marks	25 marks	25 marks

Q7	Q8
10 marks	10 marks

EACH ANSWER



Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

The paper at a glance

3 h	100	4 of 6	45 min
Total time	Total marks	Questions to answer	Per question

Time = marks × 1.8 min (180 min for 100 marks); a part with its own time (a term test) uses that. Tick the questions you will answer; each sub-part links to its solution.

Q	Sub-part	Marks	Time	Pick
Part - A · (Answer any two of the following questions)				
Q1	(a) Three Crystal Structures	6	11 min	☐
	(b) Metal–Semiconductor Junction with an Interfacial Layer and Interface States	5	9 min	
	(c) Gold on n-Type Germanium Schottky Diode	6	11 min	
	(d) Vapour-Phase Epitaxy	8	14 min	
Q2	(a) Band Diagrams of pN and nN Heterojunctions	5	9 min	☐
	(b) Depletion Width of a Heterojunction	7	13 min	
	(c) Excess Minority Electrons in the Base of a BJT	8	14 min	
	(d) n-Ge / P-GaAs Heterojunction: Doping and Depletion Width	5	9 min	
Q3	(a) Specific Contact Resistance	4	7 min	☐
	(b) Multiple Negative Differential Resistance in a Resonant Tunnelling Diode	6	11 min	
	(c) Types of Ohmic Contact	3	5 min	
	(d) Four Metal–GaAs Junctions	12	22 min	
Part - B · (Answer any two of the following questions)				
Q4	(a) Structure and Doping Profile of a Double-Heterojunction Bipolar Transistor	6	11 min	☐
	(b) Advantages of the HBT and Its Injection Ratio	7	13 min	
	(c) Charge, Field, Potential and C–V of an Ideal MOS Capacitor	6	11 min	
	(d) Flat-Band Voltage and Threshold Voltage	6	11 min	
Q5	(a) Emitter Injection Efficiency of a BJT	7	13 min	☐
	(b) Early Effect and Current Crowding	6	11 min	
	(c) The Gummel–Poon Model	7	13 min	
	(d) Base Width and Emitter Doping for a Gain of 150	5	9 min	
Q6	(a) Advantages of the HEMT	4	7 min	☐
	(b) Operation of the HEMT	7	13 min	
	(c) Ebers–Moll Equivalent Circuit	5	9 min	
	(d) Maximum Cut-Off Frequency of a MESFET and the Channel Doping	4	7 min	
	(e) Channel Thickness of a GaAs MESFET for a Threshold of 0.1 V	5	9 min	
Term Test 1 · 10 marks · 30 min				
Q7	(a) Ohmic Contacts and Their Types	4	12 min	☐
	(b) Gold on n-Type Germanium (Term Test)	6	18 min	
Term Test 2 · 10 marks · 30 min				
Q8	(a) MOS Energy-Band Diagrams Under Bias	5	15 min	☐
	(b) Threshold Voltage with a Gold Gate	5	15 min	

Formulas and constants follow Neamen, Semiconductor Physics and Devices (4th ed.); compound-semiconductor data follow Sze and Ng, Physics of Semiconductor Devices.

Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
4th Year 1st Semester Final Examination, 2022
Session: 2018-19
Course Code: EEE-459
Course Title: Compound Semiconductor and Hetero-junction Devices

Total Marks: 100 **Credit: 3.0** **Total Time: 3 Hours**

Part - A

*(Answer any **two** of the following questions)*

1. (a) Draw the following crystal structures.
   (i) Hexagonal close packed crystal structure
   (ii) Zinc blende crystal structure.
   (iii) Body centered cubic crystal structure 6

   (b) Sketch energy-band diagram of a metal-semiconductor junction with an interfacial layer and interface states. 5

   (c) Determine the theoretical barrier height, built-in potential barrier, and maximum electric field in a metal–semiconductor diode for zero applied bias. Consider a contact between gold (metal work function 5.1 V) and n-type germanium (electron affinity 4.13 V) doped to $N_d = 10^{16}$ cm^{-3} at T = 300 K. $N_C = 1.04 \times 10^{19}$ and $\varepsilon_r = 16$ 6

   (d) Explain the process of vapor phase epitaxy. 8

2. (a) Draw the band diagrams of a pN and a nN hetero-junction. 5

   (b) Derive the equation for the depletion region width of a hetero-junction. 7

   (c) For forward active mode derive the excess minority carrier concentration equation at base of a bipolar junction transistor 8

   (d) Determine the n-type doping concentration, and depletion region width in P-type side of semiconductor for an n-Ge and P-GaAs hetero-junction. Consider built in potential is 1.0 V and T = 300 K, n-type Ge electron affinity 4.13, N_{vn} = [unclear: 6 ×] 10^{18} cm^{-3}, $N_d = 10^{16}$ cm^{-3}, $n_i = 2.4 \times 10^{13}$ cm^{-3} and $\varepsilon_n = 16$. For P-type GaAs electron affinity 4.07, $N_{vP} = 7 \times 10^{18}$ cm^{-3}, $N_a = 10^{16}$ cm^{-3} and $\varepsilon_P = 13.1$ 5

3. (a) Define specific resistance. Explain which parameters affect specific resistance. 4

   (b) With proper band diagram explain multiple negative differential resistance in a Resonant Tunneling Diode. 6

   (c) Briefly explain the types of ohmic contacts? 3

   (d) Draw energy-band diagram of the following metal semiconductor junctions and mention whether it is Ohmic or Schottky then also find the ideal Schottky barrier height.
   (i) Al on top of n-GaAs
   (ii) Cu on top of n-GaAs
   (iii) Ag on top of p-GaAs
   (iv) Au on top of p-GaAs

   Here, Al, Cu, Ag, and Au have work functions of 4.06 eV, 4.98 eV, 4.74 eV, and [unclear: 5.47] eV respectively. GaAs has electron affinity of 4.07 eV and Bandgap of 1.519 eV. 12

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Part - B

*(Answer any **two** of the following questions)*

4. (a) Draw the typical structure of a Double Hetero-junction Bipolar Transistor Structure and their doping profile 6

   (b) Describe the advantages of HBT over BJT. Show that for an HBT, the ratio of J_{nE} and J_{pE} can be expressed as $\frac{J_{nE}}{J_{pE}} = \frac{N_{DE}}{N_{AB}} \cdot \frac{W_E}{W_B} \cdot \frac{D_{nE}}{D_{pE}} \exp\left(\frac{\Delta E_g}{kT}\right)$ [*sic*: should be D_{nB}]. 7

   (c) For an ideal Metal Oxide Semiconductor capacitor draw charge distribution, electric field distribution, potential distribution and capacitance versus voltage curve. 6

   (d) Show the relation between Flat band voltage and threshold voltage of a MOS device. 6

5. (a) Derive the equation for emitter injection factor of a bipolar junction transistor. 7

   (b) Briefly explain the following non-ideal effects of a bipolar junction transistor.

   (i) Early effect
   (ii) Current Crowding 6

   (c) Describe the Gummel-Poon model of a bipolar junction transistor. 7

   (d) Consider a uniformly doped silicon npn bipolar transistor at T = 300 K with the following parameters: $D_B = 23$ cm^2/s, $D_E = 8$ cm^2/s, $\tau_{B0} = 2 \times 10^{-7}$ s, $\tau_{E0} = 8 \times 10^{-8}$ s, $N_B = 2 \times 10^{16}$ cm^{-3}, and $x_E = 0.35$ μm. The recombination factor has been determined to be $\delta = 0.9975$. The required common-emitter current gain is $\beta = 150$. A minimum neutral base width of $x_B = 0.80$ μm can be fabricated. (a) Determine an appropriate neutral base width and the minimum emitter doping concentration, N_E, to meet this specification. (b) Using the results of part (a), what are the values of α_T and γ? 5

6. (a) What are the advantages of High Electron Mobility Transistor (HEMT)? 4

   (b) With proper band diagram explain how High Electron Mobility Transistor (HEMT) operates. 7

   (c) Sketch the equivalent circuit for Ebers-Moll model as used in BJT. 5

   (d) Show that maximum cut off frequency of a MESFET is related to doping concentration. 4

   (e) Determine the channel thickness of a GaAs MESFET to achieve threshold voltage $V_T = 0.1$ V. Consider an n-channel GaAs MESFET at T = 300 K with a gold Schottky barrier contact. Assume the barrier height is $\varphi_{Bn} = 0.89$ V. The n-channel doping is $N_d = 2 \times 10^{15}$ cm^{-3}. 5

Term Test 1 (its questions 1 and 2 are 7(a) and 7(b))

[Term Test 1, Course Code: EEE-459, Credit: 3.0, Session: 2018-19, Total Marks: 10, Total Time: 30 minutes]

7. (a) What is ohmic contact? Briefly explain the types of ohmic contacts. 4

   (b) Determine the theoretical barrier height, built-in potential barrier, and maximum electric field in a metal–semiconductor diode for zero applied bias. Consider a contact between gold (5.1 V) and n-type germanium (4.13 V) doped to $N_d = 10^{16}$ cm^{-3} at T = 300 K. $N_C = 1.04 \times 10^{19}$ and $\varepsilon_r = 16$. 6

Term Test 2 (its questions 1 and 2 are 8(a) and 8(b))

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[Term Test 2, Course Code: EEE-459, Credit: 3.0, Session: 2018-19, Total Marks: 10, Total Time: 30 minutes]

8. (a) Sketch the energy band diagram of the MOS capacitor under various bias conditions for both a p-type and s-type [*sic*: should be n-type] substrate. 5

   (b) A p-type Si substrate at T = 300 K was doped at $N_a = 10^{14}\ cm^{-3}$ It has $Q'_{ss} = 10^{10}\ cm^{-2}$ and $t_{ox} = 30\ nm$ Determine the threshold voltage of an MOS system with a gold gate. Consider at T = 300 K, p-type Si electron affinity 4.01, $N_v = 1.04 \times 10^{19}\ \text{cm}^{-3}$, $N_c = 2.8 \times 10^{19}\ \text{cm}^{-3}$, $n_i = 1.5 \times 10^{10}\ \text{cm}^{-3}$ and $\varepsilon_r = 11.7$ and gold p-Si work function $\varphi_{ms} = 0.07$ V. 5

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Contents

1(a) Three Crystal Structures **10**
1 Theory . . . 10
2 Answer . . . 11
3 Keyword Explanation . . . 12
4 Notes and Remarks . . . 12

1(b) Metal–Semiconductor Junction with an Interfacial Layer and Interface States **12**
1 Theory . . . 12
2 Answer . . . 13
3 Keyword Explanation . . . 14
4 Notes and Remarks . . . 14

1(c) Gold on n-Type Germanium Schottky Diode **14**
1 Theory . . . 14
2 Answer . . . 15
3 Keyword Explanation . . . 16
4 Notes and Remarks . . . 16

1(d) Vapour-Phase Epitaxy **17**
1 Theory . . . 17
2 Answer . . . 17
3 Keyword Explanation . . . 19
4 Notes and Remarks . . . 19
5 Final Concise Answer . . . 19

2(a) Band Diagrams of pN and nN Heterojunctions **20**
1 Theory . . . 20
2 Answer . . . 21
3 Keyword Explanation . . . 21
4 Notes and Remarks . . . 21

2(b) Depletion Width of a Heterojunction **22**
1 Theory . . . 22
2 Answer . . . 22
3 Keyword Explanation . . . 24
4 Notes and Remarks . . . 24

2(c) Excess Minority Electrons in the Base of a BJT **24**
1 Theory . . . 24
2 Answer . . . 25
3 Keyword Explanation . . . 26
4 Notes and Remarks . . . 27

2(d) n-Ge / P-GaAs Heterojunction: Doping and Depletion Width **27**
1 Theory . . . 27
2 Answer . . . 27
3 Keyword Explanation . . . 29
4 Notes and Remarks . . . 29
5 Final Concise Answer . . . 29

3(a) Specific Contact Resistance **30**

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1 Theory . . . 30
2 Answer . . . 30
3 Keyword Explanation . . . 31
4 Notes and Remarks . . . 32

3(b) Multiple Negative Differential Resistance in a Resonant Tunnelling Diode 32
1 Theory . . . 32
2 Answer . . . 33
3 Keyword Explanation . . . 34
4 Notes and Remarks . . . 34

3(c) Types of Ohmic Contact 34
1 Theory . . . 34
2 Answer . . . 35
3 Keyword Explanation . . . 35
4 Notes and Remarks . . . 35

3(d) Four Metal–GaAs Junctions 36
1 Theory . . . 36
2 Answer . . . 36
3 Keyword Explanation . . . 37
4 Notes and Remarks . . . 37
5 Final Concise Answer . . . 37

4(a) Structure and Doping Profile of a Double-Heterojunction Bipolar Transistor 39
1 Theory . . . 39
2 Answer . . . 40
3 Keyword Explanation . . . 41
4 Notes and Remarks . . . 41

4(b) Advantages of the HBT and Its Injection Ratio 41
1 Theory . . . 41
2 Answer . . . 42
3 Keyword Explanation . . . 43
4 Notes and Remarks . . . 43

4(c) Charge, Field, Potential and C–V of an Ideal MOS Capacitor 43
1 Theory . . . 44
2 Answer . . . 44
3 Keyword Explanation . . . 46
4 Notes and Remarks . . . 46

4(d) Flat-Band Voltage and Threshold Voltage 47
1 Theory . . . 47
2 Answer . . . 47
3 Keyword Explanation . . . 48
4 Notes and Remarks . . . 48
5 Final Concise Answer . . . 49

5(a) Emitter Injection Efficiency of a BJT 50
1 Theory . . . 50
2 Answer . . . 51

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3 Keyword Explanation . . . 52
4 Notes and Remarks . . . 52

5(b) Early Effect and Current Crowding **53**
1 Theory . . . 53
2 Answer . . . 53
3 Keyword Explanation . . . 55
4 Notes and Remarks . . . 55

5(c) The Gummel–Poon Model **56**
1 Theory . . . 56
2 Answer . . . 56
3 Keyword Explanation . . . 58
4 Notes and Remarks . . . 58

5(d) Base Width and Emitter Doping for a Gain of 150 **58**
1 Theory . . . 58
2 Answer . . . 58
3 Keyword Explanation . . . 59
4 Notes and Remarks . . . 59
5 Final Concise Answer . . . 60

6(a) Advantages of the HEMT **61**
1 Theory . . . 61
2 Answer . . . 61
3 Keyword Explanation . . . 62
4 Notes and Remarks . . . 62

6(b) Operation of the HEMT **62**
1 Theory . . . 62
2 Answer . . . 63
3 Keyword Explanation . . . 64
4 Notes and Remarks . . . 64

6(c) Ebers–Moll Equivalent Circuit **64**
1 Theory . . . 64
2 Answer . . . 64
3 Keyword Explanation . . . 65
4 Notes and Remarks . . . 65

6(d) Maximum Cut-Off Frequency of a MESFET and the Channel Doping **66**
1 Theory . . . 66
2 Answer . . . 67
3 Keyword Explanation . . . 67
4 Notes and Remarks . . . 68

6(e) Channel Thickness of a GaAs MESFET for a Threshold of 0.1 V **68**
1 Theory . . . 68
2 Answer . . . 68
3 Keyword Explanation . . . 69
4 Notes and Remarks . . . 69
5 Final Concise Answer . . . 69

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7(a) Ohmic Contacts and Their Types **70**
1 Theory . . . 70
2 Answer . . . 70
3 Keyword Explanation . . . 71
4 Notes and Remarks . . . 71

7(b) Gold on n-Type Germanium (Term Test) **71**
1 Theory . . . 71
2 Answer . . . 72
3 Keyword Explanation . . . 73
4 Notes and Remarks . . . 73
5 Final Concise Answer . . . 73

8(a) MOS Energy-Band Diagrams Under Bias **74**
1 Theory . . . 74
2 Answer . . . 75
3 Keyword Explanation . . . 75
4 Notes and Remarks . . . 76

8(b) Threshold Voltage with a Gold Gate **76**
1 Theory . . . 76
2 Answer . . . 76
3 Keyword Explanation . . . 77
4 Notes and Remarks . . . 77
5 Final Concise Answer . . . 77

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1. (a) Draw the following crystal structures.

   (i) Hexagonal close packed crystal structure
   (ii) Zinc blende crystal structure.
   (iii) Body centered cubic crystal structure Marks: 6

   (b) Sketch energy-band diagram of a metal-semiconductor junction with an interfacial layer and interface states. Marks: 5

   (c) Determine the theoretical barrier height, built-in potential barrier, and maximum electric field in a metal–semiconductor diode for zero applied bias. Consider a contact between gold (metal work function 5.1 V) and n-type germanium (electron affinity 4.13 V) doped to $N_d = 10^{16}$ cm^{-3} at T = 300 K. $N_C = 1.04 \times 10^{19}$ and $\varepsilon_r = 16$ Marks: 6

   (d) Explain the process of vapor phase epitaxy. Marks: 8

1.(a) Draw the following crystal structures. (i) Hexagonal close packed crystal structure (ii) Zinc blende crystal structure. (iii) Body centered cubic crystal structure Marks: 6

1. Theory

1. What is a crystal? (Core Idea)

A **crystal** is a solid whose atoms sit in a pattern that repeats regularly in all three directions. The pattern is described by two things:

- a **lattice**: a regular array of points in space, every point having exactly the same surroundings;
- a **basis**: the atom (or group of atoms) placed at each lattice point.

The smallest block that, repeated side by side, rebuilds the whole crystal and shows its symmetry is the **unit cell**. For cubic crystals it is a cube of side a, the **lattice constant** (Neamen, Section 1.3).

2. Useful numbers of a cell

- **Atoms per cell:** a corner atom is shared by 8 cells (counts $\frac{1}{8}$), a face atom by 2 ($\frac{1}{2}$), an edge atom by 4 ($\frac{1}{4}$), an atom inside by 1.
- **Coordination number:** how many nearest neighbours each atom has.
- **Packing fraction:** the share of the cell's volume filled by atoms treated as touching hard spheres.

3. Close packing

Equal spheres pack most densely by stacking flat hexagonal layers: a layer A, then a layer B sitting in the hollows of A, then a third layer either over A again (**ABAB…: hexagonal close packed**) or in the remaining hollows (**ABCABC…: face-centred cubic**). Both fill 74 % of space and give each atom 12 neighbours.

4. The tetrahedral bond of semiconductors

Si, Ge and the III–V compounds bond through four sp^3 bonds pointing to the corners of a tetrahedron. Each atom therefore has 4 nearest neighbours. Put two fcc lattices together, one shifted by a quarter of the body diagonal, $(\frac{a}{4}, \frac{a}{4}, \frac{a}{4})$: every atom of one lattice is then at the centre of a tetrahedron of atoms of the other. With the same atom on both lattices this is the **diamond** structure (Si, Ge); with two different atoms it is the **zinc blende** (sphalerite) structure (GaAs, InP, GaP).

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2. Answer

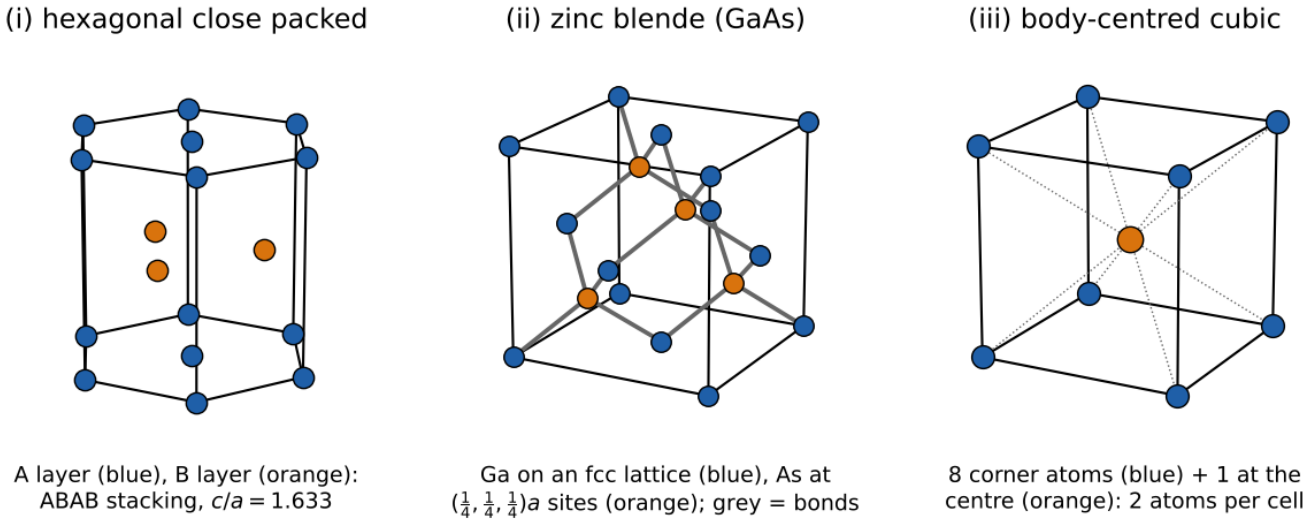


Figure 1: The three crystal structures: (i) hexagonal close packed, (ii) zinc blende (GaAs), (iii) body-centred cubic

Reading the figure. In (i) the hexagonal prism has atoms at the 12 corners and the centres of the two hexagonal faces (blue, layer A) and three atoms in the middle plane over alternate triangles (orange, layer B). In (ii) the cube has blue Ga atoms at the 8 corners and 6 face centres (an fcc lattice) and four orange As atoms inside, at $(\frac{1}{4},\frac{1}{4},\frac{1}{4})a$ and the three equivalent positions; the grey lines are the bonds, each As joined to four Ga in a tetrahedron. In (iii) the cube has blue atoms at the 8 corners and one orange atom at the body centre; the dotted lines show that the centre atom touches the corner atoms along the body diagonals.

(i) Hexagonal close packed (hcp)

- Hexagonal prism with sides a and height c; ideal $c/a = \sqrt{8/3} = 1.633$.
- Atoms: 12 corners $(\times\frac{1}{6})$ + 2 face centres $(\times\frac{1}{2})$ + 3 inside $= 2 + 1 + 3 = 6$ per prism (2 per primitive cell).
- Stacking ABAB…; coordination number 12; packing fraction 0.74.
- Examples: Mg, Zn, Ti, Co; the semiconductor analogue is the **wurtzite** structure (GaN, ZnO, hexagonal CdS), two interpenetrating hcp lattices.

(ii) Zinc blende

- Cube of side a (GaAs: $a = 5.65$ Å). Group-III atoms (Ga) on an fcc lattice: 8 corners $(\times\frac{1}{8})$ + 6 face centres $(\times\frac{1}{2}) = 4$ Ga. Group-V atoms (As) at $(\frac{1}{4},\frac{1}{4},\frac{1}{4})a$, $(\frac{3}{4},\frac{3}{4},\frac{1}{4})a$, $(\frac{3}{4},\frac{1}{4},\frac{3}{4})a$, $(\frac{1}{4},\frac{3}{4},\frac{3}{4})a$: 4 As. Total 8 atoms per cell.
- Each atom has 4 nearest neighbours of the other kind, at distance $\sqrt{3}a/4$ (2.45 Å in GaAs), in a tetrahedron.
- It is the diamond lattice with two different atoms; packing fraction 0.34. Examples: GaAs, InP, GaP, ZnS.

(iii) Body-centred cubic (bcc)

- Cube of side a with atoms at the 8 corners and 1 at the centre: $8 \times \frac{1}{8} + 1 = 2$ atoms per cell.
- Atoms touch along the body diagonal: $4r = \sqrt{3}a$. Coordination number 8; packing fraction $\dfrac{2 \cdot \frac{4}{3}\pi r^3}{a^3} = \dfrac{\sqrt{3}\pi}{8} = 0.68$.
- Examples: Na, Fe (α), W, Mo.

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Structure	Atoms per cell	Nearest neighbours	Packing fraction	Example
hcp	6 (prism)	12	0.74	Zn, Mg
zinc blende	8	4	0.34	GaAs
bcc	2	8	0.68	W, Na

Table 1: Summary

3. Keyword Explanation

1. Unit cell

↩ back

The repeating block of a crystal; translating it along its edges fills all space.

2. Coordination number

↩ back

The number of nearest neighbours of an atom: 12 in hcp, 8 in bcc, 4 in zinc blende.

3. Zinc blende

↩ back

Two fcc lattices of different atoms, displaced by a quarter of the body diagonal; the structure of most III–V compounds.

4. Notes and Remarks

- **Packing fraction of zinc blende:** atoms touch along a quarter of the body diagonal, $2r = \sqrt{3}a/4$; with 8 atoms, fraction $= 8 \cdot \frac{4}{3}\pi r^3/a^3 = \sqrt{3}\pi/16 = 0.34$ (taking equal radii). The open structure is why light atoms can move through Si and GaAs easily.
- The fcc lattice (Neamen, Figure 1.5c) is the ABCABC close packing; hcp and fcc differ only in the third layer.

1.(b) Sketch energy-band diagram of a metal-semiconductor junction with an interfacial layer and interface states. Marks: 5

1. Theory

1. The ideal metal–semiconductor junction (Core Idea)

The **vacuum level** E_0 is the energy of an electron at rest just outside a material. The **work function** $e\phi_m$ of a metal is the energy from its Fermi level to E_0; the **electron affinity** $e\chi$ of a semiconductor is the energy from its conduction-band edge E_c to E_0. In the ideal model (no interfacial layer, no interface states) a metal on n-type semiconductor gives a barrier

$$\phi_{B0} = \phi_m - \chi$$

seen by electrons in the metal (the **Schottky limit**, Neamen Eq. 9.1). The semiconductor bands bend up over a depletion layer by $V_{bi} = \phi_{B0} - \phi_n$, where $e\phi_n = E_c - E_F$ in the bulk.

2. What a real interface looks like

Measured barriers hardly follow $\phi_m - \chi$. Two things are found at a real contact:

- an **interfacial layer**: a thin oxide or disordered film, of thickness δ (a few ångström) and permittivity ϵ_i, between metal and semiconductor. Electrons tunnel through it easily, but a voltage Δ can drop across it;
- **interface states**: allowed electron levels inside the band gap at the surface, from broken bonds, defects and the oxide, with a density D_{it} per unit area per unit energy.

3. The neutral level and Fermi-level pinning

The interface states have a **neutral level** $e\phi_0$, measured up from E_v at the surface: if they are filled exactly up to $e\phi_0$, the surface is neutral. If E_F lies above $e\phi_0$, the extra filled states carry negative charge; if below, positive charge.

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In equilibrium the charges must balance: metal charge + interface-state charge + depletion charge = 0. With a large D_{it}, a tiny movement of E_F at the surface changes the interface charge a lot, so the interface states supply almost all the charge, and the surface E_F stays near $e\phi_0$ whatever the metal: **Fermi-level pinning**. The barrier then becomes

$$\phi_{B0} \approx \frac{E_g}{e} - \phi_0 \qquad \text{(Bardeen limit)}$$

and in general (Cowley and Sze, Sze & Ng)

$$\phi_{B0} = \gamma(\phi_m - \chi) + (1-\gamma)\left(\frac{E_g}{e} - \phi_0\right), \qquad \gamma = \frac{\epsilon_i}{\epsilon_i + e^2\delta D_{it}}$$

2. Answer

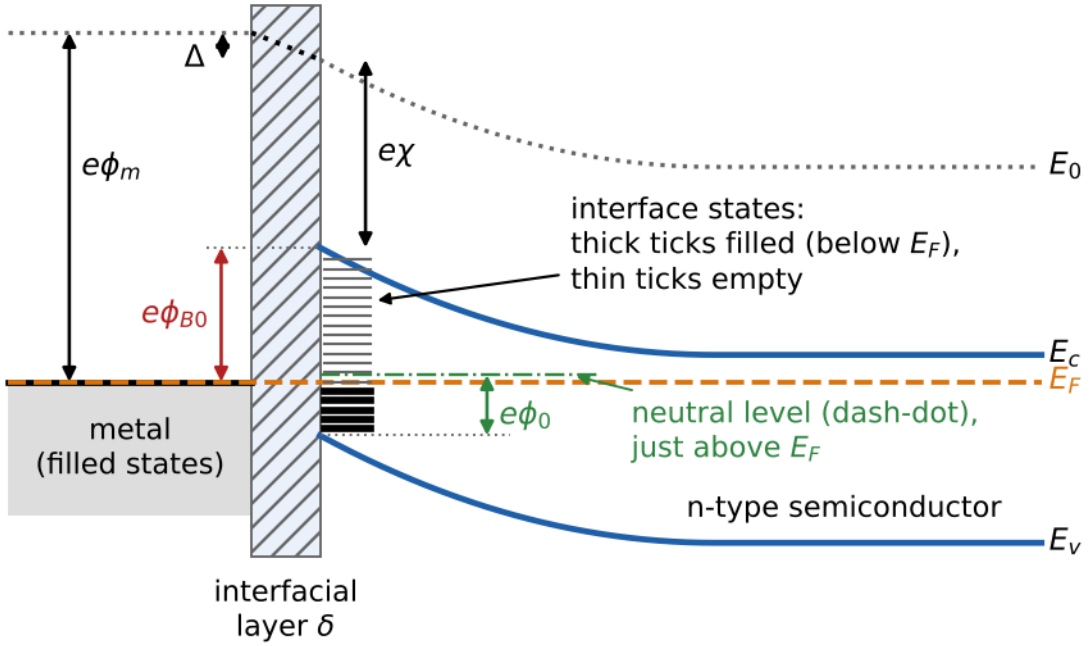


Figure 2: Band diagram of a metal on n-type semiconductor with an interfacial layer and interface states; the Fermi level at the surface is held near the neutral level (reused from the 2020–21 paper)

Reading the figure. Left: the metal, with its states filled up to E_F (grey). The hatched strip is the interfacial layer of thickness δ; the dotted vacuum level drops by Δ across it. At the semiconductor surface the short ticks are the interface states: thick black ticks below the dashed orange E_F are filled, thin ticks above it are empty. The dash-dot green line is the neutral level $e\phi_0$ above E_v; E_F sits close to it. The n-type bands (solid blue E_c and E_v) bend up towards the surface; the barrier $e\phi_{B0}$ (red arrow) runs from E_F to E_c at the surface.

Features to show in the sketch:

1. A common, flat Fermi level E_F through metal, layer and semiconductor (equilibrium).
2. The interfacial layer of thickness δ, with a potential drop Δ across it (the vacuum level is continuous but slopes in the layer).
3. Interface states at the semiconductor surface, a continuum in the gap, filled up to E_F; their neutral level $e\phi_0$ above E_v.
4. Band bending in the semiconductor (depletion layer), barrier $e\phi_{B0}$ from E_F to E_c at the surface, and eV_{bi} from the bulk E_c to the surface E_c.
5. The labels $e\phi_m$, $e\chi$, $e\phi_0$, $e\phi_{B0}$, E_g.

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Result: charge balance $Q'_m + Q'_{ss} + Q'_{sc} = 0$ gives $\phi_{B0} = \gamma(\phi_m - \chi) + (1-\gamma)(E_g/e - \phi_0)$; few states ($\gamma \to 1$) give the ideal Schottky barrier $\phi_m - \chi$; many states ($\gamma \to 0$) pin the barrier at $E_g/e - \phi_0$, nearly independent of the metal (GaAs: about 0.8–0.9 V for most metals).

3. Keyword Explanation

1. Interface states

↩ back

Electron energy levels inside the band gap at the semiconductor surface, from dangling bonds and defects.

2. Fermi-level pinning

↩ back

The surface Fermi level held near the neutral level $e\phi_0$ by a large density of interface states, so the barrier hardly depends on the metal.

3. Interfacial layer

↩ back

A thin insulating film, a few ångström thick, between metal and semiconductor; electrons tunnel through it, but a voltage Δ drops across it.

4. Notes and Remarks

- **Where γ comes from:** for n-type material the interface charge is $Q'_{ss} = -eD_{it}(E_g - e\phi_0 - e\phi_{B0})$ and the layer voltage is $\Delta = -\delta Q'_m/\epsilon_i = \phi_m - (\chi + \phi_{B0})$. With $Q'_m \approx -Q'_{ss}$ (the depletion charge is much smaller), eliminating Q'_m gives the formula above (Sze & Ng, Section 3.2).
- Measured slopes $d\phi_{B0}/d\phi_m$ are about 0.27 for Si and only 0.07 for GaAs (Sze & Ng), so GaAs is almost fully pinned.
- Also drawn in 4(b) of the 2020–21 paper (extra reading).

1.(c) Determine the theoretical barrier height, built-in potential barrier, and maximum electric field in a metal–semiconductor diode for zero applied bias. Consider a contact between gold (metal work function 5.1 V) and n-type germanium (electron affinity 4.13 V) doped to $N_d = 10^{16}$ cm^{-3} at T = 300 K. $N_C = 1.04 \times 10^{19}$ and ε_r = 16 Marks: 6

1. Theory

1. Barrier height and built-in potential (Core Idea)

When a metal of work function ϕ_m touches an n-type semiconductor of electron affinity χ with $\phi_m > \phi_s$, electrons flow from the semiconductor into the metal until the Fermi levels line up. The ideal barrier for electrons in the metal is

$$\phi_{B0} = \phi_m - \chi$$

The semiconductor's bulk Fermi level lies $e\phi_n$ below E_c, where (Neamen, Boltzmann approximation)

$$n_0 = N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right] \quad \Rightarrow \quad \phi_n = \frac{E_c - E_F}{e} = \frac{kT}{e}\ln\frac{N_c}{N_d}$$

for full ionisation, $n_0 = N_d$. The barrier seen from the semiconductor, the **built-in potential**, is

$$V_{bi} = \phi_{B0} - \phi_n$$

2. Depletion width and field

With the depletion approximation (charge $+eN_d$ for $0 < x < x_n$), Poisson's equation $dE/dx = eN_d/\epsilon_s$ and $E(x_n) = 0$ give

$$E(x) = -\frac{eN_d}{\epsilon_s}(x_n - x)$$

Integrating once more, the potential rises by $V_{bi} = eN_d x_n^2/2\epsilon_s$, so

$$x_n = \left[\frac{2\epsilon_s V_{bi}}{eN_d}\right]^{1/2}, \qquad |E_{max}| = \frac{eN_d x_n}{\epsilon_s}$$

(Neamen, Eqs. 9.1–9.7). The field is largest at the metal surface and falls linearly to zero at x_n.

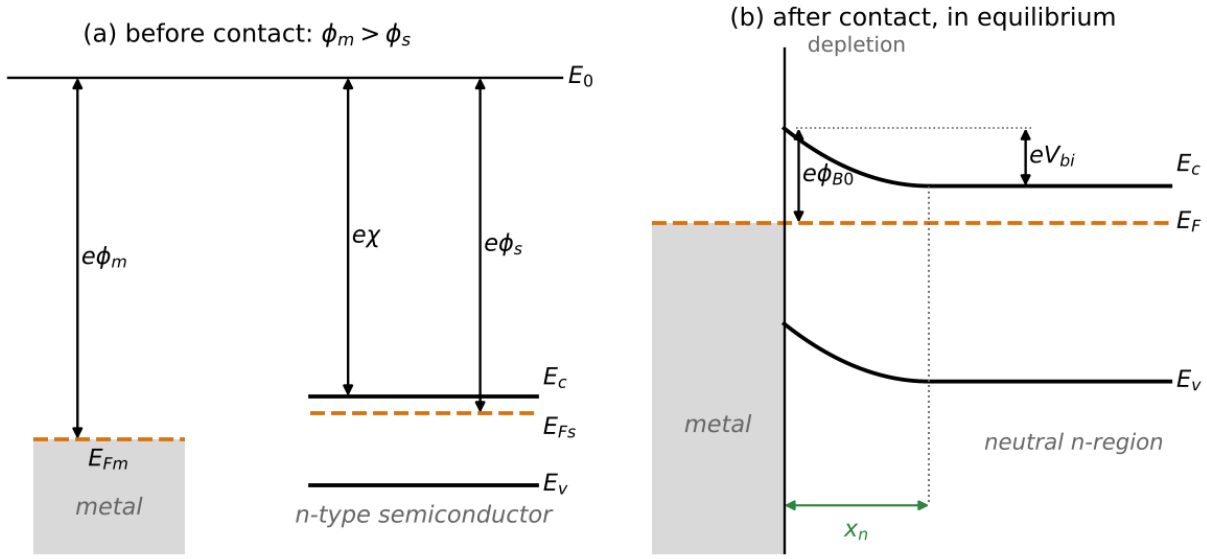


Figure 3: Metal and n-type semiconductor with $\phi_m > \phi_s$: before contact and in equilibrium (reused from the 2021–22 paper)

Reading the figure. In panel (a) both materials share the vacuum level; the n-semiconductor's E_{Fs} (dashed orange) is higher than the metal's E_{Fm}, because $\phi_s < \phi_m$. In panel (b), after contact, E_F is flat, the bands bend up over x_n, the barrier from the metal is $e\phi_{B0}$ and from the semiconductor eV_{bi}.

2. Answer

Given:

- $\phi_m = 5.1$ V (gold), $\chi = 4.13$ V (germanium)
- $N_d = 10^{16}$ cm^{-3}, $N_c = 1.04 \times 10^{19}$ cm^{-3} (Ge), $T = 300$ K, $kT/e = 0.0259$ V
- $\epsilon_s = 16\,\epsilon_0 = 16 \times 8.85 \times 10^{-14} = 1.416 \times 10^{-12}$ F/cm; $e = 1.6 \times 10^{-19}$ C

Step 1: Theoretical barrier height

$$\phi_{B0} = \phi_m - \chi = 5.1 - 4.13 = 0.97 \text{ V}$$

$$\boxed{\phi_{B0} = 0.97 \text{ V}}$$

Electrons in the gold see a 0.97 eV barrier.

Step 2: Position of the Fermi level in n-Ge

$$\phi_n = \frac{kT}{e}\ln\frac{N_c}{N_d} = 0.0259 \ln\left(\frac{1.04 \times 10^{19}}{10^{16}}\right) = 0.0259 \times 6.947 = 0.180 \text{ V}$$

Step 3: Built-in potential barrier

$$V_{bi} = \phi_{B0} - \phi_n = 0.97 - 0.180 = 0.790 \text{ V}$$

$$\boxed{V_{bi} = 0.790 \text{ V}}$$

Step 4: Depletion width at zero bias

$$x_n = \left[\frac{2\epsilon_s V_{bi}}{eN_d}\right]^{1/2} = \left[\frac{2(1.416 \times 10^{-12})(0.790)}{(1.6 \times 10^{-19})(10^{16})}\right]^{1/2} = 3.74 \times 10^{-5} \text{ cm} = 0.374\ \mu\text{m}$$

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Step 5: Maximum electric field

$$|E_{max}| = \frac{eN_d x_n}{\epsilon_s} = \frac{(1.6\times10^{-19})(10^{16})(3.74\times10^{-5})}{1.416\times10^{-12}} = 4.23\times10^4 \text{ V/cm}$$

$$\boxed{|E_{max}| = 4.23\times10^4 \text{ V/cm}}$$

It occurs at the metal surface ($x = 0$).

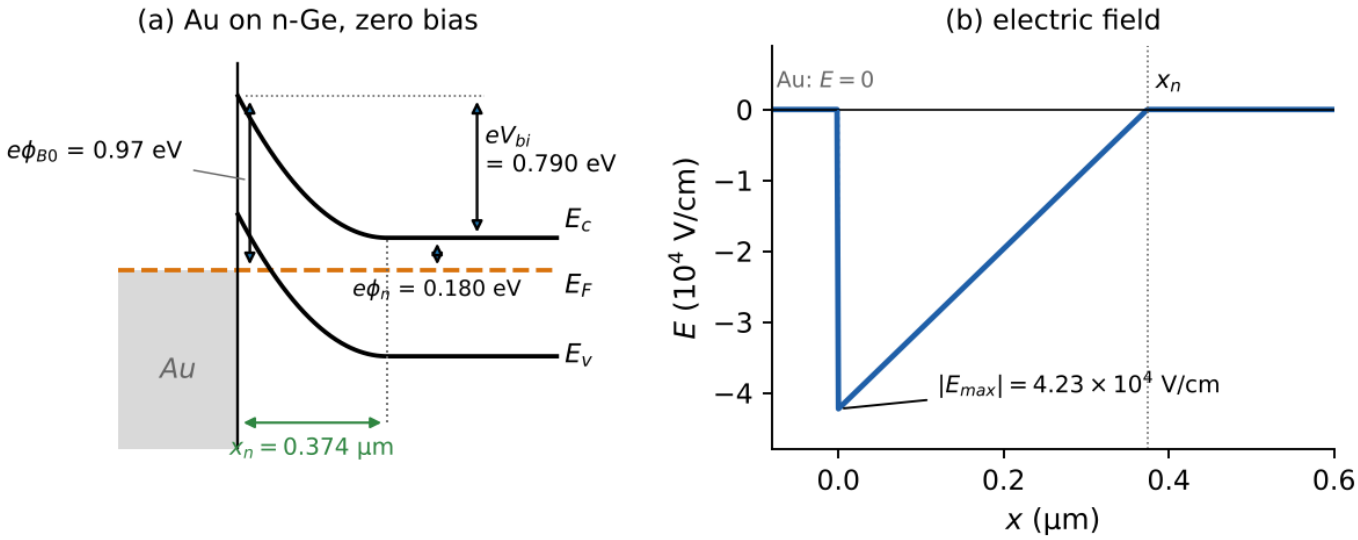


Figure 4: Au on n-Ge at zero bias: band diagram with the computed values, and the electric field

Reading the figure. Panel (a): the dashed orange E_F is flat; the solid black E_c and E_v of the Ge bend up over $x_n = 0.374$ µm (green arrow). The barrier from the gold is 0.97 eV, the built-in barrier from the Ge is 0.790 eV, and in the bulk E_c is 0.180 eV above E_F. Panel (b): the field (solid blue) is zero in the gold, -4.23×10^4 V/cm at the surface, rises linearly and reaches zero at x_n.

Final Answer:

- Barrier height $\phi_{B0} = 0.97$ V
- Built-in potential $V_{bi} = 0.790$ V
- Maximum field $|E_{max}| = 4.23\times10^4$ V/cm (with $x_n = 0.374$ µm)

3. Keyword Explanation

1. Schottky barrier height

↩ back

$\phi_{B0} = \phi_m - \chi$: the energy barrier, in volts, that electrons in the metal must cross to enter the semiconductor.

2. Built-in potential

↩ back

$V_{bi} = \phi_{B0} - \phi_n$: the band bending in the semiconductor at zero bias.

4. Notes and Remarks

- The paper writes the work function and electron affinity in volts (5.1 V, 4.13 V); as energies they are 5.1 eV and 4.13 eV.
- This is the method of Neamen's Example 9.1 (tungsten on n-Si), here with Ge data. Measured Au/n-Ge barriers are lower (about 0.6 V) because the Ge surface Fermi level is pinned near the valence band (Sze & Ng).
- The zero-bias capacitance per area is $C' = \epsilon_s/x_n = 3.79\times10^{-8}$ F/cm^2.

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1.(d) Explain the process of vapor phase epitaxy. Marks: 8

1. Theory

1. What is epitaxy? (Core Idea)

Epitaxy (Greek *epi* "upon", *taxis* "arrangement") is the growth of a thin **single-crystal** layer on a single-crystal **substrate**, so that the layer continues the substrate's crystal lattice. The substrate acts as the seed, and the growth happens far below the melting point (Neamen, Section 1.7.2).

- **Homoepitaxy:** the same material (Si on Si, GaAs on GaAs).
- **Heteroepitaxy:** a different material with a similar lattice (AlGaAs on GaAs, InGaAs on InP).

Epitaxy is how the precisely doped thin layers of transistors, lasers and HEMTs are made: their doping and composition can change sharply from layer to layer.

2. The three main methods

- **Vapour-phase epitaxy (VPE):** the elements arrive as gases (chlorides, hydrides or metal-organics) that react at the hot substrate.
- **Liquid-phase epitaxy (LPE):** the substrate is dipped in a cooling metal solution (for example Ga with dissolved As) and the crystal grows from the liquid.
- **Molecular-beam epitaxy (MBE):** beams of evaporated atoms hit a heated substrate in ultra-high vacuum.

3. Steps of growth from a vapour

A gas-phase process happens in five steps: (1) the reactants are carried by a gas to the substrate, (2) they diffuse through a thin boundary layer above it, (3) they adsorb on the surface, (4) they react and the product atoms move on the surface to sites at atomic steps, where they join the crystal, (5) the by-products (HCl, CH_4, …) leave and are swept away. The slowest step sets the rate: at lower temperature the surface reaction (rate grows steeply with temperature), at higher temperature the supply of gas (rate depends on flow).

2. Answer

Vapour-phase epitaxy (VPE) grows a single-crystal layer on a heated single-crystal substrate from chemical vapours that react or decompose at its surface; the deposited atoms take up the substrate's crystal arrangement. It is also called chemical vapour deposition (CVD) epitaxy.

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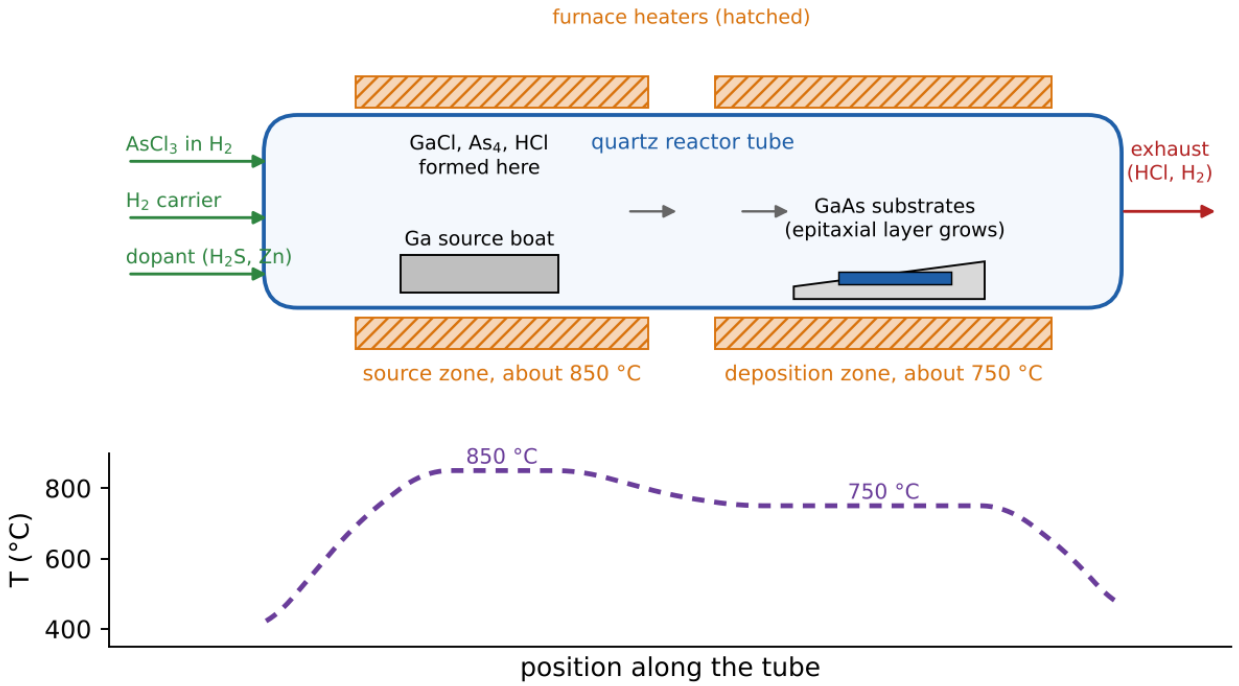


Figure 5: Horizontal chloride VPE reactor for GaAs: gas inlets, the Ga source zone, the substrates in the cooler deposition zone, and the temperature along the tube

Reading the figure. Gases enter at the left (green arrows): $AsCl_3$ in hydrogen, the hydrogen carrier and the dopant gas. In the hotter source zone (about 850 °C, hatched heaters) they pass over a boat of liquid Ga. Downstream, in the cooler deposition zone (about 750 °C), the GaAs substrates lie on a tilted holder (susceptor), where the layer grows. By-products leave at the right (red arrow). The lower plot (dashed purple) is the temperature along the tube.

Silicon (Neamen): silicon tetrachloride is reduced by hydrogen at the heated substrate (about 1200 °C):

$$SiCl_4 + 2H_2 \rightarrow Si + 4HCl$$

The Si atoms join the crystal; HCl leaves with the gas. Dopants are added as hydride gases (PH_3, AsH_3 for n-type, B_2H_6 for p-type).

GaAs, chloride process (the reactor in the figure):

1. $AsCl_3$ is reduced by H_2: $4AsCl_3 + 6H_2 \rightarrow As_4 + 12HCl$.
2. In the source zone (about 850 °C) the HCl reacts with the Ga boat: $2Ga + 2HCl \rightarrow 2GaCl + H_2$.
3. In the cooler deposition zone (about 750 °C) the reaction reverses on the substrate: $6GaCl + As_4 + 3H_2 \rightarrow 6GaAs + 6HCl$. The lower temperature makes deposition favourable.
4. The exhaust gases (HCl, H_2) are removed.

(The hydride process uses AsH_3 and HCl over Ga instead; **MOVPE/MOCVD** uses metal-organics, $Ga(CH_3)_3 + AsH_3 \rightarrow GaAs + 3CH_4$, at about 600–750 °C, and is today's main production method for III–V layers.)

Process steps: clean and etch the substrate (often an in-situ HCl etch to remove the damaged surface) → heat in H_2 → admit the reactants at controlled flows (mass-flow controllers) → growth at typically 0.1–1 µm/min → cool in the protective gas.

Control of the layer:

- **Thickness:** growth time and gas flows.
- **Doping:** dopant gas in the stream (H_2S or SiH_4 for n-type, diethylzinc for p-type GaAs); doping can be changed sharply between layers.
- **Composition:** mixing sources (for example GaCl + InCl, AsH_3 + PH_3) gives ternary and quaternary alloys.

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Advantages: high purity and crystal quality, good thickness and doping uniformity over large areas, many wafers per run, sharp doping steps; suitable for production. **Limitations:** toxic and corrosive gases (AsH_3, HCl); auto-doping from the substrate at high temperature; less sharp interfaces than MBE for very thin quantum wells.

3. Keyword Explanation

1. Substrate

↩ back

The single-crystal wafer that the layer grows on; it sets the crystal orientation of the layer.

2. Susceptor

↩ back

The heated holder (often graphite) on which the substrates lie.

3. Boundary layer

↩ back

The slow gas layer just above the substrate, through which reactants must diffuse.

4. Notes and Remarks

- **Choice of reactions:** Neamen describes Si CVD from $SiCl_4$; the GaAs chloride reactions are the standard ones of III–V VPE (Sze & Ng, Appendix; any growth text). Temperatures are typical values.
- **Lattice matching:** in heteroepitaxy the layer must have nearly the same lattice constant as the substrate, or it grows strained up to a critical thickness and then forms misfit dislocations.

5. Final Concise Answer

(a) hcp: hexagonal prism, ABAB stacking, $c/a = 1.633$, 6 atoms per prism, 12 neighbours, packing 0.74 (Zn, Mg). Zinc blende: fcc lattice of Ga plus fcc lattice of As shifted by $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})a$, 8 atoms per cell, 4 tetrahedral neighbours, packing 0.34 (GaAs). bcc: corners plus body centre, 2 atoms per cell, 8 neighbours, packing 0.68 (W, Na).

(b) Flat E_F; a thin interfacial layer δ with a voltage drop Δ; interface states in the gap at the semiconductor surface, filled up to E_F, with neutral level $e\phi_0$ above E_v; bands bent up; barrier $e\phi_{B0}$. Charge balance gives $\phi_{B0} = \gamma(\phi_m - \chi) + (1-\gamma)(E_g/e - \phi_0)$, $\gamma = \epsilon_i/(\epsilon_i + e^2\delta D_{it})$: many states pin the barrier near $E_g/e - \phi_0$.

(c) Au on n-Ge: $\phi_{B0} = 5.1 - 4.13 = 0.97$ V; $\phi_n = 0.0259\ln(1.04\times10^{19}/10^{16}) = 0.180$ V; $V_{bi} = 0.790$ V; $x_n = 0.374$ µm; $|E_{max}| = 4.23\times10^4$ V/cm.

(d) VPE grows a single-crystal layer on a heated substrate from reacting gases: Si from $SiCl_4 + H_2$; GaAs by the chloride process ($AsCl_3 + H_2$ over a Ga boat at 850 °C forms GaCl and As_4, which deposit GaAs on the substrate at 750 °C) or by MOVPE. Thickness by time and flow, doping by dopant gases, composition by mixing sources; high purity, uniform, production-scale.

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2. (a) Draw the band diagrams of a pN and a nN hetero-junction. Marks: 5
   (b) Derive the equation for the depletion region width of a hetero-junction. Marks: 7
   (c) For forward active mode derive the excess minority carrier concentration equation at base of a bipolar junction transistor Marks: 8
   (d) Determine the n-type doping concentration, and depletion region width in P-type side of semiconductor for an n-Ge and P-GaAs hetero-junction. Consider built in potential is 1.0 V and T = 300 K, n-type Ge electron affinity 4.13, N_{vn} = [unclear: 6 ×] 10^{18} cm^{-3}, $N_d = 10^{16}$ cm^{-3}, $n_i = 2.4\times10^{13}$ cm^{-3} and $\varepsilon_n = 16$. For P-type GaAs electron affinity 4.07, $N_{vP} = 7\times10^{18}$ cm^{-3}, $N_a = 10^{16}$ cm^{-3} and $\varepsilon_P = 13.1$ Marks: 5

2.(a) Draw the band diagrams of a pN and a nN hetero-junction. Marks: 5

1. Theory

1. Heterojunction and its notation (Core Idea)

A **heterojunction** joins two different semiconductors with different band gaps, for example GaAs ($E_g = 1.42$ eV) and $Al_{0.3}Ga_{0.7}As$ ($E_g \approx 1.80$ eV), which have almost the same lattice constant. Neamen's notation: a **small letter** marks the narrow-gap material and a **capital letter** the wide-gap one. So

- **pN:** p-type narrow gap (p-GaAs) with N-type wide gap (N-AlGaAs): an **anisotype** junction (opposite doping types), a diode;
- **nN:** n-type narrow gap with N-type wide gap: an **isotype** junction (the same type).

2. Lining up the bands: Anderson's rule

Before contact both materials are drawn on a common **vacuum level** E_0. Each E_c is its electron affinity χ below E_0, so at the interface

$$\Delta E_c = e(\chi_n - \chi_N), \qquad \Delta E_v = \Delta E_g - \Delta E_c$$

(Neamen, Eq. 9.34). For GaAs/AlGaAs the narrow gap lies inside the wide gap (**straddling**, type I).

3. Rules for the equilibrium diagram

1. E_F is flat through the whole structure.
2. Far from the junction each side has its own bulk band positions relative to E_F.
3. The vacuum level is continuous; E_c and E_v follow it in parallel on each side.
4. At the interface E_c jumps by ΔE_c and E_v by ΔE_v.
5. The total bending of E_0 is eV_{bi}, the difference of the two work functions.

Electrons move from the side with the higher E_F (lower work function) to the other side. In a pN junction both sides are **depleted**; in an nN junction the electrons that leave the wide-gap N side **accumulate** in the narrow-gap n side, against the interface.

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2. Answer

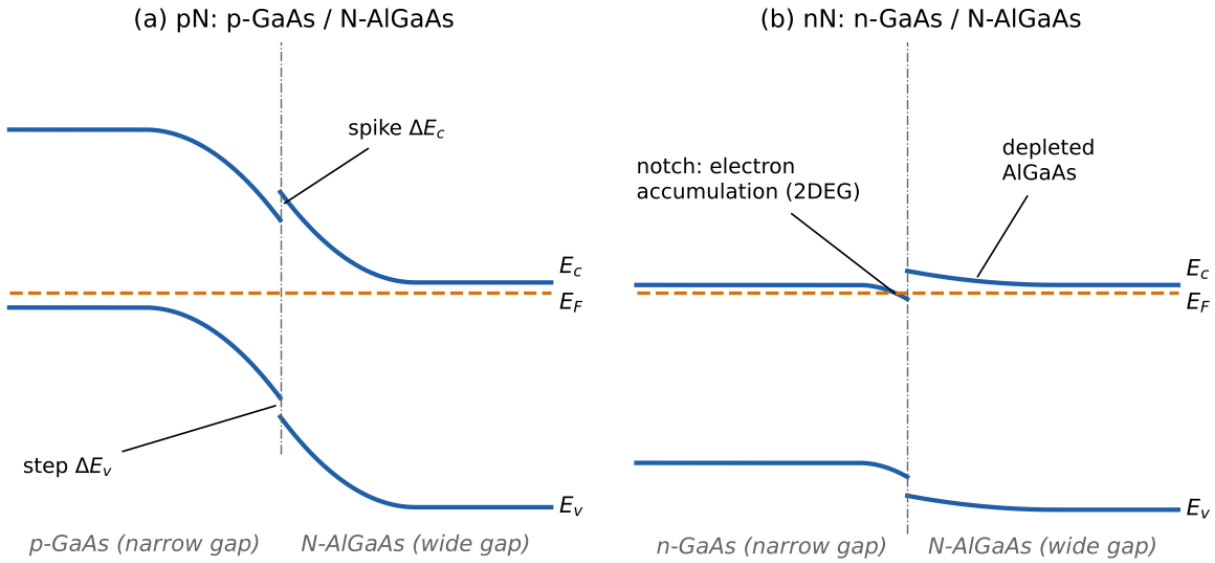


Figure 6: Equilibrium band diagrams (solid blue E_c and E_v, dashed orange E_F): (a) pN, p-GaAs / N-AlGaAs; (b) nN, n-GaAs / N-AlGaAs

Reading the figure. In both panels the vertical dash-dot line is the interface, the narrow-gap GaAs is on the left and the wide-gap AlGaAs on the right.

- **(a) pN:** in the p-GaAs E_F is near E_v; in the N-AlGaAs near E_c. Holes and electrons leave the junction, so both sides are depleted and the bands bend *down* from left to right by eV_{bi} in total. At the interface E_c jumps *up* by ΔE_c (a **spike**, an extra barrier for electrons going from N to p) and E_v steps *down* by ΔE_v (a larger barrier for holes going from p to N: the wide-gap N side injects electrons but receives few holes).
- **(b) nN:** electrons flow from the N-AlGaAs into the n-GaAs. The AlGaAs next to the interface is depleted (positive donors, bands bend up towards the interface); the GaAs side gains electrons, and its E_c dips below E_F at the interface: a narrow **notch** (potential well) in which the electrons form a two-dimensional electron gas. E_c jumps up by ΔE_c at the interface.

Values for GaAs / $Al_{0.3}Ga_{0.7}As$: $\Delta E_g = 1.247 \times 0.3 = 0.374$ eV; with $\Delta E_c \approx 0.6\,\Delta E_g$: $\Delta E_c \approx 0.22$ eV, $\Delta E_v \approx 0.15$ eV.

Key points for the sketch: flat E_F; bulk positions far away; continuous vacuum level; discontinuities ΔE_c, ΔE_v at the interface; depletion on both sides for pN, depletion (N) plus accumulation (n) for nN.

3. Keyword Explanation

1. Anisotype and isotype

↩ back

A heterojunction between opposite doping types (pN, nP) or the same type (nN, pP).

2. Spike and notch

↩ back

The sharp step up of E_c at the interface (spike) and the potential well next to it on the narrow-gap side (notch).

4. Notes and Remarks

- The offset ratio $\Delta E_c \approx 0.6\,\Delta E_g$ is the measured GaAs/AlGaAs value (Sze & Ng); Anderson's rule with tabulated χ gives a different split. The shapes do not depend on it.
- The nN notch is the basis of the HEMT (see 6(b)).

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2.(b) Derive the equation for the depletion region width of a hetero-junction. Marks: 7

1. Theory

1. Charges in an abrupt nP heterojunction (Core Idea)

Take an **nP** junction: n-type narrow-gap material (doping N_{dn}, permittivity ϵ_n) for $-x_n < x < 0$ and P-type wide-gap material (doping N_{aP}, permittivity ϵ_P) for $0 < x < x_P$. In the **depletion approximation** the region $-x_n < x < x_P$ has no free carriers: the n side holds the positive donor ions, $\rho = +eN_{dn}$; the P side the negative acceptor ions, $\rho = -eN_{aP}$. Outside, the material is neutral and the field is zero. Assume no interface charge.

2. The two physics rules used

- **Poisson's equation** in each material, with its own permittivity: $\dfrac{dE}{dx} = \dfrac{\rho}{\epsilon}$, and $E = -\dfrac{d\phi}{dx}$.
- At the interface (no surface charge) the **electric flux density** $D = \epsilon E$ is continuous, and the **potential** is continuous. The field itself jumps, since $\epsilon_n \neq \epsilon_P$.

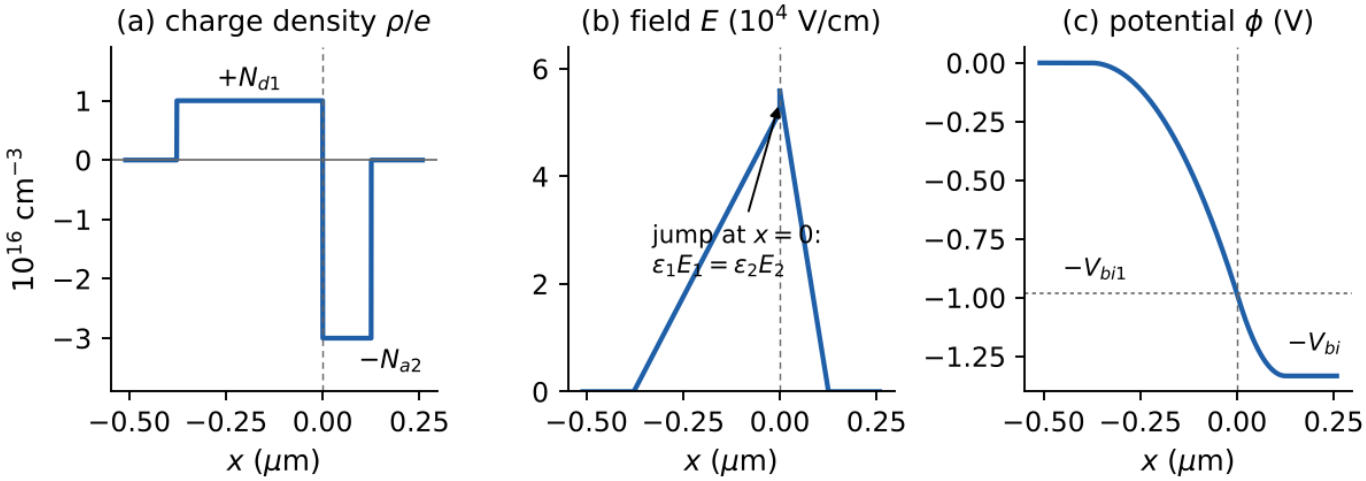


Figure 7: Charge density, field and potential of a straddling heterojunction: the field jumps at x = 0 because the permittivities differ (reused from the 2020–21 paper)

Reading the figure. Panel (a): a positive block of charge on one side and a negative block on the other, with equal areas. Panel (b): the field is triangular on each side and jumps at $x = 0$ (where $\epsilon_1 E_1 = \epsilon_2 E_2$). Panel (c): the potential changes as a parabola on each side and is continuous at $x = 0$; its total change is V_{bi}.

2. Answer

Step 1: Field in the n region

Poisson with $\rho = +eN_{dn}$:

$$\frac{dE_n}{dx} = \frac{eN_{dn}}{\epsilon_n} \quad \Rightarrow \quad E_n = \frac{eN_{dn}}{\epsilon_n}x + C_1$$

The field is zero at the depletion edge, $E_n(-x_n) = 0$, so $C_1 = eN_{dn}x_n/\epsilon_n$:

$$E_n(x) = \frac{eN_{dn}}{\epsilon_n}(x_n + x), \qquad -x_n \le x \le 0$$

Step 2: Field in the P region

Poisson with $\rho = -eN_{aP}$ and $E_P(x_P) = 0$:

$$E_P = -\frac{eN_{aP}}{\epsilon_P}x + C_2, \quad C_2 = \frac{eN_{aP}x_P}{\epsilon_P} \quad \Rightarrow \quad E_P(x) = \frac{eN_{aP}}{\epsilon_P}(x_P - x), \qquad 0 \le x \le x_P$$

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(Neamen, Eq. 9.41.)

Step 3: Boundary condition at the interface

Continuity of D at $x = 0$: $\epsilon_n E_n(0) = \epsilon_P E_P(0)$:

$$\epsilon_n \frac{eN_{dn}x_n}{\epsilon_n} = \epsilon_P \frac{eN_{aP}x_P}{\epsilon_P} \quad \Rightarrow \quad N_{dn}x_n = N_{aP}x_P$$

The positive charge per area on the n side equals the negative charge per area on the P side (Eq. 9.42b).

Step 4: Potential drop on each side

Integrate $E = -d\phi/dx$ across each region. On the n side the drop is the area under the triangle E_n:

$$V_{bin} = \int_{-x_n}^{0} E_n\,dx = \frac{eN_{dn}}{\epsilon_n}\int_{-x_n}^{0}(x_n + x)\,dx = \frac{eN_{dn}}{\epsilon_n}\cdot\frac{x_n^2}{2} = \frac{eN_{dn}x_n^2}{2\epsilon_n}$$

and in the same way on the P side

$$V_{biP} = \int_{0}^{x_P} E_P\,dx = \frac{eN_{aP}}{\epsilon_P}\int_{0}^{x_P}(x_P - x)\,dx = \frac{eN_{aP}x_P^2}{2\epsilon_P}$$

Step 5: The total built-in potential

The potential is continuous at $x = 0$, so the two drops add:

$$V_{bi} = V_{bin} + V_{biP} = \frac{eN_{dn}x_n^2}{2\epsilon_n} + \frac{eN_{aP}x_P^2}{2\epsilon_P} \qquad (9.46)$$

Step 6: Solve for x_n

From Step 3, $x_P = N_{dn}x_n/N_{aP}$. Substitute:

$$V_{bi} = \frac{eN_{dn}x_n^2}{2\epsilon_n} + \frac{eN_{aP}}{2\epsilon_P}\cdot\frac{N_{dn}^2x_n^2}{N_{aP}^2} = \frac{eN_{dn}x_n^2}{2}\left(\frac{1}{\epsilon_n} + \frac{N_{dn}}{\epsilon_P N_{aP}}\right) = \frac{eN_{dn}x_n^2}{2}\cdot\frac{\epsilon_P N_{aP} + \epsilon_n N_{dn}}{\epsilon_n\epsilon_P N_{aP}}$$

$$\boxed{x_n = \left[\frac{2\epsilon_n\epsilon_P N_{aP}V_{bi}}{eN_{dn}(\epsilon_n N_{dn} + \epsilon_P N_{aP})}\right]^{1/2}} \qquad (9.47a)$$

Step 7: Solve for x_P

By symmetry (substitute $x_n = N_{aP}x_P/N_{dn}$ into Step 5):

$$\boxed{x_P = \left[\frac{2\epsilon_n\epsilon_P N_{dn}V_{bi}}{eN_{aP}(\epsilon_n N_{dn} + \epsilon_P N_{aP})}\right]^{1/2}} \qquad (9.47b)$$

Step 8: Total depletion width

$$W = x_n + x_P = x_n\left(1 + \frac{N_{dn}}{N_{aP}}\right) = x_n\frac{N_{aP} + N_{dn}}{N_{aP}}$$

$$\boxed{W = \left[\frac{2\epsilon_n\epsilon_P(N_{dn} + N_{aP})^2V_{bi}}{eN_{dn}N_{aP}(\epsilon_n N_{dn} + \epsilon_P N_{aP})}\right]^{1/2}} \qquad (9.48)$$

With bias: replace V_{bi} by $V_{bi}+V_R$ (reverse) or $V_{bi}-V_a$ (forward). **Check:** with $\epsilon_n = \epsilon_P = \epsilon_s$ it reduces to the homojunction result $W = [2\epsilon_s V_{bi}(N_a + N_d)/(eN_aN_d)]^{1/2}$.

Steps to remember

Poisson on each side with its own $\epsilon \to E = 0$ at both edges $\to \epsilon_n E_n(0) = \epsilon_P E_P(0)$ gives $N_{dn}x_n = N_{aP}x_P \to$ triangle areas give V_{bin}, $V_{biP} \to$ add $\to$ eliminate one width $\to x_n$, x_P, W.

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3. Keyword Explanation

1. Depletion approximation

↩ back

Treating the junction region as fully emptied of free carriers, with sharp edges, so only fixed ions carry charge there.

2. Continuity of D

↩ back

With no interface charge, $\epsilon_n E_n = \epsilon_P E_P$ at the interface; the field jumps when the permittivities differ.

4. Notes and Remarks

- **Sharing of V_{bi}:** $V_{bin}/V_{biP} = \epsilon_P N_{aP}/(\epsilon_n N_{dn})$ (Neamen Eq. 9.45): the lower-doped side takes most of the potential.
- The junction capacitance follows as $C_j' = \left[\dfrac{eN_{dn}N_{aP}\epsilon_n\epsilon_P}{2(\epsilon_n N_{dn} + \epsilon_P N_{aP})(V_{bi} + V_R)}\right]^{1/2}$ (Eq. 9.49).
- Interface states would add a sheet charge, making D discontinuous; the ideal result neglects them.

2.(c) For forward active mode derive the excess minority carrier concentration equation at base of a bipolar junction transistor Marks: 8

1. Theory

1. The npn transistor in forward-active mode (Core Idea)

An npn bipolar transistor is an n^+ emitter, a thin p base and an n collector. In **forward-active** mode the base–emitter (B–E) junction is forward biased ($V_{BE} > 0$) and the base–collector (B–C) junction reverse biased. The emitter injects electrons into the base; they **diffuse** across it (there is no field in the neutral base) and are swept into the collector by the B–C field. In the base the electrons are **minority carriers**; their thermal-equilibrium value is $n_{B0} = n_i^2/N_B$.

2. The two physical laws used

- **Law of the junction** (Neamen, Chapter 8): at the edge of a junction with voltage V across it, the minority-carrier concentration is the equilibrium value times $\exp(eV/kT)$.
- **Ambipolar (minority-carrier) diffusion equation** in a neutral region with no field and no generation, in steady state:

$$D_B \frac{d^2(\delta n_B)}{dx^2} - \frac{\delta n_B}{\tau_{B0}} = 0$$

where $\delta n_B = n_B - n_{B0}$ is the excess electron concentration, D_B the electron diffusion coefficient in the base and τ_{B0} the electron lifetime. Physically: diffusion brings electrons in, recombination removes them.

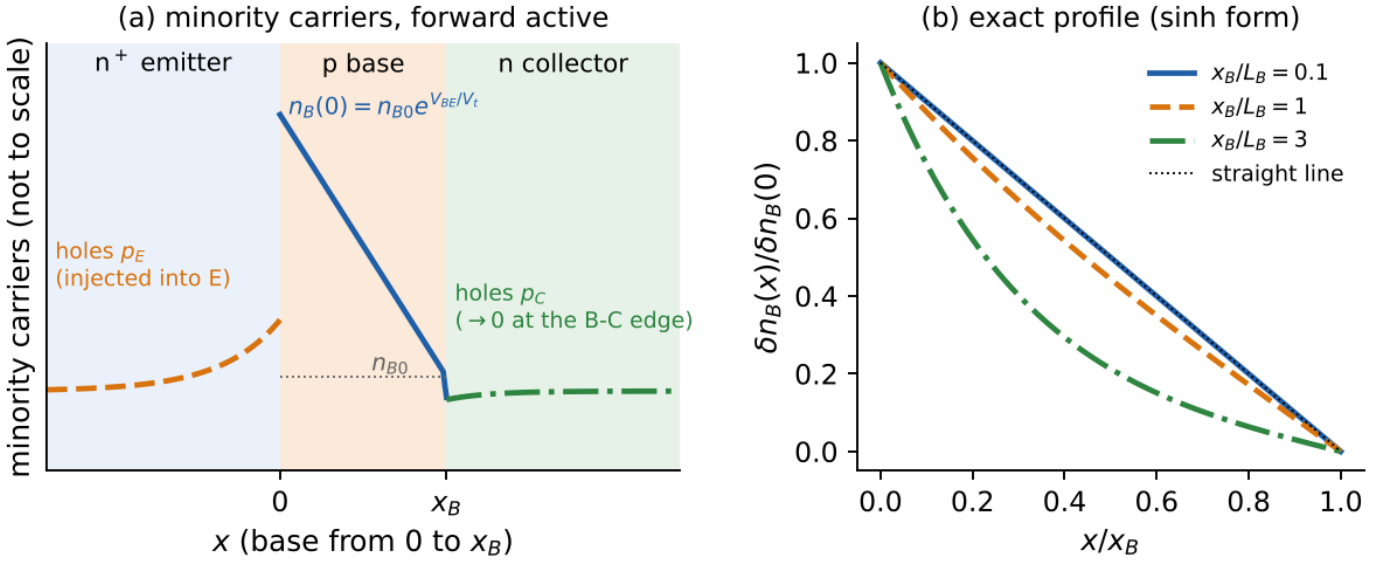


Figure 8: Minority-carrier distributions in an npn transistor in forward-active mode, and the exact base profile for several base widths

Reading the figure. Panel (a): the shaded regions are the n^+ emitter, p base and n collector. In the base (solid blue) the electrons start at $n_B(0) = n_{B0}e^{V_{BE}/V_t}$, far above n_{B0} (dotted grey), and fall almost linearly to nearly zero at the reverse-biased collector edge. The dashed orange curve is the holes injected back into the emitter; the dash-dot green curve the holes in the collector, pulled to zero at the B–C edge. Panel (b): the exact base profile for three base widths; for $x_B \ll L_B$ (solid blue) it is the straight dotted line; for wider bases it sags (more recombination).

2. Answer

Step 1: Coordinates and the equation

Let $x = 0$ be the edge of the B–E depletion region and $x = x_B$ the edge of the B–C depletion region (the neutral base width). In the neutral base, with $L_B^2 = D_B\tau_{B0}$:

$$\frac{d^2(\delta n_B)}{dx^2} - \frac{\delta n_B}{L_B^2} = 0$$

Step 2 (Basic): General solution

Try $\delta n_B = e^{sx}$: $s^2 - 1/L_B^2 = 0$, so $s = \pm 1/L_B$. For a base of finite width the convenient form is

$$\delta n_B(x) = A\cosh\frac{x}{L_B} + B\sinh\frac{x}{L_B}$$

(since $\cosh u = (e^u + e^{-u})/2$ and $\sinh u = (e^u - e^{-u})/2$ are combinations of the two exponentials).

Step 3: Boundary conditions

At $x = 0$ the B–E junction is forward biased; by the law of the junction

$$\delta n_B(0) = n_{B0}\left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1\right]$$

At $x = x_B$ the B–C junction is reverse biased; the law of the junction gives $n_B = n_{B0}\exp(-eV_{CB}/kT) \approx 0$, so

$$\delta n_B(x_B) = n_{B0}\left[\exp\left(\frac{-eV_{CB}}{kT}\right) - 1\right] \approx -n_{B0}$$

Step 4: Find A and B

At $x = 0$: $\cosh 0 = 1$, $\sinh 0 = 0$, so $A = \delta n_B(0)$.

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At $x = x_B$: $A\cosh(x_B/L_B) + B\sinh(x_B/L_B) = \delta n_B(x_B)$, so

$$B = \frac{\delta n_B(x_B) - \delta n_B(0)\cosh(x_B/L_B)}{\sinh(x_B/L_B)}$$

Step 5: Combine into the standard form

$$\delta n_B(x) = \delta n_B(0)\cosh\frac{x}{L_B} + \frac{\delta n_B(x_B) - \delta n_B(0)\cosh(x_B/L_B)}{\sinh(x_B/L_B)}\sinh\frac{x}{L_B}$$

Put everything over $\sinh(x_B/L_B)$ and use the identity $\sinh(a-b) = \sinh a\cosh b - \cosh a\sinh b$ with $a = x_B/L_B$, $b = x/L_B$:

$$\delta n_B(0)\frac{\sinh(x_B/L_B)\cosh(x/L_B) - \cosh(x_B/L_B)\sinh(x/L_B)}{\sinh(x_B/L_B)} = \delta n_B(0)\frac{\sinh[(x_B-x)/L_B]}{\sinh(x_B/L_B)}$$

so

$$\boxed{\delta n_B(x) = \frac{n_{B0}\left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1\right]\sinh\left(\frac{x_B - x}{L_B}\right) - n_{B0}\sinh\left(\frac{x}{L_B}\right)}{\sinh\left(\frac{x_B}{L_B}\right)}}$$

(Neamen, Eq. 12.15b, with V_{CB} large so that $\exp(-eV_{CB}/kT) - 1 = -1$).

Step 6: The narrow-base approximation

A useful transistor has $x_B \ll L_B$. Then $\sinh u \approx u$ for small u:

$$\delta n_B(x) \approx \frac{n_{B0}\left[\exp(eV_{BE}/kT) - 1\right](x_B - x) - n_{B0}x}{x_B}$$

and since $\exp(eV_{BE}/kT) \gg 1$:

$$\boxed{\delta n_B(x) \approx n_{B0}\exp\left(\frac{eV_{BE}}{kT}\right)\left(1 - \frac{x}{x_B}\right)}$$

a **straight line** from $n_{B0}e^{eV_{BE}/kT}$ at the emitter to zero at the collector. Its constant slope gives the diffusion current

$$J_n = eD_B\frac{d(\delta n_B)}{dx} = -\frac{eD_B n_{B0}}{x_B}\exp\left(\frac{eV_{BE}}{kT}\right)$$

so the collector current grows exponentially with V_{BE}, which is the transistor action.

Steps to remember

Diffusion equation in the base → general cosh/sinh solution → law of the junction at both edges → solve for A, B → sinh identity → narrow base gives a straight line.

3. Keyword Explanation

1. Minority-carrier diffusion length

↩ back

$L_B = \sqrt{D_B\tau_{B0}}$: the average distance a minority electron diffuses in the base before recombining.

2. Law of the junction

↩ back

$n = n_0\exp(eV/kT)$ at the edge of a junction biased by V.

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4. Notes and Remarks

- The emitter has the matching result for the injected holes, $\delta p_E(x') = p_{E0}[\exp(eV_{BE}/kT) - 1]\sinh[(x_E - x')/L_E]/\sinh(x_E/L_E)$ for an emitter of width x_E with an ohmic contact (Neamen, Eq. 12.21).
- The small sag of the exact curve is due to recombination in the base; it sets the base transport factor $\alpha_T = 1/\cosh(x_B/L_B)$.

2.(d) Determine the n-type doping concentration, and depletion region width in P-type side of semiconductor for an n-Ge and P-GaAs hetero-junction. Consider built in potential is 1.0 V and T = 300 K, n-type Ge electron affinity 4.13, N_{vn} = [unclear: 6 ×] 10^{18} cm^{-3}, $N_d = 10^{16}$ cm^{-3}, $n_i = 2.4 \times 10^{13}$ cm^{-3} and ε_n = 16. For P-type GaAs electron affinity 4.07, $N_{vP} = 7 \times 10^{18}$ cm^{-3}, $N_a = 10^{16}$ cm^{-3} and ε_P = 13.1 Marks: 5

1. Theory

1. Built-in potential of an nP heterojunction (Core Idea)

For an nP heterojunction the built-in potential is the difference of the two work functions. Writing each Fermi level through the hole concentration, $p = N_v \exp[-(E_F - E_v)/kT]$, on both sides and using the valence-band offset gives (Neamen, Eq. 9.39)

$$eV_{bi} = \Delta E_v + kT \ln\left(\frac{p_{P0}}{p_{n0}} \cdot \frac{N_{vn}}{N_{vP}}\right)$$

where $p_{P0} = N_{aP}$ is the hole concentration in the P material, $p_{n0} = n_i^2/N_d$ the hole (minority) concentration in the n material, and N_{vn}, N_{vP} the valence-band effective densities of states. The offsets follow Anderson's rule: $\Delta E_c = e(\chi_n - \chi_P)$ and $\Delta E_v = \Delta E_g - \Delta E_c$.

2. Depletion widths

From 2(b):

$$x_P = \left[\frac{2\epsilon_n\epsilon_P N_{dn} V_{bi}}{eN_{aP}(\epsilon_n N_{dn} + \epsilon_P N_{aP})}\right]^{1/2}, \qquad N_{dn}x_n = N_{aP}x_P$$

2. Answer

Given:

- n-Ge: $\chi_n = 4.13$ eV, $E_{gn} = 0.67$ eV (Neamen), $N_{vn} = 6 \times 10^{18}$ cm^{-3}, $n_i = 2.4 \times 10^{13}$ cm^{-3}, $\epsilon_n = 16\epsilon_0 = 1.416 \times 10^{-12}$ F/cm
- P-GaAs: $\chi_P = 4.07$ eV, $E_{gP} = 1.43$ eV, $N_{vP} = 7 \times 10^{18}$ cm^{-3}, $N_{aP} = 10^{16}$ cm^{-3}, $\epsilon_P = 13.1\epsilon_0 = 1.159 \times 10^{-12}$ F/cm
- $V_{bi} = 1.0$ V, $kT = 0.0259$ eV; the paper also lists $N_d = 10^{16}$ cm^{-3}

Assumption: the n-type doping is found from the given $V_{bi} = 1.0$ V (the paper's $N_d = 10^{16}$ cm^{-3} is the rounded design value; see the Notes).

Step 1: Band offsets

$$\Delta E_c = \chi_n - \chi_P = 4.13 - 4.07 = 0.06 \text{ eV}$$

$$\Delta E_v = (E_{gP} - E_{gn}) - \Delta E_c = (1.43 - 0.67) - 0.06 = 0.70 \text{ eV}$$

Step 2: Hole concentration in the n-Ge from V_{bi}

Rearrange $eV_{bi} = \Delta E_v + kT \ln\left(\frac{N_{aP}N_{vn}}{p_{n0}N_{vP}}\right)$:

$$\ln\left(\frac{N_{aP}N_{vn}}{p_{n0}N_{vP}}\right) = \frac{1.0 - 0.70}{0.0259} = 11.58 \quad \Rightarrow \quad \frac{N_{aP}N_{vn}}{p_{n0}N_{vP}} = 1.073 \times 10^5$$

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$$p_{n0} = \frac{(10^{16})(6\times10^{18})}{(7\times10^{18})(1.073\times10^{5})} = 7.99\times10^{10}\ \text{cm}^{-3}$$

Step 3: n-type doping

$$N_d = \frac{n_i^2}{p_{n0}} = \frac{(2.4\times10^{13})^2}{7.99\times10^{10}} = 7.21\times10^{15}\ \text{cm}^{-3}$$

$$\boxed{N_d \approx 7.2\times10^{15}\ \text{cm}^{-3}\ (\approx 10^{16}\ \text{cm}^{-3})}$$

Check: with $N_d = 10^{16}$ cm^{-3}, $p_{n0} = 5.76\times10^{10}$ cm^{-3} and $V_{bi} = 0.70+0.0259\ln[(10^{16})(6\times10^{18})/((5.76\times10^{10})(7\times10^{18}))] = 1.0085$ V ≈ 1.0 V, as in Neamen's Example 9.8.

Step 4: Depletion width on the P (GaAs) side

$$\epsilon_n N_{dn} + \epsilon_P N_{aP} = (1.416\times10^{-12})(7.21\times10^{15}) + (1.159\times10^{-12})(10^{16}) = 2.180\times10^{4}$$

$$x_P = \left[\frac{2(1.416\times10^{-12})(1.159\times10^{-12})(7.21\times10^{15})(1.0)}{(1.6\times10^{-19})(10^{16})(2.180\times10^{4})}\right]^{1/2} = 2.61\times10^{-5}\ \text{cm}$$

$$\boxed{x_P = 0.261\ \mu\text{m}}$$

For completeness: $x_n = N_{aP}x_P/N_{dn} = 0.361$ µm, $W = 0.622$ µm, and the potential splits as $V_{bin} = 0.532$ V, $V_{biP} = 0.468$ V.

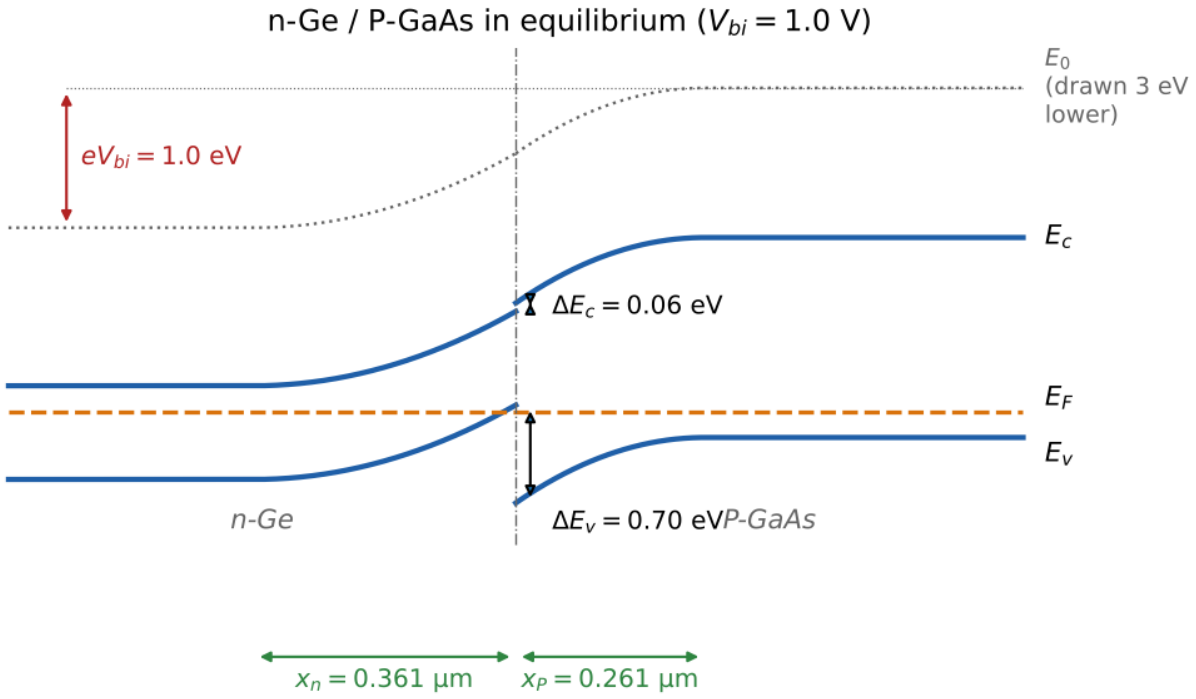


Figure 9: n-Ge / P-GaAs heterojunction in equilibrium with the computed values (vacuum level drawn 3 eV lower)

Reading the figure. The dashed orange E_F is flat. On the left the n-Ge bands (solid blue) bend up over $x_n = 0.361$ µm; on the right the P-GaAs bands bend up over $x_P = 0.261$ µm. At the interface E_c jumps by only $\Delta E_c = 0.06$ eV but E_v drops by $\Delta E_v = 0.70$ eV. The dotted vacuum level (drawn 3 eV lower to fit) rises by $eV_{bi} = 1.0$ eV (red arrow).

Final Answer: $N_d \approx 7.2\times10^{15}$ cm^{-3} (consistent with the given 10^{16} cm^{-3}); $x_P = 0.261$ µm (0.282 µm if $N_d = 10^{16}$ cm^{-3} is used).

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3. Keyword Explanation

1. Electron-affinity rule

↩ back

$\Delta E_c = e(\chi_n - \chi_P)$: the conduction-band offset from the difference of the electron affinities.

2. Effective density of states

↩ back

N_v (or N_c): the number of states per cm^3 that, put at the band edge, give the same carrier count as the real band.

4. Notes and Remarks

- **Reading of the question:** the paper gives both $V_{bi} = 1.0$ V and $N_d = 10^{16}$ cm^{-3} and asks for "the n-type doping". The two agree to rounding (Neamen's Example 9.8 computes $V_{bi} \approx 1.0$ V from $N_d = 10^{16}$). The answer finds N_d from V_{bi}, as asked; with the printed $N_d = 10^{16}$ cm^{-3} instead, $x_P = x_n = 0.282$ µm ($N_d = N_a$) and $V_{bin} = 0.450$ V, $V_{biP} = 0.550$ V.
- The paper's N_{vn} is faint on the scan; 6×10^{18} cm^{-3} is Neamen's value for Ge and is used.
- The large ΔE_v and small ΔE_c make this junction very asymmetric: electrons cross it easily, holes hardly at all.

5. Final Concise Answer

(a) Draw with a flat E_F, bulk band positions far away, continuous vacuum level and band offsets $\Delta E_c = e(\chi_1 - \chi_2)$, $\Delta E_v = \Delta E_g - \Delta E_c$ at the interface. pN (p-GaAs/N-AlGaAs): both sides depleted, bands bend down towards N, E_c spike and E_v step at the interface. nN (n-GaAs/N-AlGaAs): AlGaAs depleted, GaAs accumulated; E_c forms a notch (2DEG) on the GaAs side and a spike ΔE_c.

(b) Poisson on each side, $E = 0$ at the edges, $\epsilon_n E_n(0) = \epsilon_P E_P(0)$ gives $N_{dn} x_n = N_{aP} x_P$; $V_{bi} = eN_{dn}x_n^2/2\epsilon_n + eN_{aP}x_P^2/2\epsilon_P$; hence $x_n = [2\epsilon_n\epsilon_P N_{aP} V_{bi}/(eN_{dn}(\epsilon_n N_{dn} + \epsilon_P N_{aP}))]^{1/2}$, x_P likewise, and $W = [2\epsilon_n\epsilon_P(N_{dn} + N_{aP})^2 V_{bi}/(eN_{dn}N_{aP}(\epsilon_n N_{dn} + \epsilon_P N_{aP}))]^{1/2}$.

(c) $D_B\, d^2\delta n_B/dx^2 - \delta n_B/\tau_{B0} = 0$ with $\delta n_B(0) = n_{B0}(e^{eV_{BE}/kT} - 1)$ and $\delta n_B(x_B) = -n_{B0}$ gives $\delta n_B(x) = \{n_{B0}(e^{eV_{BE}/kT} - 1)\sinh[(x_B - x)/L_B] - n_{B0}\sinh(x/L_B)\}/\sinh(x_B/L_B)$; for $x_B \ll L_B$, $\delta n_B \approx n_{B0}e^{eV_{BE}/kT}(1 - x/x_B)$.

(d) $\Delta E_c = 0.06$ eV, $\Delta E_v = 0.70$ eV; from $V_{bi} = 1.0$ V: $p_{n0} = 7.99 \times 10^{10}$ cm^{-3}, $N_d = 7.2 \times 10^{15}$ cm^{-3} ($\approx 10^{16}$); $x_P = 0.261$ µm ($x_n = 0.361$ µm).

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3. (a) Define specific resistance. Explain which parameters affect specific resistance. Marks: 4

(b) With proper band diagram explain multiple negative differential resistance in a Resonant Tunneling Diode. Marks: 6

(c) Briefly explain the types of ohmic contacts? Marks: 3

(d) Draw energy-band diagram of the following metal semiconductor junctions and mention whether it is Ohmic or Schottky then also find the ideal Schottky barrier height.

(i) Al on top of n-GaAs
(ii) Cu on top of n-GaAs
(iii) Ag on top of p-GaAs
(iv) Au on top of p-GaAs

Here, Al, Cu, Ag, and Au have work functions of 4.06 eV, 4.98 eV, 4.74 eV, and [unclear: 5.47] eV respectively. GaAs has electron affinity of 4.07 eV and Bandgap of 1.519 eV. Marks: 12

3.(a) Define specific resistance. Explain which parameters affect specific resistance. Marks: 4

1. Theory

1. What is an ohmic contact? (Core Idea)

Every device needs metal wires joined to the semiconductor. Such a joint should pass current equally easily in both directions with only a small voltage drop: its current–voltage line should be straight and steep. This is an **ohmic contact** (Neamen, Section 9.2). A rectifying (Schottky) contact is the opposite.

2. Why a resistance per area?

The resistance of a contact falls as its area grows, so a single number in ohms says little. The useful figure of merit is the resistance **times** the area, the **specific contact resistance** R_c, in $\Omega{\cdot}\text{cm}^2$. Dividing it by the contact area A gives the actual contact resistance, $R = R_c/A$.

3. How electrons cross a metal–semiconductor barrier

- **Thermionic emission:** electrons with enough thermal energy go **over** the barrier $e\phi_{Bn}$. The current density is $J = A^*T^2e^{-e\phi_{Bn}/kT}(e^{eV/kT} - 1)$, where A^* is the effective Richardson constant (Neamen, Eq. 9.23).
- **Tunnelling (field emission):** if the semiconductor is very heavily doped, the depletion layer is only a few nanometres thick, and electrons pass **through** the barrier. The tunnelling current grows as $\exp(-e\phi_{Bn}/E_{00})$ with $E_{00} = \dfrac{e\hbar}{2}\sqrt{\dfrac{N_d}{\epsilon_s m_n^*}}$ (Eqs. 9.29–9.30).

2. Answer

Definition: the **specific contact resistance** is the reciprocal of the slope of the current density–voltage curve of the contact at zero bias (Neamen, Eq. 9.31):

$$R_c = \left(\frac{\partial J}{\partial V}\right)^{-1}_{V=0} \quad (\Omega \cdot \text{cm}^2)$$

A good ohmic contact has a small R_c (below about 10^{-6} $\Omega{\cdot}\text{cm}^2$ for modern devices); the contact resistance is R_c/A.

Parameters that affect it:

1. **Barrier height** ϕ_{Bn} (set by the metal work function, the electron affinity and the interface states). For low to moderate doping thermionic emission dominates, and differentiating J at

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$V = 0$ gives

$$R_c = \frac{kT/e}{A^* T^2} \exp\left(\frac{e\phi_{Bn}}{kT}\right) \qquad (9.32)$$

R_c rises **exponentially** with the barrier: a lower barrier gives a much better contact.

2. **Semiconductor doping N_d.** For $N_d \gtrsim 10^{19}$ cm^{-3} tunnelling dominates and

$$R_c \propto \exp\left(\frac{2\sqrt{\epsilon_s m_n^*}}{\hbar} \cdot \frac{\phi_{Bn}}{\sqrt{N_d}}\right) \qquad (9.33)$$

so R_c falls **exponentially** with $\sqrt{N_d}$. This is why contacts are made on n$^+$ or p$^+$ layers.

3. **Temperature T:** through kT in the thermionic term (higher T, lower R_c); tunnelling is nearly independent of T.
4. **Material constants:** the effective mass m_n^* and the Richardson constant A^*, and the permittivity ϵ_s (a smaller m^* and a larger ϵ_s help tunnelling). Wide-gap materials, with high barriers, are harder to contact.
5. **Interface quality:** oxides, contamination and alloying (for example annealed Au–Ge–Ni on n-GaAs) change the effective barrier and doping at the surface.

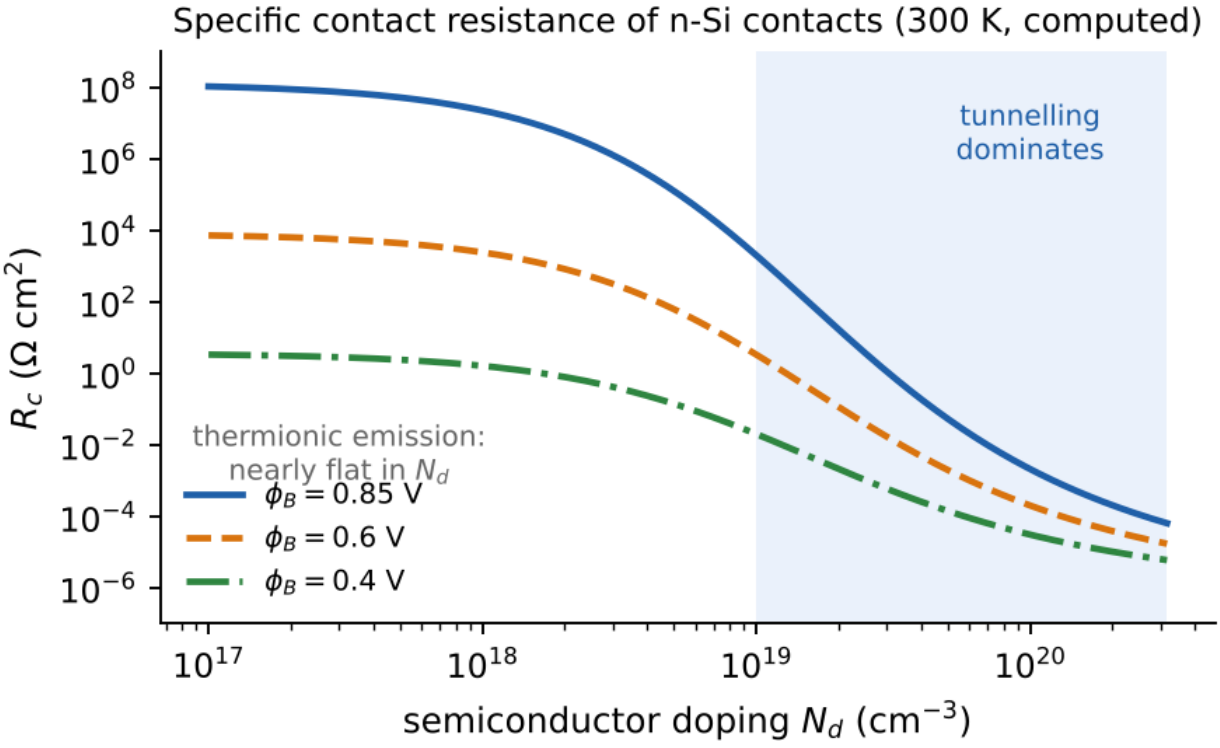


Figure 10: Specific contact resistance of n-Si contacts against doping for three barrier heights, computed with the thermionic-field-emission formula

Reading the figure. Each curve is one barrier height (solid blue 0.85 V, dashed orange 0.6 V, dash-dot green 0.4 V). At low doping the curves are flat: R_c depends only on the barrier (thermionic emission), and a 0.45 V lower barrier gives about 10^7 times less resistance. Above about 10^{19} cm^{-3} (shaded) tunnelling takes over and R_c falls steeply with doping: for $\phi_B = 0.85$ V from 2×10^3 at 10^{19} to 2×10^{-3} Ω·cm^2 at 10^{20} cm^{-3}.

3. Keyword Explanation

1. Richardson constant

↩ back

$A^* = 4\pi e m^* k^2/h^3$: the constant of thermionic emission (120 A/cm^{2}K^2 for free electrons; about 8 for n-GaAs).

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2. Characteristic tunnelling energy E_{00}

↩ back

$E_{00} = (e\hbar/2)\sqrt{N_d/(\epsilon_s m^*)}$; when $E_{00} \gg kT$ tunnelling dominates (for n-Si, $E_{00} = 34$ meV at 10^{19} cm^{-3}, above $kT = 26$ meV).

4. Notes and Remarks

- **Curves of the figure:** they use $R_c = \dfrac{k}{eA^*T}\exp\left(\dfrac{e\phi_B}{E_0}\right)$ with $E_0 = E_{00}\coth(E_{00}/kT)$ (thermionic-field emission, Sze & Ng, Section 3.6), which goes to Eq. 9.32 at low doping and to the form of Eq. 9.33 at high doping; $m^* = 0.26\,m_0$, $A^* = 120$ A/cm^{2}K^2. Neamen's Figure 9.15 shows the same shape.
- "Specific resistance" in the paper is read as **specific contact resistance**, the quantity Neamen defines for ohmic contacts; the resistivity ρ of a material is a different quantity.

3.(b) With proper band diagram explain multiple negative differential resistance in a Resonant Tunneling Diode. Marks: 6

1. Theory

1. Tunnelling (Core Idea)

In classical physics a particle with less energy than a barrier can never cross it. In quantum mechanics an electron is a wave, and its wave decays inside a barrier but does not stop at once. If the barrier is thin (a few nanometres), part of the wave comes out on the other side: the electron **tunnels** through, with a probability that falls exponentially with the barrier's width and height.

2. A quantum well between two barriers

A **resonant tunnelling diode** (RTD) has a thin undoped GaAs layer (the well, about 5 nm) between two thin AlAs or AlGaAs barriers (about 2–3 nm), with heavily doped n$^+$ GaAs on both sides (emitter and collector). An electron in the well is trapped between the barriers like a wave on a string, so only certain energies fit: the **quasi-bound levels** $E_1, E_2, \ldots$ For a well of width L_w, roughly

$$E_n \approx \frac{\hbar^2}{2m^*}\left(\frac{n\pi}{L_w}\right)^2$$

measured from the bottom of the well.

3. Resonance

A single barrier lets through only a small fraction of electrons. But when an incoming electron has the same energy as a level in the well, the waves reflected inside the well add up in step, and the transmission through *both* barriers rises to nearly 100 % (like light through a Fabry–Pérot filter). This is **resonant tunnelling**. Away from a level, the transmission is very small.

4. What negative differential resistance means

Differential resistance is $r = dV/dI$. It is negative where the current **falls** as the voltage rises. A device with negative resistance can cancel the losses of a resonant circuit, so it can oscillate or amplify, up to very high frequencies.

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2. Answer

Structure: n^+ GaAs emitter / AlAs barrier / GaAs quantum well / AlAs barrier / n^+ GaAs collector. The emitter's electrons fill the conduction band up to its Fermi level E_F (a narrow energy band, a few tens of meV). The well has quasi-bound levels $E_1 < E_2$.

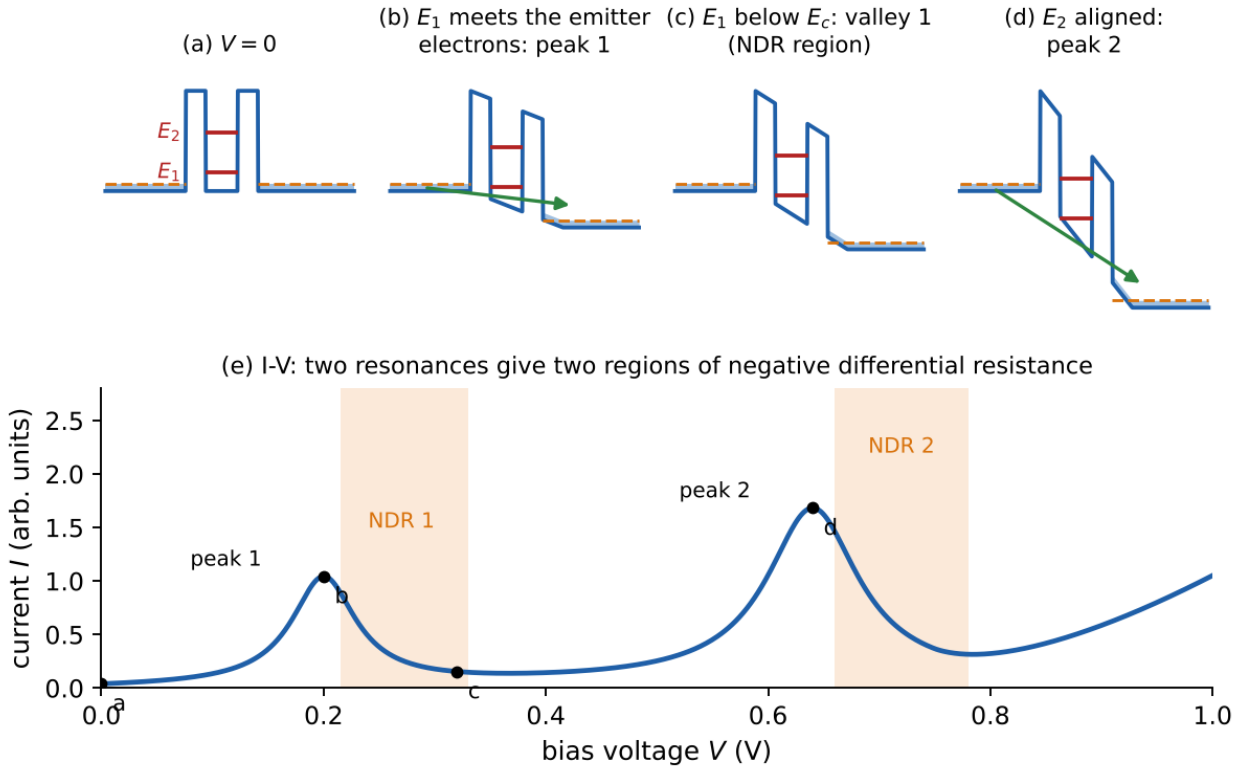


Figure 11: Band diagrams of a double-barrier resonant tunnelling diode at four biases, and its current-voltage curve with two peaks

Reading the figure. In panels (a)–(d) the solid blue line is the conduction-band edge with the two barriers; the shaded strips at left and right are the electrons of the emitter and collector up to their Fermi levels (dashed orange); the solid red lines are the levels E_1 and E_2 in the well; a green arrow marks strong resonant current. Panel (e) is the current against bias, with the points a–d marked and the two negative-resistance regions shaded.

Operation, bias step by step:

1. **(a) $V = 0$:** E_1 lies above the emitter's electrons; no net current.
2. **Rising V:** the bias drops across the barriers and the well, so the well's levels move *down* relative to the emitter by about half of eV. When E_1 enters the band of filled emitter states, electrons tunnel resonantly and the current rises.
3. **(b) Peak 1:** E_1 lies within the filled emitter states (just above the emitter E_c); the current is at its first maximum.
4. **(c) Valley 1:** at higher bias E_1 falls **below** the emitter's conduction-band edge. No emitter electron has that energy any more (energy and transverse momentum are conserved), so resonance is lost and the current **drops** although V rises: the **first negative differential resistance** region.
5. **(d) Peak 2:** raising V further brings the second level E_2 down into the emitter's electron band: a second resonance and a second current peak.
6. **Beyond:** E_2 also falls below the emitter E_c; the current drops again (**second NDR region**), then rises steadily from thermionic emission over the barriers and non-resonant tunnelling.

So each quasi-bound level gives one peak–valley pair: n levels give n regions of negative differential resistance. The bias of each peak is roughly $V_n \approx 2E_n/e$ (since about half the bias drops before the well).

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Uses: very fast oscillators (hundreds of GHz to above 1 THz, since tunnelling is fast), multi-valued (multi-state) logic and memory, using the several stable points that multiple NDR gives.

3. Keyword Explanation

1. Quasi-bound state

↩ back

A level in the well that is not truly bound because electrons can tunnel out through the thin barriers; it has a finite width (lifetime).

2. Peak-to-valley current ratio

↩ back

I_{peak}/I_{valley}: the figure of merit of an RTD (about 2–4 for GaAs/AlAs at room temperature, much higher for InGaAs/AlAs).

4. Notes and Remarks

- Neamen does not treat the RTD; the description follows Sze & Ng, Chapter 8 (Tunnel Devices).
- The valley current is not zero: scattering, phonon-assisted tunnelling and thermionic current over the barriers keep it finite.
- A tunnel (Esaki) diode also has NDR, but only one region; the RTD's multiple levels give several.

3.(c) Briefly explain the types of ohmic contacts? Marks: 3

1. Theory

1. Ohmic or rectifying? (Core Idea)

Whether a metal–semiconductor contact passes current freely depends on the barrier between them. With ϕ_m the metal work function and ϕ_s the semiconductor work function:

- **n-type:** if $\phi_m > \phi_s$ electrons leave the semiconductor, a depletion barrier forms: **rectifying**. If $\phi_m < \phi_s$ electrons enter the semiconductor, an accumulation layer forms: **no barrier**.
- **p-type:** the reverse: $\phi_m < \phi_s$ is rectifying for holes; $\phi_m > \phi_s$ gives hole accumulation, no barrier.

Even with a barrier, a contact can conduct freely if the barrier is so **thin** that electrons tunnel through it.

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2. Answer

An **ohmic contact** has a linear, symmetric I–V curve and a voltage drop small compared with that across the device. There are two types (Neamen, Section 9.2):

1. **Ideal non-rectifying (barrier-free) contact.** Choose the metal so that no barrier forms: $\phi_m < \phi_s$ on **n-type** (electrons flow into the semiconductor and accumulate at the surface; the bands bend down) or $\phi_m > \phi_s$ on **p-type** (holes accumulate; the bands bend up). Carriers flow freely either way.
2. **Tunnelling contact.** Use any metal (a barrier exists) but dope the semiconductor surface very heavily (n^+ or p^+, $\gtrsim 10^{19}$ cm^{-3}). The depletion width $x_n = (2\epsilon_s V_{bi}/eN_d)^{1/2}$ becomes a few nanometres, and carriers tunnel through the barrier in both directions; R_c falls exponentially with $\sqrt{N_d}$. This is the usual practical contact, because interface states often pin the barrier whatever the metal (Au–Ge–Ni alloyed on n-GaAs, Al on p^+ Si, silicides on n^+ Si).

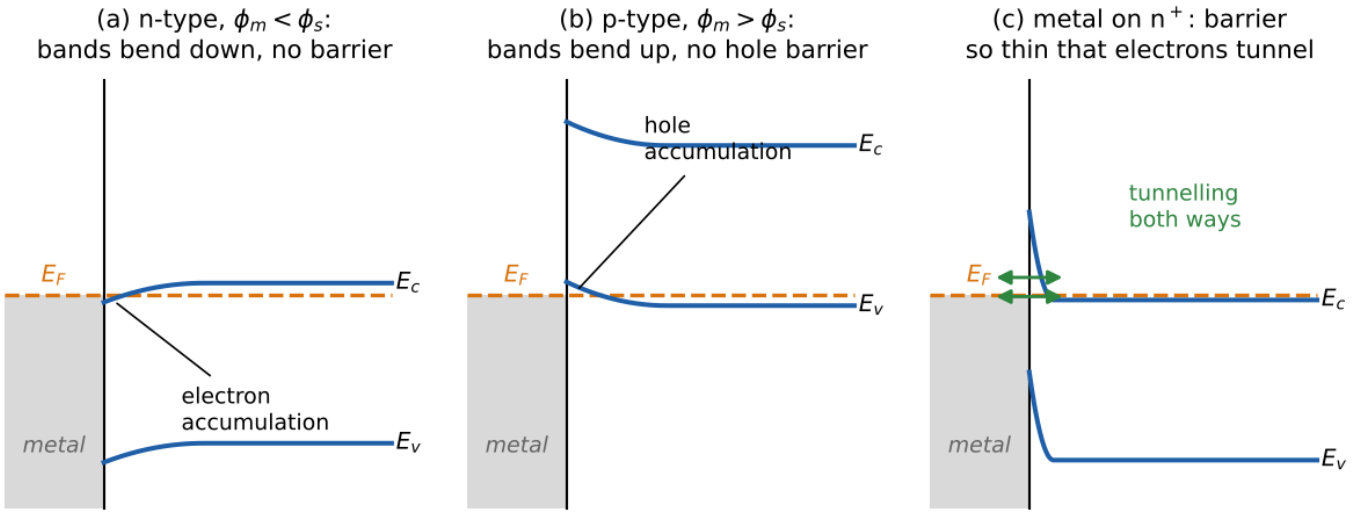


Figure 12: Ohmic contacts: (a) n-type with $\phi_m < \phi_s$, (b) p-type with $\phi_m > \phi_s$, (c) metal on n^+ with a thin tunnelling barrier

Reading the figure. In each panel the grey block is the metal, the dashed orange line the flat Fermi level and the solid blue lines E_c and E_v. (a) The n-type bands bend *down* at the surface: electrons collect there and meet no barrier. (b) The p-type bands bend *up*: holes collect at the surface, no barrier for holes. (c) A barrier exists, but on the heavily doped n^+ side it is only a few nanometres thick, so electrons tunnel through it both ways (green arrows).

3. Keyword Explanation

1. Accumulation layer

↩ back

A thin surface layer with more majority carriers than the bulk, formed when the bands bend towards the majority-carrier band.

2. Tunnelling contact

↩ back

A contact whose barrier is thin enough, because of heavy doping, for carriers to tunnel through it.

4. Notes and Remarks

- The same question is 7(a) (Term Test 1).
- Surface states can spoil type 1: if they pin the Fermi level, a barrier forms even with $\phi_m < \phi_s$ on n-type. That is why the tunnelling type dominates in practice.

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3.(d) Draw energy-band diagram of the following metal semiconductor junctions and mention whether it is Ohmic or Schottky then also find the ideal Schottky barrier height. (i) Al on top of n-GaAs (ii) Cu on top of n-GaAs (iii) Ag on top of p-GaAs (iv) Au on top of p-GaAs Here, Al, Cu, Ag, and Au have work functions of 4.06 eV, 4.98 eV, 4.74 eV, and [unclear: 5.47] eV respectively. GaAs has electron affinity of 4.07 eV and Bandgap of 1.519 eV. Marks: 12

1. Theory

1. Ideal barrier heights (Core Idea)

For an ideal metal–semiconductor contact (no interface states) on a semiconductor of electron affinity χ and band gap E_g:

- **n-type:** electrons in the metal see the barrier $\phi_{Bn} = \phi_m - \chi$ (Neamen, Eq. 9.1).
- **p-type:** holes in the metal see the barrier from E_F down to E_v at the surface,

$$\phi_{Bp} = \frac{E_g}{e} + \chi - \phi_m$$

since E_v lies $(\chi + E_g/e)$ below the vacuum level and E_F of the metal ϕ_m below it.

The two add up to the gap: $\phi_{Bn} + \phi_{Bp} = E_g/e$ for the same metal.

2. Schottky or ohmic?

The contact is a **Schottky (rectifying)** contact when there is a positive barrier for the majority carriers, which happens for $\phi_m > \phi_s$ on n-type and $\phi_m < \phi_s$ on p-type, where $\phi_s = \chi + (E_c - E_F)/e$ (n) or $\chi + E_g/e - (E_F - E_v)/e$ (p). If the barrier is zero or negative, the bands bend towards the majority carriers (accumulation) and the contact is **ohmic**. Since the doping is not given, we use the barrier itself as the test: $\phi_{Bn} = \phi_m - \chi > 0$ (n-type) or $\phi_{Bp} = \chi + E_g/e - \phi_m > 0$ (p-type) means Schottky.

2. Answer

Given: χ(GaAs) = 4.07 eV, E_g = 1.519 eV; ϕ_m: Al 4.06, Cu 4.98, Ag 4.74, Au 5.47 eV.

Junction	Barrier formula	Barrier	Type
(i) Al on n-GaAs	$\phi_{Bn} = \phi_m - \chi = 4.06 - 4.07$	-0.01 V	**Ohmic**
(ii) Cu on n-GaAs	$\phi_{Bn} = 4.98 - 4.07$	**0.91 V**	**Schottky**
(iii) Ag on p-GaAs	$\phi_{Bp} = 4.07 + 1.519 - 4.74$	**0.849 V**	**Schottky**
(iv) Au on p-GaAs	$\phi_{Bp} = 4.07 + 1.519 - 5.47$	**0.119 V**	**Schottky** (very low barrier)

Table 2: Summary

(i) Al on n-GaAs: $\phi_m = 4.06$ V is below $\chi = 4.07$ V, so it is also below ϕ_s (which is larger than χ). Electrons flow from the metal into the GaAs, the bands bend *down* at the surface, and there is no barrier: **ohmic**; the "barrier" is -0.01 V.

(ii) Cu on n-GaAs: $\phi_m > \phi_s$; electrons leave the GaAs, a depletion layer forms, bands bend *up*: **Schottky**, $\phi_{Bn} = 0.91$ V.

(iii) Ag on p-GaAs: $\phi_m = 4.74$ V is well below ϕ_s of p-GaAs (about 5.4–5.5 V), so electrons flow from the metal into the p-GaAs, holes are depleted, bands bend *down*: **Schottky** for holes, $\phi_{Bp} = 0.849$ V.

(iv) Au on p-GaAs: $\phi_{Bp} = 0.119$ V is positive, so in the ideal model it is a **Schottky** contact, but a very weak one (only about $4.6\,kT$). Its current at zero bias is large, and for p-GaAs doped below about 10^{17} cm^{-3} (where $\phi_s < 5.47$ V) the bands would bend the other way and the contact becomes ohmic. In practice it behaves as a nearly ohmic contact.

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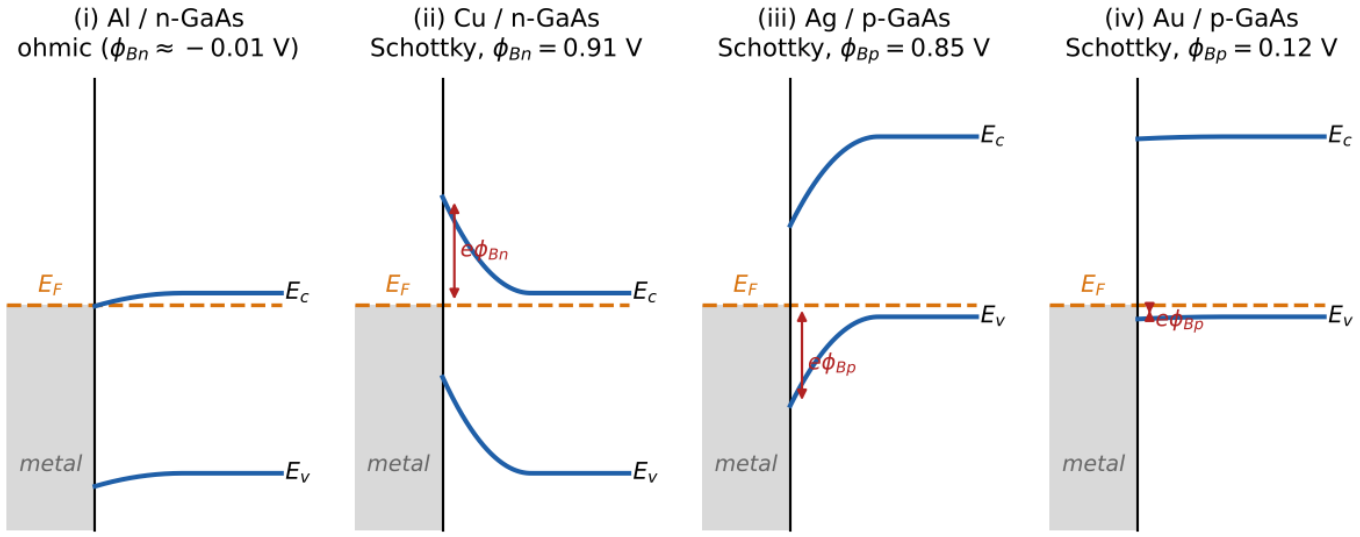


Figure 13: Equilibrium band diagrams of the four junctions (barriers from the ideal model, bulk doping assumed)

Reading the figure. In each panel the grey block is the metal, the dashed orange line E_F and the solid blue lines E_c and E_v of the GaAs. (i) The bands bend slightly down at the surface (no barrier). (ii) The bands bend up; the red arrow is $e\phi_{Bn} = 0.91$ eV from E_F to E_c at the surface. (iii) The p-type bands bend down; the red arrow $e\phi_{Bp} = 0.85$ eV runs from E_F down to E_v at the surface. (iv) Almost flat bands; the barrier $e\phi_{Bp}$ is only 0.12 eV.

Final Answer: (i) Al/n-GaAs: ohmic ($\phi_{Bn} \approx -0.01$ V); (ii) Cu/n-GaAs: Schottky, $\phi_{Bn} = 0.91$ V; (iii) Ag/p-GaAs: Schottky, $\phi_{Bp} = 0.849$ V; (iv) Au/p-GaAs: Schottky with a very low barrier, $\phi_{Bp} = 0.119$ V (nearly ohmic).

3. Keyword Explanation

1. Schottky barrier for holes

↩ back

$\phi_{Bp} = E_g/e + \chi - \phi_m$: the barrier between the metal's Fermi level and the valence-band edge at the surface of a p-type semiconductor.

2. Ohmic contact

↩ back

A metal–semiconductor contact with no barrier (or one thin enough to tunnel through) for majority carriers.

4. Notes and Remarks

- **Real GaAs contacts:** GaAs has many interface states, so its Fermi level is pinned near 0.8 eV below E_c; measured barriers on n-GaAs are about 0.7–0.9 V for almost every metal (even Al), and ohmic contacts are made by alloying Au–Ge–Ni to dope the surface n^+. The ideal answers above are what the question asks for.
- **The band gap 1.519 eV** given in the paper is the GaAs gap at low temperature (Sze & Ng); at 300 K it is 1.42 eV, which would lower ϕ_{Bp} by 0.1 V: (iii) 0.75 V, (iv) 0.02 V.
- **Au work function:** the last digit of "5.47" is faint on the scan; with Neamen's 5.1 V, (iv) would be a clear Schottky contact with $\phi_{Bp} = 0.489$ V.

5. Final Concise Answer

(a) Specific contact resistance $R_c = (\partial J/\partial V)^{-1}_{V=0}$ ($\Omega\cdot\text{cm}^2$). It depends on the barrier height (thermionic, $R_c = (kT/e)e^{e\phi_{Bn}/kT}/(A^*T^2)$, exponential in ϕ_{Bn}), the doping (tunnelling, $R_c \propto \exp\left[2\sqrt{\epsilon_s m^*}\phi_{Bn}/(\hbar\sqrt{N_d})\right]$, falls fast above 10^{19} cm^{-3}), temperature, effective mass and permittivity, and the interface quality.

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(b) RTD: n^+ emitter, two thin barriers and a quantum well with levels E_1, E_2. As bias lowers the levels, the current peaks when E_1 lies in the band of emitter electrons (resonant tunnelling) and drops when E_1 falls below the emitter E_c (NDR 1); E_2 then gives a second peak and a second NDR region. Each level gives one peak–valley pair.

(c) (1) Barrier-free contact: $\phi_m < \phi_s$ on n-type or $\phi_m > \phi_s$ on p-type, majority carriers accumulate, no barrier. (2) Tunnelling contact: heavy surface doping ($\gtrsim 10^{19}$ cm^{-3}) makes the barrier a few nm thick, carriers tunnel through; the practical type.

(d) (i) Al/n-GaAs: $\phi_{Bn} = -0.01$ V, ohmic. (ii) Cu/n-GaAs: Schottky, 0.91 V. (iii) Ag/p-GaAs: Schottky, $\phi_{Bp} = 4.07 + 1.519 - 4.74 = 0.849$ V. (iv) Au/p-GaAs: $\phi_{Bp} = 0.119$ V, a very low Schottky barrier, nearly ohmic.

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4. (a) Draw the typical structure of a Double Hetero-junction Bipolar Transistor Structure and their doping profile Marks: 6

(b) Describe the advantages of HBT over BJT. Show that for an HBT, the ratio of J_{nE} and J_{pE} can be expressed as $\frac{J_{nE}}{J_{pE}} = \frac{N_{DE}}{N_{AB}} \cdot \frac{W_E}{W_B} \cdot \frac{D_{nE}}{D_{pE}} \exp\left(\frac{\Delta E_g}{kT}\right)$ [*sic*: should be D_{nB}]. Marks: 7

(c) For an ideal Metal Oxide Semiconductor capacitor draw charge distribution, electric field distribution, potential distribution and capacitance versus voltage curve. Marks: 6

(d) Show the relation between Flat band voltage and threshold voltage of a MOS device. Marks: 6

4.(a) Draw the typical structure of a Double Hetero-junction Bipolar Transistor Structure and their doping profile Marks: 6

1. Theory

1. The bipolar transistor in one picture (Core Idea)

An npn bipolar transistor has three layers: an n-type **emitter** that injects electrons, a thin p-type **base** that they cross by diffusion, and an n-type **collector** that gathers them. Its current gain is limited by holes that the base injects back into the emitter: they make base current without collector current.

2. What "heterojunction" adds

A **heterojunction bipolar transistor (HBT)** makes the emitter from a **wider-gap** material than the base (AlGaAs emitter, GaAs base). The extra band gap ΔE_g forms a high barrier for holes going from the base into the emitter, so the emitter can stay moderately doped while the base is doped very heavily (low base resistance) and the gain stays high (details in 4(b)).

A **single-heterojunction** HBT (SHBT) has only the base–emitter junction as a heterojunction. A **double-heterojunction** HBT (DHBT) also makes the **collector** from the wide-gap material, so the base–collector junction is a heterojunction too.

3. Why a wide-gap collector?

- **Higher breakdown voltage:** impact ionisation needs carriers with energy above about E_g; a wider gap in the collector depletion region raises the breakdown voltage.
- **No hole injection into the collector in saturation:** the valence-band barrier at the B–C junction stops holes, so less charge is stored and switching is faster.
- **Lower offset voltage:** the two junctions are nearly symmetric, so V_{CE} at $I_C = 0$ is small.
- **Symmetric operation:** the device works almost equally well in the inverse mode.

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2. Answer

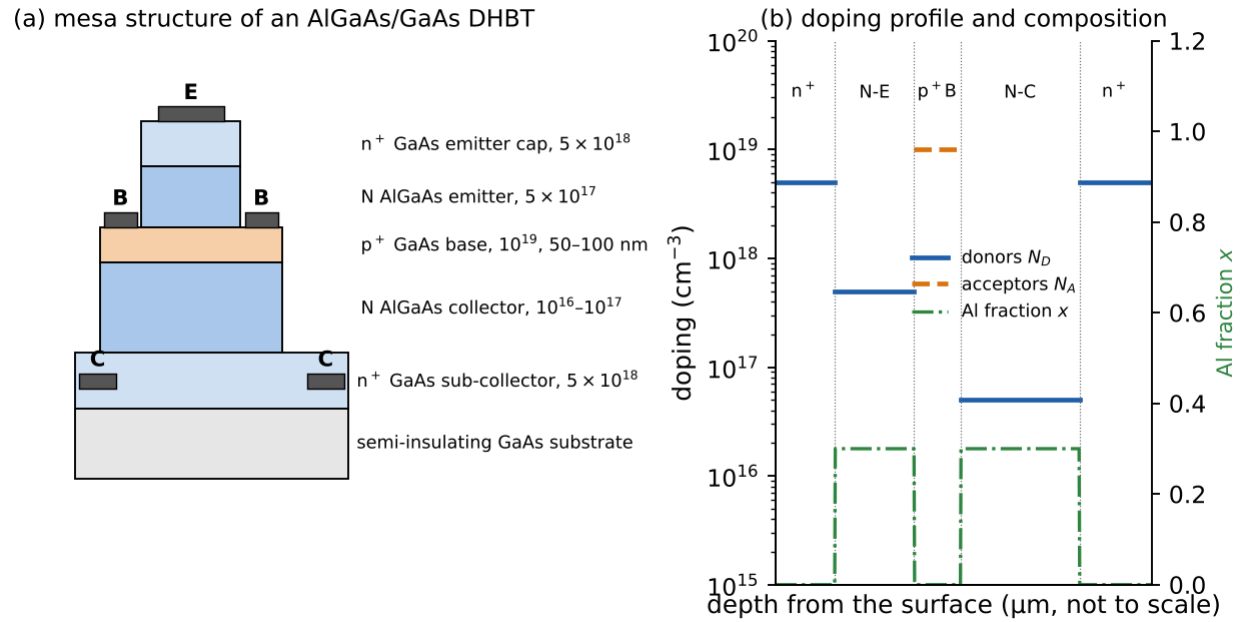


Figure 14: (a) Mesa cross-section of an AlGaAs/GaAs double-heterojunction bipolar transistor; (b) its doping profile (solid blue donors, dashed orange acceptors) and aluminium fraction (dash-dot green)

Reading the figure. Panel (a), from top to bottom: an n^+ GaAs cap under the emitter contact (E), the N-AlGaAs emitter, the thin p^+ GaAs base with its two contacts (B), the N-AlGaAs collector, the n^+ GaAs sub-collector with the collector contacts (C), all on a semi-insulating GaAs substrate. The layers are etched into steps (mesas) so that each contact reaches its own layer. Panel (b): doping against depth on a log scale; the base is the highest-doped layer (10^{19} cm^{-3} acceptors), the emitter and collector are moderately doped; the Al fraction is 0.3 in the emitter and collector and 0 in the base and the contact layers.

Typical layers (grown by MBE or MOCVD):

Layer	Material	Doping (cm^{-3})	Thickness
Emitter cap (contact)	n^+ GaAs (or InGaAs)	5×10^{18} – 10^{19}	0.1–0.2 µm
Emitter	N $Al_{0.3}Ga_{0.7}As$ (graded at its edges)	5×10^{17}	0.1–0.2 µm
Base	p^+ GaAs	10^{19} – 5×10^{19}	0.05–0.1 µm
Collector	N $Al_{0.3}Ga_{0.7}As$ (graded near the base)	10^{16} – 10^{17}	0.3–1 µm
Sub-collector	n^+ GaAs	5×10^{18}	0.5 µm
Substrate	semi-insulating GaAs	—	500 µm

Table 3: Summary

Key features of the profile:

1. **Base doping higher than emitter doping** (the opposite of a Si BJT): possible because the valence-band barrier, not the doping ratio, stops back-injection. It gives a thin base with low resistance and no punch-through.
2. **Wide-gap emitter and collector** (Al fraction 0.3) on both sides of the narrow-gap base: two heterojunctions.
3. **Compositional grading** (a few tens of nm where x changes smoothly) at the junctions removes the conduction-band spikes, so electrons are not blocked.
4. **Heavily doped contact layers** (n^+ cap and sub-collector) give low-resistance tunnelling ohmic contacts.

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3. Keyword Explanation

1. Mesa structure

↩ back

A layered device etched into steps so that contacts can be made to buried layers.

2. Sub-collector

↩ back

The heavily doped layer below the collector that carries the collector current sideways to its contact with low resistance.

4. Notes and Remarks

- InP-based DHBTs (InP emitter and collector, InGaAs base) are the main high-speed versions; their InP collector gives much higher breakdown than an InGaAs one.
- Neamen shows a single-heterojunction AlGaAs/GaAs HBT (Figure 12.50); the double version follows Sze & Ng, Chapter 5.

4.(b) Describe the advantages of HBT over BJT. Show that for an HBT, the ratio of J_{nE} and J_{pE} can be expressed as $\frac{J_{nE}}{J_{pE}} = \frac{N_{DE}}{N_{AB}} \cdot \frac{W_E}{W_B} \cdot \frac{D_{nE}}{D_{pE}} \exp\left(\frac{\Delta E_g}{kT}\right)$ [*sic:* should be D_{nB}]. Marks: 7

1. Theory

1. Emitter injection efficiency (Core Idea)

In an npn transistor the B–E junction carries two currents: electrons injected from the emitter into the base (J_{nE}, useful, it becomes the collector current) and holes injected from the base into the emitter (J_{pE}, wasted, it is base current). The **emitter injection efficiency** is

$$\gamma = \frac{J_{nE}}{J_{nE} + J_{pE}} = \frac{1}{1 + J_{pE}/J_{nE}}$$

and the current gain β can be at most about J_{nE}/J_{pE}. So the ratio J_{nE}/J_{pE} must be large.

2. Minority carriers and the intrinsic concentration

Each current is a diffusion current of minority carriers, proportional to their equilibrium concentration: $n_{B0} = n_{iB}^2/N_{AB}$ in the base and $p_{E0} = n_{iE}^2/N_{DE}$ in the emitter. The intrinsic concentration depends on the band gap (Neamen, Eq. 4.23):

$$n_i^2 = N_c N_v \exp\left(-\frac{E_g}{kT}\right)$$

In a homojunction BJT $n_{iB} = n_{iE}$, so the only way to make $J_{nE} \gg J_{pE}$ is $N_{DE} \gg N_{AB}$; and heavy emitter doping causes bandgap narrowing, which spoils it again. A wide-gap emitter makes n_{iE}^2 much smaller instead.

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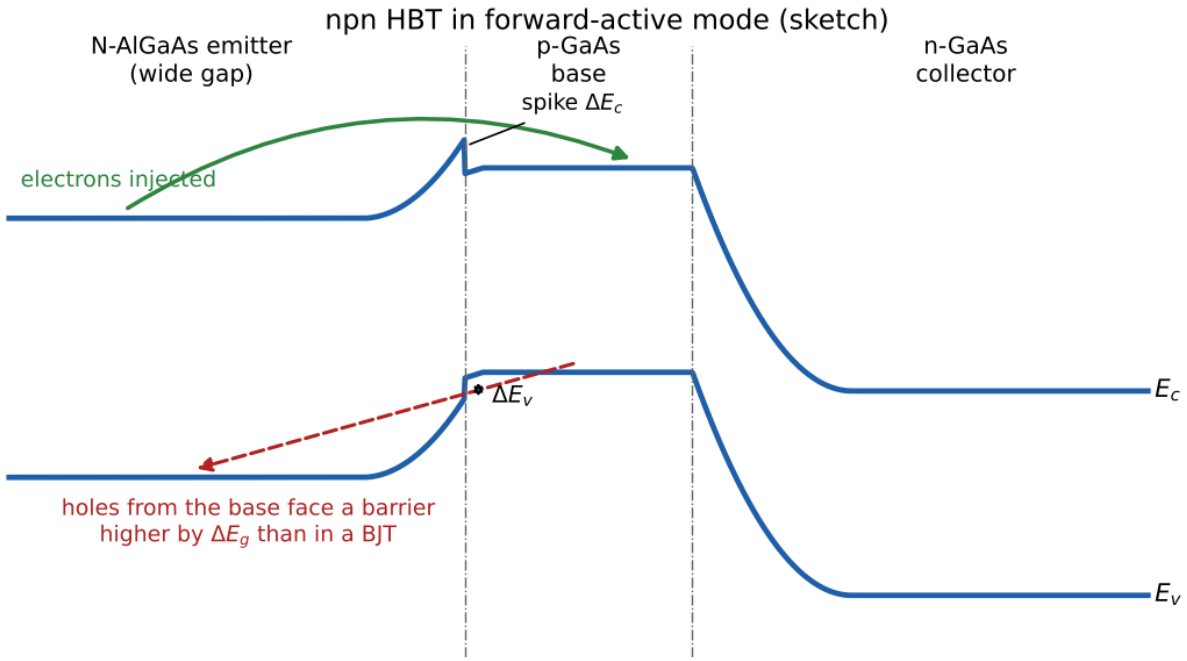


Figure 15: npn HBT in forward-active mode: the wide-gap emitter gives holes in the base an extra barrier

Reading the figure. Solid blue lines are E_c (upper) and E_v (lower). Electrons (green arrow) cross from the N-AlGaAs emitter over the small barrier into the p-GaAs base. Holes in the base (dashed red arrow) would have to climb down the valence band into the emitter, but its edge lies much lower because the emitter gap is wider: the barrier for holes is higher than that for electrons by about ΔE_g. The small step ΔE_v at the interface and the spike in E_c are the band offsets.

2. Answer

Advantages of the HBT over a BJT:

1. **High injection efficiency and gain** independent of the doping ratio: the wide-gap emitter cuts hole back-injection by $\exp(\Delta E_g/kT)$ (about 10^5 for $\Delta E_g = 0.3$ eV).
2. **Very heavily doped base** (10^{19}–10^{20} cm^{-3}): low base resistance, thin base without punch-through, small Early effect.
3. **Lightly doped emitter:** small B–E capacitance and no bandgap narrowing in the emitter.
4. **High speed:** thin base, high electron mobility in GaAs or InGaAs (short base transit time), low r_b and C_{je} give f_T and f_{max} of tens to hundreds of GHz.
5. **Design freedom:** grading the base composition gives a built-in field that speeds up electrons; DHBTs add high breakdown voltage.

Proof of the ratio.

Step 1: Electron current injected into the base

With a short base ($W_B \ll L_{nB}$), the excess electrons fall linearly from $n_{B0}e^{eV_{BE}/kT}$ to zero (see 2(c)), so the diffusion current is

$$J_{nE} = eD_{nB}\frac{n_{B0}}{W_B}\left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1\right], \qquad n_{B0} = \frac{n_{iB}^2}{N_{AB}}$$

Step 2: Hole current injected into the emitter

With a short emitter ($W_E \ll L_{pE}$) and an ohmic contact at its far end, the injected holes also fall linearly:

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$$J_{pE} = eD_{pE}\frac{p_{E0}}{W_E}\left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1\right], \qquad p_{E0} = \frac{n_{iE}^2}{N_{DE}}$$

Step 3: Take the ratio

The bias factor $[\exp(eV_{BE}/kT) - 1]$ and e cancel:

$$\frac{J_{nE}}{J_{pE}} = \frac{D_{nB}\, n_{iB}^2/(N_{AB}W_B)}{D_{pE}\, n_{iE}^2/(N_{DE}W_E)} = \frac{N_{DE}}{N_{AB}} \cdot \frac{W_E}{W_B} \cdot \frac{D_{nB}}{D_{pE}} \cdot \frac{n_{iB}^2}{n_{iE}^2}$$

Step 4: Intrinsic concentrations of the two materials

$$\frac{n_{iB}^2}{n_{iE}^2} = \frac{N_{cB}N_{vB}\exp(-E_{gB}/kT)}{N_{cE}N_{vE}\exp(-E_{gE}/kT)} \approx \exp\left(\frac{E_{gE} - E_{gB}}{kT}\right) = \exp\left(\frac{\Delta E_g}{kT}\right)$$

assuming the effective densities of states are nearly equal in the two materials ($N_{cB}N_{vB} \approx N_{cE}N_{vE}$).

Step 5: Result

$$\boxed{\frac{J_{nE}}{J_{pE}} = \frac{N_{DE}}{N_{AB}} \cdot \frac{W_E}{W_B} \cdot \frac{D_{nB}}{D_{pE}} \exp\left(\frac{\Delta E_g}{kT}\right)}$$

which is the required expression (with D_{nB}, the electron diffusion coefficient in the base, for the paper's D_{nE}). Even with $N_{DE} < N_{AB}$ the exponential makes the ratio huge: for $\Delta E_g = 0.374$ eV ($Al_{0.3}Ga_{0.7}As$), $\exp(0.374/0.0259) = 1.9 \times 10^6$.

Steps to remember

Short-base electron current $\propto D_{nB}n_{iB}^2/(N_{AB}W_B)$ → short-emitter hole current $\propto D_{pE}n_{iE}^2/(N_{DE}W_E)$ → ratio → $n_{iB}^2/n_{iE}^2 = e^{\Delta E_g/kT}$.

3. Keyword Explanation

1. Emitter injection efficiency

↩ back

$\gamma = J_{nE}/(J_{nE} + J_{pE})$: the fraction of emitter current carried by the useful electrons.

2. Short diode

↩ back

A region much thinner than the diffusion length, so the minority-carrier profile is a straight line.

4. Notes and Remarks

- **Note on the paper's formula:** it prints D_{nE}; the electron current is a diffusion current in the **base**, so the coefficient is D_{nB}. With D_{nE} the units still work but the physics does not.
- If the emitter and base are wide (long diodes), W_E and W_B are replaced by the diffusion lengths L_{pE} and L_{nB}.
- The conduction-band spike at an abrupt junction reduces the electron injection a little; a graded junction removes it (Sze & Ng).

4.(c) For an ideal Metal Oxide Semiconductor capacitor draw charge distribution, electric field distribution, potential distribution and capacitance versus voltage curve. Marks: 6

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1. Theory

1. The MOS capacitor (Core Idea)

A **metal–oxide–semiconductor (MOS) capacitor** is a metal gate, a thin insulating oxide (SiO_2, thickness t_{ox}) and a semiconductor substrate. "Ideal" means: no charge in the oxide or at the interface, no current through the oxide, and a work-function difference of zero ($\phi_{ms} = 0$), so the bands are flat at $V_G = 0$.

2. The three regimes for a p-type substrate

- **Accumulation ($V_G < 0$):** the negative gate attracts holes to the surface; a thin positive layer of holes.
- **Depletion ($0 < V_G < V_T$):** the positive gate pushes holes away and leaves negative acceptor ions in a depletion layer of width x_d.
- **Inversion ($V_G > V_T$):** the bands bend so far that electrons collect at the surface, an n-type **inversion layer**. At threshold the surface potential is $\phi_s = 2\phi_{fp}$ and the depletion width reaches its maximum,

$$x_{dT} = \left(\frac{4\epsilon_s \phi_{fp}}{eN_a}\right)^{1/2}, \qquad \phi_{fp} = V_t \ln \frac{N_a}{n_i}$$

(Neamen, Eq. 10.7). Any further charge is added to the inversion layer, not to the depletion layer.

3. The tools

- **Charge neutrality:** $Q'_m + Q'_n + Q'_{SD} = 0$ (per area).
- **Gauss's law and Poisson's equation:** the field is constant in the charge-free oxide, jumps at the oxide–semiconductor interface ($\epsilon_{ox} E_{ox} = \epsilon_s E_s$), and falls linearly through the uniform depletion charge.
- **Potential:** the gate voltage divides as $V_G = V_{ox} + \phi_s$ (ideal case).

2. Answer

Example used for the curves

p-Si, $N_a = 10^{16}$ cm^{-3}, $t_{ox} = 20$ nm, so $\phi_{fp} = 0.347$ V, $x_{dT} = 0.300$ µm, $C_{ox} = 1.73 \times 10^{-7}$ F/cm^2.

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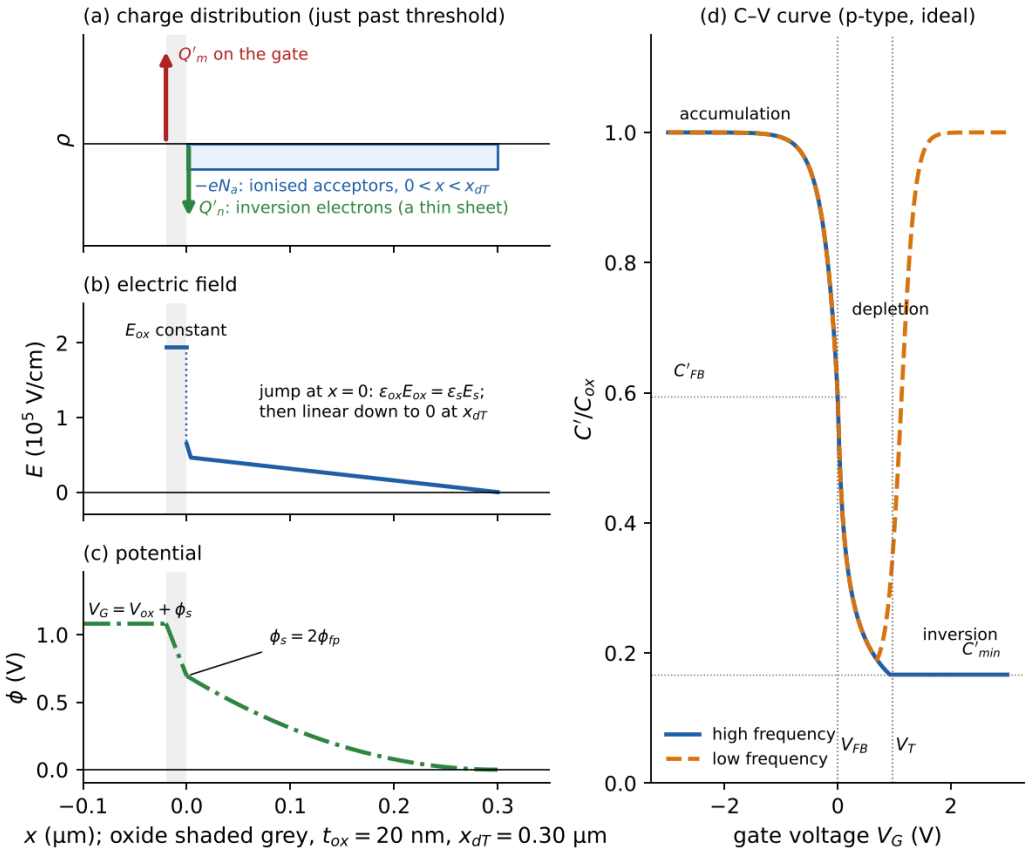


Figure 16: Ideal MOS capacitor on p-Si: (a) charge, (b) field and (c) potential at inversion, and (d) the C-V curves

Reading the figure. The grey strip is the oxide. **(a) Charge:** a positive sheet Q'_m on the gate (red arrow), the negative inversion-electron sheet Q'_n at the surface (green arrow), and the uniform negative acceptor charge $-eN_a$ of the depletion layer, width x_{dT} (blue block); $Q'_m = |Q'_n| + eN_a x_{dT}$. **(b) Field:** constant in the oxide (no charge there), a jump down at the interface by the factor $\epsilon_{ox}/\epsilon_s = 0.33$, a small drop across the inversion sheet, then a linear fall to zero at x_{dT}. **(c) Potential:** linear in the oxide (drop V_{ox}), parabolic in the depletion layer, from $\phi_s = 2\phi_{fp}$ at the surface to 0 in the bulk; $V_G = V_{ox} + \phi_s$. **(d) C–V:** high-frequency (solid blue) and low-frequency (dashed orange) capacitance against V_G.

Charge distribution: gate charge Q'_m (sheet) $= -(Q'_n + Q'_{SD})$, with $Q'_{SD} = -eN_a x_d$ (uniform block) and Q'_n (thin sheet at the surface, present only in inversion).

Electric field:

- In the oxide: $E_{ox} = Q'_m/\epsilon_{ox}$, constant.
- At the interface: $\epsilon_{ox}E_{ox} = \epsilon_s E_s$, so the field drops (Si: $\epsilon_s = 3\epsilon_{ox}$).
- In the depletion layer: $E(x) = \dfrac{eN_a}{\epsilon_s}(x_d - x)$, linear, zero at x_d.

Potential: $V_{ox} = E_{ox}t_{ox}$ (linear); in the semiconductor $\phi(x) = \phi_s(1 - x/x_d)^2$ (parabolic), with $\phi_s = eN_a x_d^2/2\epsilon_s$.

Capacitance–voltage curve (the capacitance per area $C' = dQ'_m/dV_G$ is the oxide capacitance in series with the semiconductor's):

1. **Accumulation:** holes at the surface respond at once; $C' = C_{ox} = \epsilon_{ox}/t_{ox}$.
2. **Flat band ($V_G = V_{FB}$):** $C'_{FB} = \dfrac{\epsilon_{ox}}{t_{ox} + (\epsilon_{ox}/\epsilon_s)L_D}$ with the Debye length $L_D = \sqrt{\epsilon_s V_t/(eN_a)}$ ($C'_{FB}/C_{ox} = 0.59$ in the example).
3. **Depletion:** the depletion capacitance ϵ_s/x_d is in series with C_{ox}: $C' = \dfrac{\epsilon_{ox}}{t_{ox} + (\epsilon_{ox}/\epsilon_s)x_d}$, falling

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as x_d grows.

4. **Inversion:** at **low frequency** the inversion electrons follow the signal and C' rises back to C_{ox}. At **high frequency** they cannot (they are generated slowly), x_d stays at x_{dT} and C' stays at the minimum $C'_{min} = \dfrac{\epsilon_{ox}}{t_{ox} + (\epsilon_{ox}/\epsilon_s)x_{dT}}$ ($C'_{min}/C_{ox} = 0.17$ in the example).

3. Keyword Explanation

1. Threshold inversion point

↩ back

The gate voltage at which the surface electron density equals the bulk hole density, $\phi_s = 2\phi_{fp}$.

2. Debye length

↩ back

$L_D = \sqrt{\epsilon_s V_t / eN_a}$: the screening length that sets the semiconductor capacitance at flat band.

4. Notes and Remarks

- For an n-type substrate everything is mirrored: accumulation for $V_G > 0$, inversion (holes) for $V_G < V_T < 0$.
- The C–V curves follow Neamen, Section 10.2 (Figures 10.24–10.27); the numbers are from the calc script.

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4.(d) Show the relation between Flat band voltage and threshold voltage of a MOS device. Marks: 6

1. Theory

1. Flat-band voltage (Core Idea)

In a real MOS structure two things bend the bands even at $V_G = 0$:

- the **work-function difference** $\phi_{ms} = \phi'_m - \left(\chi' + \frac{E_g}{2e} + \phi_{fp}\right)$ between gate and semiconductor (Neamen, Eq. 10.15);
- a fixed positive **oxide charge** Q'_{ss} (per area) at the oxide–semiconductor interface.

The **flat-band voltage** V_{FB} is the gate voltage that cancels both, so that the semiconductor bands are flat (no space charge).

2. Threshold voltage

The **threshold voltage** V_T is the gate voltage that brings the surface to the threshold inversion point, $\phi_s = 2\phi_{fp}$, where the depletion charge is at its maximum, $|Q'_{SD}(\max)| = eN_a x_{dT}$.

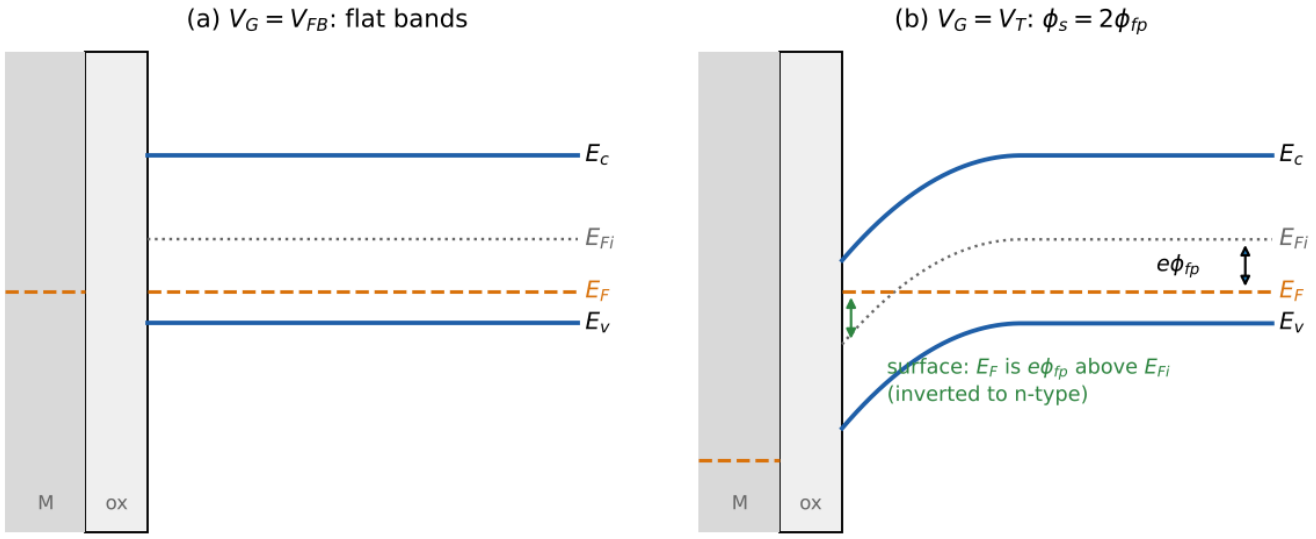


Figure 17: MOS on p-type silicon: (a) at the flat-band voltage, (b) at the threshold voltage (ideal oxide, metal Fermi level shown on the left)

Reading the figure. The grey block is the gate (M), the light block the oxide (ox). Solid blue lines are E_c and E_v, the dotted grey line the intrinsic level E_{Fi}, the dashed orange line the semiconductor E_F. (a) At V_{FB} the bands are flat up to the oxide. (b) At V_T the bands bend down by $2e\phi_{fp}$: in the bulk E_{Fi} is $e\phi_{fp}$ above E_F, at the surface it is $e\phi_{fp}$ below it, so the surface is as n-type as the bulk is p-type; the gate Fermi level has moved down by eV_T.

2. Answer

Step 1: Charge neutrality at flat band

At flat band there is no space charge in the semiconductor, so the gate charge only balances the oxide charge:

$$Q'_m + Q'_{ss} = 0$$

Step 2: Voltage across the oxide at flat band

$$V_{ox} = \frac{Q'_m}{C_{ox}} = -\frac{Q'_{ss}}{C_{ox}}$$

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Step 3: Voltage loop at flat band

Going round the circuit (gate → oxide → semiconductor), $V_G = V_{ox} + \phi_s + \phi_{ms}$ (Neamen, Eq. 10.23). With $\phi_s = 0$ at flat band:

$$\boxed{V_{FB} = \phi_{ms} - \frac{Q'_{ss}}{C_{ox}}} \qquad (10.25)$$

Step 4: Charge neutrality at threshold

At threshold the semiconductor holds the maximum depletion charge $-|Q'_{SD}(\max)|$ (the inversion charge is still negligible):

$$Q'_{mT} + Q'_{ss} = |Q'_{SD}(\max)|$$

Step 5: Oxide voltage at threshold

$$V_{oxT} = \frac{Q'_{mT}}{C_{ox}} = \frac{|Q'_{SD}(\max)| - Q'_{ss}}{C_{ox}}$$

Step 6: Voltage loop at threshold

With $\phi_s = 2\phi_{fp}$:

$$V_{TN} = V_{oxT} + 2\phi_{fp} + \phi_{ms} = \frac{|Q'_{SD}(\max)|}{C_{ox}} - \frac{Q'_{ss}}{C_{ox}} + \phi_{ms} + 2\phi_{fp} \qquad (10.31a)$$

Step 7: The relation

The middle two terms are exactly V_{FB} from Step 3, so

$$\boxed{V_{TN} = V_{FB} + \frac{|Q'_{SD}(\max)|}{C_{ox}} + 2\phi_{fp}} \qquad (10.31c)$$

with $|Q'_{SD}(\max)| = eN_a x_{dT}$, $x_{dT} = (4\epsilon_s\phi_{fp}/eN_a)^{1/2}$ and $C_{ox} = \epsilon_{ox}/t_{ox}$.

Meaning: first the gate must undo the built-in band bending (V_{FB}), then supply the depletion charge (voltage $|Q'_{SD}(\max)|/C_{ox}$ across the oxide) and bend the bands by $2\phi_{fp}$. For an n-type substrate (p-channel), $V_{TP} = V_{FB} - |Q'_{SD}(\max)|/C_{ox} - 2\phi_{fn}$.

Steps to remember

Flat band: $Q'_m = -Q'_{ss}$, $\phi_s = 0 \to V_{FB} = \phi_{ms} - Q'_{ss}/C_{ox}$ → threshold: $Q'_{mT} = |Q'_{SD}(\max)| - Q'_{ss}$, $\phi_s = 2\phi_{fp} \to V_T = V_{FB} + |Q'_{SD}(\max)|/C_{ox} + 2\phi_{fp}$.

3. Keyword Explanation

1. Work-function difference

↩ back

ϕ_{ms}: the difference between the gate's and the semiconductor's work functions (measured through the oxide), which bends the bands at zero gate voltage.

2. Fixed oxide charge

↩ back

Q'_{ss}: positive charge trapped in the oxide near the interface; it acts like an extra gate voltage.

4. Notes and Remarks

- A numerical use of this relation is 8(b) (Term Test 2).
- A positive Q'_{ss} and a negative ϕ_{ms} (aluminium on p-Si) can make V_{TN} negative: a depletion-mode device (Neamen, Example 10.4).

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5. Final Concise Answer

(a) DHBT (mesa): n^+ GaAs cap / N-$Al_{0.3}Ga_{0.7}As$ emitter (5×10^{17}) / thin p^+ GaAs base (10^{19}, 50–100 nm) / N-AlGaAs collector (10^{16}–10^{17}) / n^+ GaAs sub-collector / semi-insulating substrate. Base doped higher than the emitter; wide-gap emitter and collector (two heterojunctions); graded junctions remove spikes; n^+ contact layers.

(b) Advantages: high gain independent of the doping ratio, heavily doped thin base (low r_b, no punch-through, small Early effect), lightly doped emitter (low C_{je}), high f_T and f_{max}. Short-diode currents $J_{nE} = eD_{nB}n_{iB}^2e^{eV_{BE}/kT}/(N_{AB}W_B)$ and $J_{pE} = eD_{pE}n_{iE}^2e^{eV_{BE}/kT}/(N_{DE}W_E)$ with $n_{iB}^2/n_{iE}^2 = e^{\Delta E_g/kT}$ give $J_{nE}/J_{pE} = (N_{DE}/N_{AB})(W_E/W_B)(D_{nB}/D_{pE})e^{\Delta E_g/kT}$.

(c) At inversion: gate sheet $+Q'_m$, inversion sheet Q'_n, depletion block $-eN_a$ to x_{dT}; field constant in the oxide, drops by ϵ_{ox}/ϵ_s at the interface, falls linearly to 0 at x_d; potential linear in the oxide, parabolic in the silicon ($\phi_s = 2\phi_{fp}$ at threshold). C–V: C_{ox} in accumulation, falls through depletion ($C' = \epsilon_{ox}/[t_{ox} + (\epsilon_{ox}/\epsilon_s)x_d]$), at high frequency stays at C'_{min} in inversion, at low frequency returns to C_{ox}.

(d) $V_{FB} = \phi_{ms} - Q'_{ss}/C_{ox}$ and $V_T = V_{FB} + |Q'_{SD}(\text{max})|/C_{ox} + 2\phi_{fp}$, with $|Q'_{SD}(\text{max})| = eN_ax_{dT}$.

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5. (a) Derive the equation for emitter injection factor of a bipolar junction transistor. Marks: 7

(b) Briefly explain the following non-ideal effects of a bipolar junction transistor.

(i) Early effect
(ii) Current Crowding Marks: 6

(c) Describe the Gummel-Poon model of a bipolar junction transistor. Marks: 7

(d) Consider a uniformly doped silicon npn bipolar transistor at T = 300 K with the following parameters: $D_B = 23\ \text{cm}^2/\text{s}$, $D_E = 8\ \text{cm}^2/\text{s}$, $\tau_{B0} = 2\times10^{-7}$ s, $\tau_{E0} = 8\times10^{-8}$ s, $N_B = 2\times10^{16}$ cm^{-3}, and $x_E = 0.35$ µm. The recombination factor has been determined to be $\delta = 0.9975$. The required common-emitter current gain is $\beta = 150$. A minimum neutral base width of $x_B = 0.80$ µm can be fabricated. (a) Determine an appropriate neutral base width and the minimum emitter doping concentration, N_E, to meet this specification. (b) Using the results of part (a), what are the values of α_T and γ? Marks: 5

5.(a) Derive the equation for emitter injection factor of a bipolar junction transistor. Marks: 7

1. Theory

1. Current components at the emitter (Core Idea)

In an npn transistor in forward-active mode the forward-biased B–E junction carries:

- J_{nE}: electrons injected from the emitter into the base (they diffuse to the collector: the useful current);
- J_{pE}: holes injected from the base back into the emitter (they only add to the base current);
- J_R: recombination inside the B–E depletion region (treated separately as the recombination factor δ).

The **emitter injection efficiency factor** (Neamen, Eq. 12.32) is the share of the emitter current carried by the useful electrons:

$$\gamma = \frac{J_{nE}}{J_{nE} + J_{pE}}$$

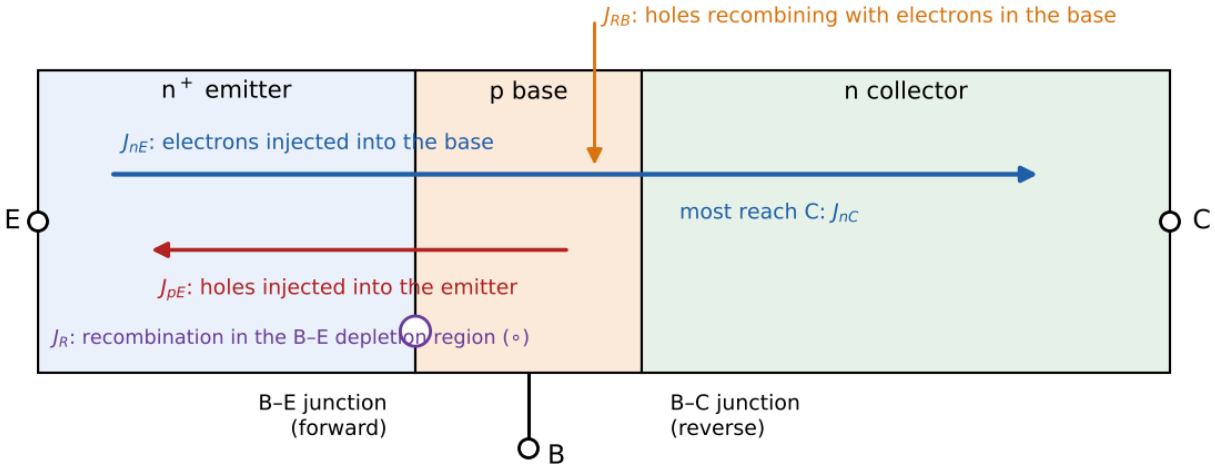


Figure 18: Current components in an npn transistor in forward-active mode

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Reading the figure. The blue arrow is the electron flow from the emitter through the base to the collector (J_{nE}, most of which becomes J_{nC}). The red arrow is the hole flow from the base into the emitter (J_{pE}). The orange arrow marks holes supplied by the base contact to recombine with electrons in the base. The purple circle at the B–E junction marks recombination in the depletion region (J_R).

2. Tools

- The **minority-carrier diffusion equation** in each neutral region: $D\,d^2(\delta p)/dx^2 - \delta p/\tau = 0$, with diffusion length $L = \sqrt{D\tau}$.
- The **law of the junction**: at the edge of the forward-biased junction the minority concentration is the equilibrium value times $e^{eV_{BE}/kT}$.
- **Diffusion current:** $J = eD\,d(\delta n)/dx$ (electrons), $J = -eD\,d(\delta p)/dx$ (holes), evaluated at the junction edge.

2. Answer

Coordinates: in the base x runs from 0 (B–E edge) to x_B; in the emitter x' runs from 0 (B–E edge) to x_E (the emitter contact). Equilibrium minority values $n_{B0} = n_i^2/N_B$, $p_{E0} = n_i^2/N_E$.

Step 1: Holes in the emitter

Diffusion equation: $\dfrac{d^2\delta p_E}{dx'^2} - \dfrac{\delta p_E}{L_E^2} = 0$, general solution $\delta p_E = A\cosh(x'/L_E) + B\sinh(x'/L_E)$.

Boundary conditions: $\delta p_E(0) = p_{E0}[\exp(eV_{BE}/kT) - 1]$ (law of the junction) and $\delta p_E(x_E) = 0$ (ohmic contact, infinite recombination).

Solving as in 2(c) (with the identity $\sinh(a-b) = \sinh a\cosh b - \cosh a\sinh b$):

$$\delta p_E(x') = p_{E0}\left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1\right]\frac{\sinh[(x_E - x')/L_E]}{\sinh(x_E/L_E)}$$

Step 2: Hole current at the junction

$$J_{pE} = -eD_E\left.\frac{d\delta p_E}{dx'}\right|_{x'=0} = \frac{eD_E p_{E0}}{L_E}\cdot\frac{\cosh(x_E/L_E)}{\sinh(x_E/L_E)}\left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1\right]$$
$$= \frac{eD_E p_{E0}}{L_E\tanh(x_E/L_E)}\left[\exp\left(\frac{eV_{BE}}{kT}\right) - 1\right]$$

(the derivative of $\sinh[(x_E - x')/L_E]$ is $-\cosh[(x_E - x')/L_E]/L_E$).

Step 3: Electrons in the base

From 2(c), with the B–C junction reverse biased:

$$\delta n_B(x) = \frac{n_{B0}[\exp(eV_{BE}/kT) - 1]\sinh[(x_B - x)/L_B] - n_{B0}\sinh(x/L_B)}{\sinh(x_B/L_B)}$$

Step 4: Electron current at the junction

$$J_{nE} = eD_B\left|\frac{d\delta n_B}{dx}\right|_{x=0}$$
$$= \frac{eD_B n_{B0}}{L_B}\left\{\frac{\exp(eV_{BE}/kT) - 1}{\tanh(x_B/L_B)} + \frac{1}{\sinh(x_B/L_B)}\right\}$$

(the derivative of $\sinh[(x_B - x)/L_B]$ at 0 is $-\cosh(x_B/L_B)/L_B$, and of $\sinh(x/L_B)$ it is $1/L_B$).

Step 5: The efficiency

$$\gamma = \frac{J_{nE}}{J_{nE} + J_{pE}} = \frac{1}{1 + J_{pE}/J_{nE}}$$

In forward-active mode $\exp(eV_{BE}/kT) \gg 1$, so the "-1" and the small $1/\sinh$ term can be dropped:

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$$\frac{J_{pE}}{J_{nE}} = \frac{p_{E0} D_E L_B}{n_{B0} D_B L_E} \cdot \frac{\tanh(x_B/L_B)}{\tanh(x_E/L_E)}$$

$$\boxed{\gamma = \frac{1}{1 + \dfrac{p_{E0} D_E L_B}{n_{B0} D_B L_E} \cdot \dfrac{\tanh(x_B/L_B)}{\tanh(x_E/L_E)}}} \qquad (12.35a)$$

Step 6: Short emitter and base

If $x_B \ll L_B$ and $x_E \ll L_E$, then $\tanh u \approx u$, and with $p_{E0}/n_{B0} = N_B/N_E$:

$$\boxed{\gamma \approx \frac{1}{1 + \dfrac{N_B}{N_E} \cdot \dfrac{D_E}{D_B} \cdot \dfrac{x_B}{x_E}}} \qquad (12.36)$$

Meaning: $\gamma \to 1$ needs $N_E \gg N_B$ (emitter much more heavily doped than the base), and a narrow base helps.

Steps to remember

Diffusion equation in the emitter with $\delta p_E(x_E) = 0 \to J_{pE}$ with $1/\tanh(x_E/L_E) \to$ base solution $\to J_{nE}$ with $1/\tanh(x_B/L_B) \to$ ratio, drop small terms $\to \gamma$; short regions give $1/(1 + N_B D_E x_B / N_E D_B x_E)$.

3. Keyword Explanation

1. Emitter injection efficiency

↩ back

γ: the fraction of emitter current due to carriers injected into the base; one of the three factors of $\alpha = \gamma \alpha_T \delta$.

2. Short diode

↩ back

A neutral region much thinner than its diffusion length; its minority-carrier profile is a straight line.

4. Notes and Remarks

- The derivation follows Neamen, Section 12.3.1; the other two factors are the base transport factor $\alpha_T = 1/\cosh(x_B/L_B)$ and the recombination factor δ.
- With heavy emitter doping, bandgap narrowing raises p_{E0} ($p'_{E0} = (n_i^2/N_E)e^{\Delta E_g/kT}$) and lowers γ; HBTs avoid this with a wide-gap emitter.

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5.(b) Briefly explain the following non-ideal effects of a bipolar junction transistor. (i) Early effect (ii) Current Crowding Marks: 6

1. Theory

1. The ideal output curve (Core Idea)

In an ideal transistor the collector current in forward-active mode depends only on V_{BE}: $I_C \propto (D_B n_{B0}/x_B)e^{eV_{BE}/kT}$. So the output curves ($I_C$ against V_{CE} at fixed I_B) would be flat lines. Real transistors depart from this; two of the non-ideal effects are below.

2. Why the base width can change

The B–C junction is reverse biased. Its depletion region spreads into both the base and the collector, by an amount that grows with V_{CB} (as $\sqrt{V_{bi}+V_{CB}}$). So the **neutral** base width x_B, the part of the base outside the depletion regions, shrinks as V_{CB} grows.

3. Why the base current runs sideways

The base contact is beside the emitter, so the base current flows **sideways** (parallel to the junction) under the emitter, through the thin base layer, which has resistance. This lateral current makes a voltage drop along the base.

2. Answer

(i) Early effect (base-width modulation)

- As V_{CE} (hence V_{CB}) rises, the B–C depletion region widens into the base and the **neutral base width x_B shrinks**.
- With the same $n_B(0) = n_{B0}e^{eV_{BE}/kT}$, the minority-electron profile becomes **steeper**, so the diffusion current $I_C \propto eD_B n_B(0)/x_B$ **increases**. Recombination in the base also falls, so α_T and β rise.
- The output curves slope upward. Extended backwards, they meet the V_{CE} axis at about the same point, $-V_A$, where V_A is the **Early voltage** (typically 100–300 V; Neamen):

$$\frac{dI_C}{dV_{CE}} = \frac{I_C}{V_{CE}+V_A}, \qquad r_o = \frac{V_A}{I_C}$$

- **Consequences:** a finite output resistance r_o (lower voltage gain), and in the extreme **punch-through**, when the B–C depletion region reaches the B–E one.
- **Reduced by:** heavier base doping (the depletion spreads less into it), lighter collector doping, wider base (at the cost of gain); HBTs, with very heavily doped bases, have high V_A.

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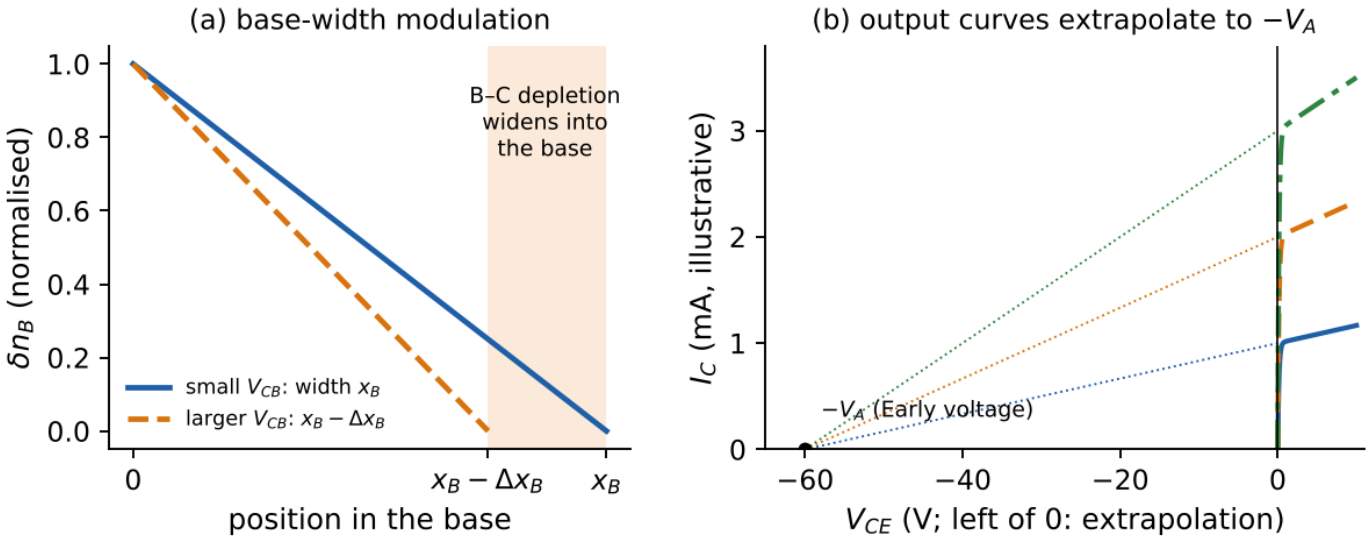


Figure 19: (a) Base-width modulation: the electron profile gets steeper as the B-C depletion region widens; (b) output curves extrapolate to the Early voltage

Reading the figure. Panel (a): the solid blue line is the electron profile at small V_{CB}; at larger V_{CB} the shaded B–C depletion region takes a part Δx_B of the base, and the dashed orange profile, starting from the same value, falls more steeply. Panel (b): three output curves at increasing base current (solid blue, dashed orange, dash-dot green), each sloping up in the active region; their dotted extensions all meet the axis at $-V_A$ (here 60 V, illustrative).

(ii) Current crowding (emitter crowding)

- The lateral base current I_B flowing under the emitter causes an IR drop in the thin base, so the actual B–E voltage is **largest at the emitter edges** (nearest the base contacts) and smaller at the centre.
- Since the injected current goes as $\exp(eV_{BE}/kT)$, even a drop of a few kT/e makes the emitter current **crowd at the edges**; the centre carries little.
- **Consequences:** high local current density at the edges (local high-level injection, lower β), local heating (hot spots, second breakdown), and a larger effective base resistance.
- **Reduced by:** narrow emitter **stripes** interleaved with base contacts (interdigitated layout, high perimeter-to-area ratio), low base sheet resistance, emitter ballast resistors in power transistors.

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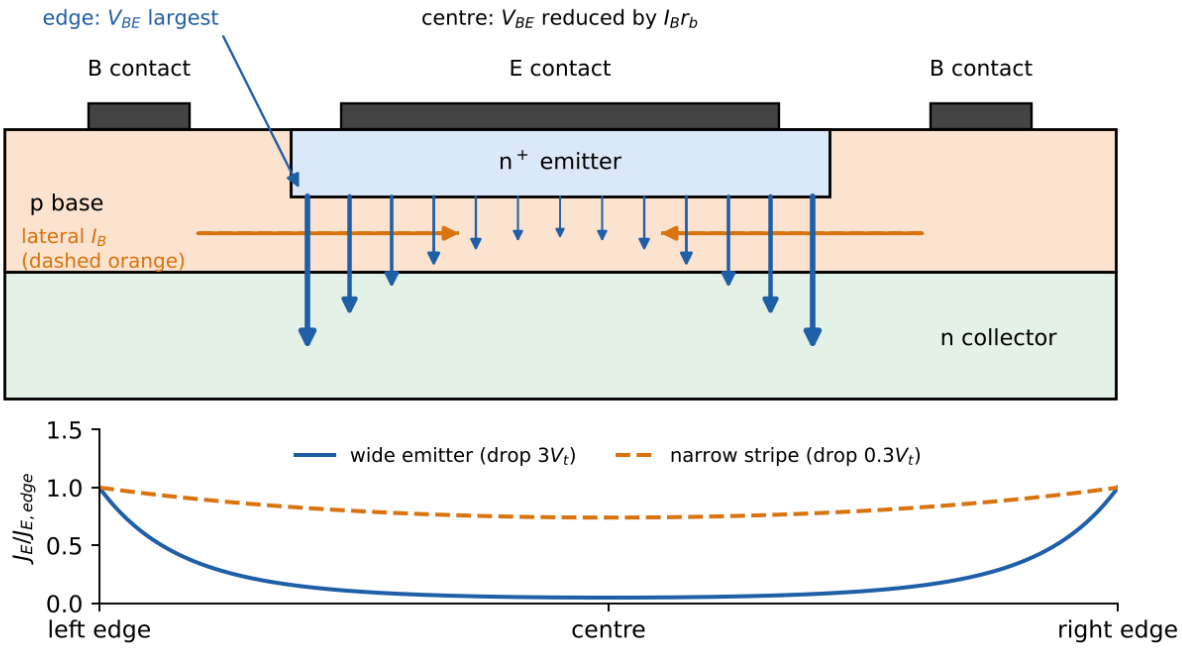


Figure 20: Emitter current crowding: the lateral base current makes V_{BE} largest at the emitter edges (reused from the 2020–21 paper)

Reading the figure. Top: the n^+ emitter sits on the p base with base contacts on both sides; the lateral base current (dashed orange) flows inwards under the emitter, and the injected emitter current (blue arrows) is largest at the edges. Bottom: the emitter current density across the emitter, normalised to its edge value: a wide emitter (solid blue) has almost no current at its centre, while a narrow stripe (dashed orange) is nearly uniform.

3. Keyword Explanation

1. Early voltage

↩ back

V_A: the magnitude of the intercept of the extrapolated output curves on the V_{CE} axis; a large V_A means flat curves.

2. Punch-through

↩ back

Breakdown that occurs when the neutral base vanishes and the collector depletion region touches the emitter.

4. Notes and Remarks

- **Early's formula** (Neamen, Section 12.4.1): from $I_C \propto 1/x_B$, $dI_C/dV_{CE} = -(I_C/x_B)(dx_B/dV_{CE})$, which with a constant slope gives the straight lines meeting at $-V_A$.
- The current-crowding figure and a full treatment are also in 6(b) of the 2020–21 paper (extra reading).

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5.(c) Describe the Gummel-Poon model of a bipolar junction transistor. Marks: 7

1. Theory

1. Why another model? (Core Idea)

The **Ebers–Moll** model treats the transistor as two coupled diodes with constant parameters. It cannot describe a base with **non-uniform doping**, the **Early effect**, **high-level injection** (gain falling at high current) or **recombination** at low current. The **Gummel–Poon** model (1970) fixes this by writing the collector current in terms of the **total charge** in the base. It is the BJT model used in SPICE.

2. Drift and diffusion with a non-uniform base

If the base doping $N_A(x)$ varies, the holes set up a built-in field (Neamen, Eq. 12.79)

$$E = \frac{kT}{e} \cdot \frac{1}{p(x)} \cdot \frac{dp}{dx}$$

which acts on the electrons crossing the base.

2. Answer

Step 1: Electron current with drift and diffusion

In the base of an npn transistor (Neamen, Eq. 12.78):

$$J_n = e\mu_n n(x)E + eD_n \frac{dn}{dx}$$

Step 2: Put in the built-in field

Using E above and Einstein's relation $\mu_n kT/e = D_n$:

$$J_n = \frac{eD_n}{p(x)}\left[n(x)\frac{dp}{dx} + p(x)\frac{dn}{dx}\right] = \frac{eD_n}{p(x)} \cdot \frac{d(pn)}{dx}$$

Step 3: Integrate across the base

J_n and D_n are nearly constant across a thin base, so multiply by $p(x)/eD_n$ and integrate from 0 to x_B:

$$\frac{J_n}{eD_n}\int_0^{x_B} p(x)\,dx = p(x_B)n(x_B) - p(0)n(0)$$

Step 4: Boundary values

Forward-active: $n(x_B) \approx 0$ (reverse-biased B–C) and $p(0)n(0) = n_i^2 \exp(eV_{BE}/kT)$ (law of the junction). Hence

$$\boxed{J_n = \frac{-eD_n n_i^2 \exp(eV_{BE}/kT)}{\displaystyle\int_0^{x_B} p(x)\,dx}} \qquad (12.84)$$

The denominator is the **base Gummel number** Q_B (the total hole charge per area in the base, divided by e). Likewise the hole current into the emitter is set by the **emitter Gummel number** $Q_E = \int_0^{x_E} n(x')\,dx'$ (Eq. 12.85).

Step 5: The full model

The collector current is written with the **normalised base charge** $q_b = Q_B/Q_{B0}$:

$$I_C = \frac{I_S}{q_b}\left[\exp\left(\frac{V_{BE}}{V_t}\right) - \exp\left(\frac{V_{BC}}{V_t}\right)\right] - (\text{B–C diode currents})$$

$$q_b = \frac{q_1}{2}\left(1 + \sqrt{1+4q_2}\right), \quad q_1 = \left(1 - \frac{V_{BC}}{V_A} - \frac{V_{BE}}{V_B}\right)^{-1}, \quad q_2 = \frac{I_S}{I_{KF}}e^{V_{BE}/V_t} + \frac{I_S}{I_{KR}}e^{V_{BC}/V_t}$$

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- q_1 describes the change of the depletion charges with bias: the **Early effect** (V_A, forward Early voltage; V_B, reverse).
- q_2 describes the extra injected charge: **high-level injection**; above the knee current I_{KF}, $q_b \approx \sqrt{q_2}$ and $I_C \propto \exp(V_{BE}/2V_t)$.
- The **base current** has ideal terms I_S/β_F and I_S/β_R and **non-ideal recombination** terms with emission coefficients n_{EL}, n_{CL} (C_2, C_4), which make β fall at low current.
- Series resistances r_B (bias-dependent, current crowding), r_E, r_C and junction and diffusion capacitances complete the model.

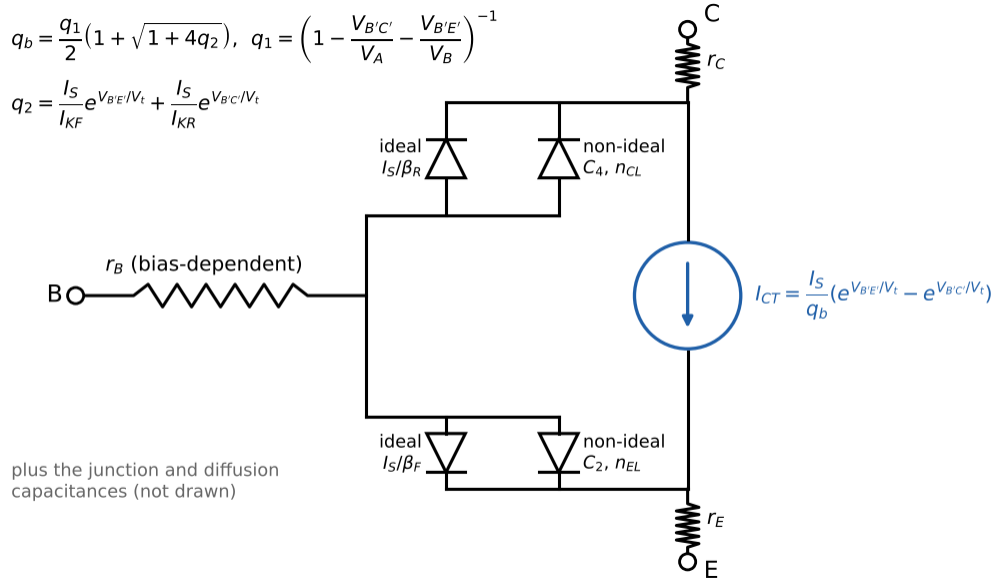


Figure 21: Gummel-Poon equivalent circuit of an npn transistor (SPICE form)

Reading the figure. Between the internal collector and emitter nodes sits the transport current source I_{CT} (the circle with an arrow), divided by q_b. From the internal base, two diodes go to the emitter (ideal I_S/β_F and non-ideal recombination) and two to the collector (ideal I_S/β_R and non-ideal). The external terminals connect through r_B, r_C and r_E. The formulas for q_b, q_1 and q_2 are at the top.

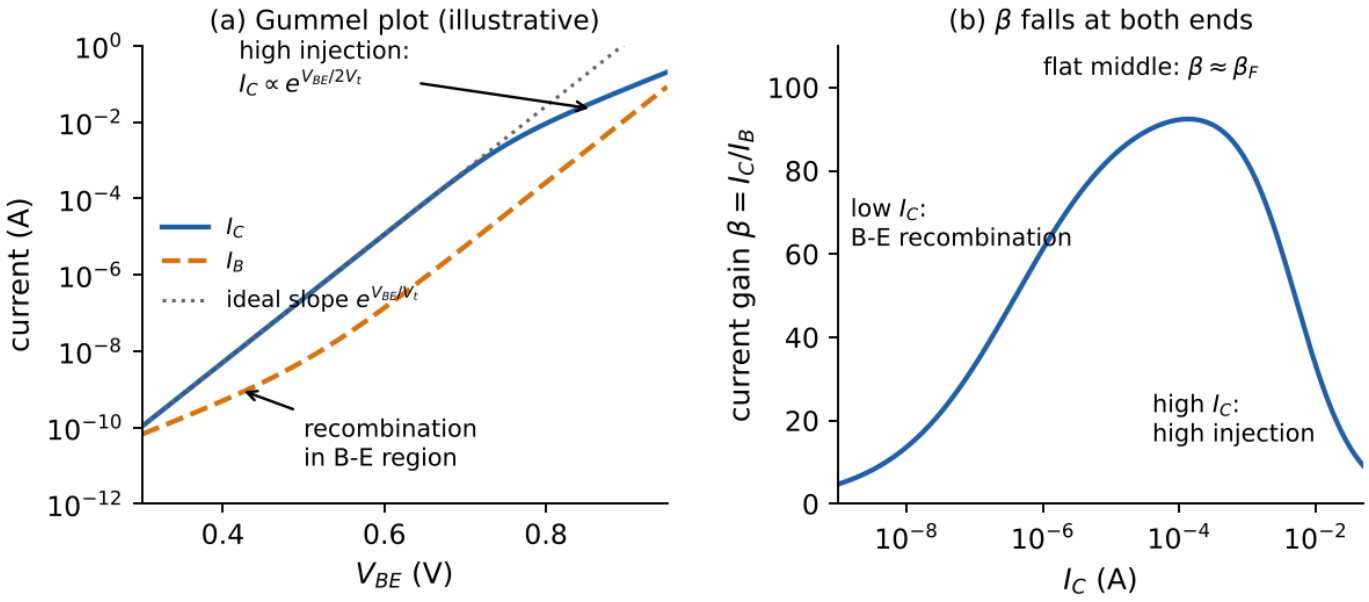


Figure 22: Gummel plot and current gain against collector current (reused from the 2020–21 paper)

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Reading the figure. Panel (a): I_C (solid blue) and I_B (dashed orange) on a log scale against V_{BE}. At low V_{BE} the base current has the slower slope of B–E recombination ($n_{EL} \approx 2$); at high V_{BE} the collector current bends to the $e^{V_{BE}/2V_t}$ slope of high-level injection. Panel (b): the gain $\beta = I_C/I_B$ is low at small and large currents and flat in between. These are exactly the effects the Gummel–Poon terms model.

3. Keyword Explanation

1. Gummel number

↩ back

The integrated majority-carrier charge per area of the base (Q_B) or emitter (Q_E); the transistor currents are inversely proportional to them.

2. Knee current

↩ back

I_{KF}: the collector current above which high-level injection makes the gain fall.

4. Notes and Remarks

- The derivation of Steps 1–4 is Neamen, Section 12.5.2; the full parameter set (q_1, q_2, I_{KF}, C_2, n_{EL}) follows Sze & Ng, Chapter 5, and the SPICE BJT model.
- For a uniformly doped base at low injection, $\int p\,dx = N_B x_B$ and Eq. 12.84 reduces to the ideal $J_n = eD_n n_{B0} e^{eV_{BE}/kT}/x_B$.

5.(d) Consider a uniformly doped silicon npn bipolar transistor at T = 300 K with the following parameters: $D_B = 23$ cm^2/s, $D_E = 8$ cm^2/s, $\tau_{B0} = 2 \times 10^{-7}$ s, $\tau_{E0} = 8 \times 10^{-8}$ s, $N_B = 2 \times 10^{16}$ cm^{-3}, and $x_E = 0.35$ µm. The recombination factor has been determined to be $\delta = 0.9975$. The required common-emitter current gain is $\beta = 150$. A minimum neutral base width of $x_B = 0.80$ µm can be fabricated. (a) Determine an appropriate neutral base width and the minimum emitter doping concentration, N_E, to meet this specification. (b) Using the results of part (a), what are the values of α_T and γ? Marks: 5

1. Theory

1. The current gain from its three factors (Core Idea)

The common-base gain is the product of three efficiencies (Neamen, Eq. 12.46):

$$\alpha = \gamma\,\alpha_T\,\delta, \qquad \beta = \frac{\alpha}{1-\alpha}$$

- γ (emitter injection efficiency) from 5(a);
- $\alpha_T = 1/\cosh(x_B/L_B)$ (base transport factor): the share of injected electrons that cross the base without recombining (Eq. 12.39a);
- δ (recombination factor): the share of emitter current not lost in the B–E depletion region.

2. Design logic

A thinner base gives a larger α_T; a larger α_T means γ may be smaller, so the emitter doping can be lower. So the **minimum emitter doping** goes with the **minimum base width** that can be made.

2. Answer

Given: $D_B = 23$ cm^2/s, $D_E = 8$ cm^2/s, $\tau_{B0} = 2 \times 10^{-7}$ s, $\tau_{E0} = 8 \times 10^{-8}$ s, $N_B = 2 \times 10^{16}$ cm^{-3}, $x_E = 0.35$ µm $= 3.5 \times 10^{-5}$ cm, $\delta = 0.9975$, $\beta = 150$, minimum $x_B = 0.80$ µm $= 8 \times 10^{-5}$ cm.

Step 1: Diffusion lengths

$$L_B = \sqrt{D_B \tau_{B0}} = \sqrt{(23)(2 \times 10^{-7})} = 2.145 \times 10^{-3}\ \text{cm} = 21.45\ \mu\text{m}$$

$$L_E = \sqrt{D_E \tau_{E0}} = \sqrt{(8)(8 \times 10^{-8})} = 8.0 \times 10^{-4}\ \text{cm} = 8.0\ \mu\text{m}$$

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Step 2: Required α

$$\alpha = \frac{\beta}{1+\beta} = \frac{150}{151} = 0.993377$$

Step 3: Choose the base width and find α_T

Choose the **minimum** fabricable width, $x_B = 0.80$ µm, to get the highest α_T and so the lowest N_E:

$$\alpha_T = \frac{1}{\cosh(x_B/L_B)} = \frac{1}{\cosh(0.80/21.45)} = \frac{1}{\cosh(0.03730)} = \frac{1}{1.000696} = 0.999305$$

$$\boxed{x_B = 0.80\ \mu\text{m}, \quad \alpha_T = 0.99930}$$

Step 4: Required injection efficiency

$$\gamma = \frac{\alpha}{\alpha_T \delta} = \frac{0.993377}{(0.999305)(0.9975)} = 0.996560$$

$$\boxed{\gamma = 0.99656}$$

Step 5: Solve the γ formula for N_E

From Eq. 12.35a with $p_{E0}/n_{B0} = N_B/N_E$:

$$\frac{1}{\gamma} - 1 = \frac{N_B}{N_E} \cdot \frac{D_E L_B}{D_B L_E} \cdot \frac{\tanh(x_B/L_B)}{\tanh(x_E/L_E)}$$

The pieces:

$$\frac{D_E L_B}{D_B L_E} = \frac{(8)(21.45)}{(23)(8.0)} = 0.9325, \qquad \frac{\tanh(0.03730)}{\tanh(0.04375)} = \frac{0.037283}{0.043722} = 0.8527$$

so the product is $0.9325 \times 0.8527 = 0.7952$, and $1/\gamma - 1 = 1/0.996560 - 1 = 0.0034519$. Then

$$N_E = \frac{N_B \times 0.7952}{0.0034519} = \frac{(2\times10^{16})(0.7952)}{0.0034519} = 4.61 \times 10^{18}\ \text{cm}^{-3}$$

$$\boxed{N_E(\text{min}) = 4.6 \times 10^{18}\ \text{cm}^{-3}}$$

Check: $\alpha = (0.99656)(0.99930)(0.9975) = 0.993377$, $\beta = 150.0$.

Final Answer:

- (a) Neutral base width $x_B = 0.80$ µm (the minimum, which allows the lowest emitter doping); minimum emitter doping $N_E = 4.6 \times 10^{18}$ cm^{-3}.
- (b) $\alpha_T = 0.99930$ and $\gamma = 0.99656$.

3. Keyword Explanation

1. Base transport factor

↩ back

$\alpha_T = J_{nC}/J_{nE} = 1/\cosh(x_B/L_B)$: the share of electrons injected into the base that reach the collector.

2. Recombination factor

↩ back

δ: the share of the emitter current that is not lost to recombination in the B–E depletion region.

4. Notes and Remarks

- A wider base would need more emitter doping: at $x_B = 1.0$ µm, $\alpha_T = 0.99891$ and $N_E = 6.5\times10^{18}$ cm^{-3} (calc script). So $x_B = 0.80$ µm gives the minimum N_E.
- The short-diode form (Eq. 12.36) gives almost the same value, 4.61×10^{18} cm^{-3}, because $x_B \ll L_B$ and $x_E \ll L_E$.

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- Bandgap narrowing at $N_E \approx 5 \times 10^{18}$ cm^{-3} (about 50 meV in Si) would raise p_{E0} and call for a higher N_E in practice; the question's ideal model leaves it out.

5. Final Concise Answer

(a) Solving the diffusion equation in the emitter ($\delta p_E(x_E) = 0$) and the base gives $J_{pE} = [eD_E p_{E0}/(L_E \tanh(x_E/L_E))](e^{eV_{BE}/kT} - 1)$ and $J_{nE} \approx [eD_B n_{B0}/(L_B \tanh(x_B/L_B))]e^{eV_{BE}/kT}$; $\gamma = J_{nE}/(J_{nE} + J_{pE}) = 1/[1 + (p_{E0}D_E L_B/n_{B0}D_B L_E)\tanh(x_B/L_B)/\tanh(x_E/L_E)]$, or for short regions $1/(1 + N_B D_E x_B/N_E D_B x_E)$.

(b) (i) Early effect: the B–C depletion region widens with V_{CB}, the neutral base narrows, the electron gradient and I_C rise, output curves slope up and extrapolate to $-V_A$; $r_o = V_A/I_C$; extreme case punch-through. (ii) Current crowding: the lateral base current's IR drop makes V_{BE} largest at the emitter edges, so emitter current crowds there (hot spots, high injection); cure: narrow interdigitated emitter stripes, low base resistance, ballast resistors.

(c) Gummel–Poon: $J_n = eD_n\, d(pn)/dx/p$ integrated over the base gives $J_n = -eD_n n_i^2 e^{eV_{BE}/kT}/\int_0^{x_B} p\, dx$ (base Gummel number); $I_C = (I_S/q_b)(e^{V_{BE}/V_t} - e^{V_{BC}/V_t})$ with $q_b = (q_1/2)(1+\sqrt{1+4q_2})$, q_1 for the Early effect, q_2 for high injection (I_{KF}), plus ideal and non-ideal base diodes, r_B, r_E, r_C and capacitances.

(d) $L_B = 21.45$ µm, $L_E = 8.0$ µm, $\alpha = 0.993377$; with $x_B = 0.80$ µm, $\alpha_T = 0.99930$, $\gamma = 0.99656$, and $N_E = 4.6 \times 10^{18}$ cm^{-3}.

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6. (a) What are the advantages of High Electron Mobility Transistor (HEMT)? Marks: 4
   (b) With proper band diagram explain how High Electron Mobility Transistor (HEMT) operates. Marks: 7
   (c) Sketch the equivalent circuit for Ebers-Moll model as used in BJT. Marks: 5
   (d) Show that maximum cut off frequency of a MESFET is related to doping concentration. Marks: 4
   (e) Determine the channel thickness of a GaAs MESFET to achieve threshold voltage $V_T = 0.1$ V. Consider an n-channel GaAs MESFET at T = 300 K with a gold Schottky barrier contact. Assume the barrier height is $\varphi_{Bn} = 0.89$ V. The n-channel doping is $N_d = 2 \times 10^{15}$ cm^{-3}. Marks: 5

6.(a) What are the advantages of High Electron Mobility Transistor (HEMT)? Marks: 4

1. Theory

1. What is a HEMT? (Core Idea)

A **high electron mobility transistor** (HEMT, also MODFET, modulation-doped FET) is a field-effect transistor whose channel is a thin sheet of electrons at a heterojunction, for example between doped N-AlGaAs and undoped GaAs. The donors sit in the wide-gap AlGaAs; their electrons fall into the lower conduction band of the GaAs and collect in a narrow potential well at the interface: a **two-dimensional electron gas (2DEG)**. A Schottky gate on top controls how many electrons the 2DEG holds, as in a MESFET (operation in 6(b)).

2. Why separating electrons from donors matters

In an ordinary doped channel (MESFET) the electrons move among the ionised donors that supplied them and are scattered by their Coulomb fields (**ionised-impurity scattering**). In a HEMT the electrons travel in **undoped** GaAs, separated from their donors by the interface and by an undoped spacer layer. This idea is **modulation doping**. Ionised-impurity scattering almost disappears, so the mobility is much higher: about 8000–9000 cm^2/V s at 300 K, and above 10^5 cm^2/V s at 77 K (Neamen, Section 13.5).

2. Answer

Advantages of the HEMT (over MESFETs and Si MOSFETs):

1. **High electron mobility and velocity:** the 2DEG is in undoped material, away from the donors, so impurity scattering is very weak: about 8500 cm^2/V s at 300 K (against about 4000–5000 in a doped GaAs MESFET channel), much higher at low temperature.
2. **High speed and frequency:** high mobility, a short gate and a thin channel give a short transit time; f_T and f_{max} of 100 GHz to over 500 GHz (InP-based), used in mm-wave circuits.
3. **High transconductance:** the 2DEG lies very close to the gate (tens of nanometres), so a small gate voltage changes the charge a lot: typically 250 mS/mm or more, larger than JFETs and MESFETs (Neamen).
4. **Low noise:** low scattering, low gate resistance and high g_m give the lowest noise figure of any transistor: used in satellite receivers, radio-telescope front ends and mobile-phone LNAs.
5. **High sheet density at low doping in the channel:** about 10^{12} cm^{-2} per interface, and more with multiple heterojunctions, for more current and power.
6. **Low-voltage, low-power operation**, and enhancement- or depletion-mode devices by choosing the AlGaAs thickness.

Limitations (for balance): complex epitaxy (MBE or MOCVD), parallel conduction in the AlGaAs at high gate bias, DX centres in AlGaAs (fixed by pseudomorphic InGaAs channels).

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3. Keyword Explanation

1. Two-dimensional electron gas

↩ back

A layer of electrons free to move in a plane but confined to a few nanometres in the third direction, at the heterojunction.

2. Modulation doping

↩ back

Doping the wide-gap layer only, so that the carriers it supplies move in the undoped narrow-gap layer, away from their donors.

4. Notes and Remarks

- Pseudomorphic HEMTs (pHEMTs, with a thin strained InGaAs channel) and InP-based HEMTs give still higher mobility and ΔE_c; GaN HEMTs (AlGaN/GaN) give high power from the polarisation-induced 2DEG.

6.(b) With proper band diagram explain how High Electron Mobility Transistor (HEMT) operates. Marks: 7

1. Theory

1. The heterojunction and its notch (Core Idea)

When N-type AlGaAs (wide gap) meets undoped GaAs (narrow gap), the conduction band steps down by ΔE_c (about 0.22 eV for $Al_{0.3}Ga_{0.7}As$, Neamen) going into the GaAs. Electrons from the AlGaAs donors move to the lower-energy GaAs side and pile up against the interface. Their charge and the positive donor ions left in the AlGaAs bend the bands: the GaAs conduction band forms a narrow **triangular potential well** (the notch) just next to the interface (see 2(a), nN junction).

2. Why the electrons are "two-dimensional"

The well is only about 10 nm wide, comparable with the electron wavelength. Motion across it is quantised into subband levels $E_0, E_1, \ldots$; the electrons occupy the lowest one and move freely only in the plane of the interface: a **2DEG**. Its density per unit area is n_s (cm^{-2}).

3. Gate control

The gate is a Schottky contact on the AlGaAs. Its barrier $e\phi_B$ and the gate voltage set the band bending in the AlGaAs, and so how far the GaAs well dips below E_F, and with it n_s (Neamen, Eqs. 13.68–13.71):

$$n_s = \frac{\epsilon_N}{e(d+\Delta d)}(V_g - V_{off}), \qquad V_{off} = \phi_B - \frac{\Delta E_c}{e} - V_{p2}, \qquad V_{p2} = \frac{eN_d d_d^2}{2\epsilon_N}$$

where ϵ_N is the AlGaAs permittivity, $d = d_d + d_i$ the doped plus undoped (spacer) AlGaAs thickness, $\Delta d \approx 80$ Å a correction for the width of the 2DEG, and V_{off} the threshold voltage. This is like a capacitor of plate distance $d + \Delta d$ between the gate and the 2DEG.

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2. Answer

Structure (from the top): metal Schottky gate between n^+ GaAs source and drain contacts; heavily doped N-AlGaAs (30–50 nm); thin undoped AlGaAs spacer (2–5 nm); undoped GaAs channel and buffer; semi-insulating GaAs substrate.

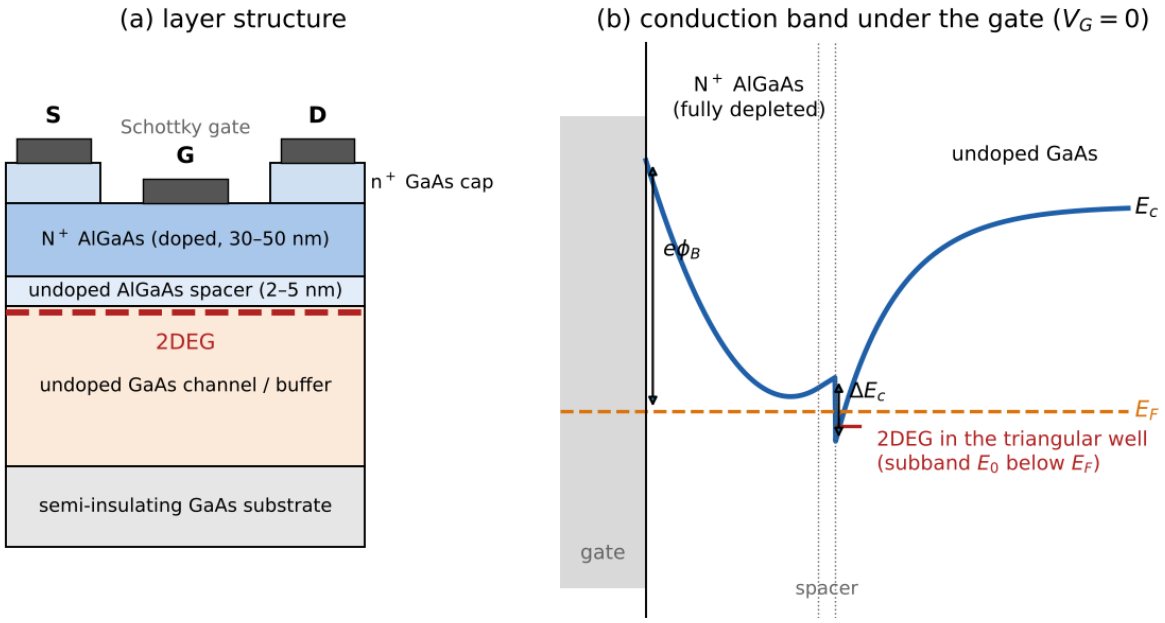


Figure 23: (a) HEMT layer structure with the 2DEG at the AlGaAs/GaAs interface; (b) the conduction band under the gate at zero gate voltage

Reading the figure. Panel (a): the layers; the red dashed line marks the 2DEG just below the spacer, in the undoped GaAs; source and drain sit on the n^+ cap, the gate on the AlGaAs in a recess. Panel (b): the conduction band (solid blue) from the gate (grey, left) into the GaAs; the dashed orange line is E_F. In the doped AlGaAs the band is a parabola (the donor charge), rising from its minimum through the spacer; at the interface it drops by ΔE_c, and in the GaAs it rises again away from the interface, forming a triangular well that dips below E_F (shaded red): the 2DEG, with its lowest subband E_0 below E_F.

Operation:

1. **Channel formation (modulation doping):** electrons from the AlGaAs donors transfer across the heterojunction into the GaAs, where E_c is lower by ΔE_c, and are held in the triangular well: a 2DEG of $n_s \approx 10^{12}$ cm^{-2}, spatially separated from its donors (the spacer adds distance), so its mobility is very high.
2. **Conduction:** a drain–source voltage drives the 2DEG electrons along the interface from source to drain. For small V_{DS} (constant mobility) $I_D = \dfrac{\epsilon_N \mu W}{2L(d+\Delta d)}[2(V_g - V_{off})V_{DS} - V_{DS}^2]$; at large V_{DS} the electrons reach the saturation velocity, $I_D(\text{sat}) = \dfrac{\epsilon_N W}{d+\Delta d}(V_g - V_{off} - V_0)v_{sat}$ (Eqs. 13.74–13.75).
3. **Gate control:** a **negative** gate voltage raises the bands under the gate, lifts the well towards E_F and reduces n_s; at $V_g = V_{off}$ the well rises above E_F and the channel is cut off. A **positive** gate voltage deepens the well and increases n_s, until the AlGaAs conduction band reaches E_F and a parallel (low-mobility) channel forms in the AlGaAs; then the gate loses control.
4. **Modes:** a thick doped AlGaAs layer gives $V_{off} < 0$ (**depletion mode**, normally on); a thin one gives $V_{off} > 0$ (**enhancement mode**, normally off).

Numerical illustration (Neamen, Example 13.10 data): $N_d = 10^{18}$ cm^{-3}, $d_d = 500$ Å, spacer 20 Å, $\phi_B = 0.85$ V, $\Delta E_c/e = 0.22$ V, $\epsilon_r = 12.2$: $V_{p2} = 1.85$ V, $V_{off} = -1.22$ V, and at $V_g = 0$, $n_s = 1.37 \times 10^{12}$ cm^{-2} (a depletion-mode device).

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3. Keyword Explanation

1. Spacer layer

↩ back

The undoped AlGaAs between the doped AlGaAs and the GaAs; it moves the ionised donors further from the 2DEG and raises the mobility, at the cost of a little sheet density.

2. Off voltage

↩ back

V_{off}: the gate voltage at which the 2DEG disappears (the threshold voltage of the HEMT).

4. Notes and Remarks

- The charge-control equations are Neamen's Eqs. 13.68–13.75; the illustration's numbers are from the calc script (the same values are worked in 5(d) of the 2017–18 paper, extra reading).
- The 2DEG density is limited to about 10^{12} cm^{-2} per interface by ΔE_c; multi-channel HEMTs stack several interfaces.

6.(c) Sketch the equivalent circuit for Ebers-Moll model as used in BJT. Marks: 5

1. Theory

1. The transistor as two coupled diodes (Core Idea)

A bipolar transistor has two pn junctions, B–E and B–C, back to back. Each behaves like a diode. But the junctions are so close (thin base) that carriers injected by one are collected by the other. The **Ebers–Moll model** (Neamen, Section 12.5.1) describes any bias condition (active, saturation, cut-off, inverse) with:

- two diodes, with currents $I_F = I_{ES}[\exp(V_{BE}/V_t) - 1]$ (B–E) and $I_R = I_{CS}[\exp(V_{BC}/V_t) - 1]$ (B–C);
- two current sources that model the coupling: a fraction α_F of the forward diode current reaches the collector, and a fraction α_R of the reverse diode current reaches the emitter.

2. Terminal currents

With all currents taken into the device (Neamen, Eqs. 12.64–12.66):

$$I_C = \alpha_F I_F - I_R, \qquad I_E = -I_F + \alpha_R I_R \text{ (into the emitter)}, \qquad I_E + I_B + I_C = 0$$

and the **reciprocity relation** $\alpha_F I_{ES} = \alpha_R I_{CS}$ ties the parameters together.

2. Answer

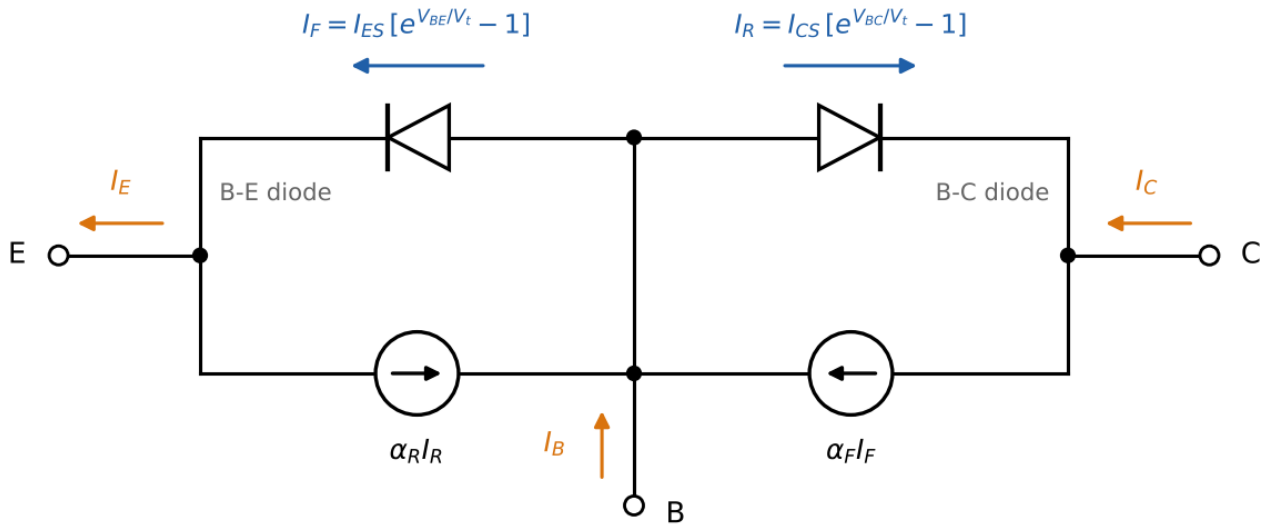


Figure 24: Ebers-Moll equivalent circuit of an npn transistor (reused from the 2020–21 paper)

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Reading the figure. Between E and the internal base node sits the B–E diode carrying $I_F = I_{ES}[e^{V_{BE}/V_t} - 1]$; between the base node and C sits the B–C diode carrying $I_R = I_{CS}[e^{V_{BC}/V_t} - 1]$ (both drawn pointing from the p base to the n regions). In parallel with each diode is a dependent current source: $\alpha_F I_F$ across the B–C diode (current that the emitter injects and the collector collects) and $\alpha_R I_R$ across the B–E diode (the same in the inverse direction). The base terminal B joins the node between the diodes.

Elements and equations (npn):

- B–E diode: $I_F = I_{ES}\left[\exp\left(\frac{V_{BE}}{V_t}\right) - 1\right]$; B–C diode: $I_R = I_{CS}\left[\exp\left(\frac{V_{BC}}{V_t}\right) - 1\right]$.
- Current sources: $\alpha_F I_F$ (collector side), $\alpha_R I_R$ (emitter side).
- Terminal currents: $I_C = \alpha_F I_F - I_R$, $I_E = I_F - \alpha_R I_R$ (out of the emitter), $I_B = I_E - I_C = (1-\alpha_F)I_F + (1-\alpha_R)I_R$.
- Reciprocity: $\alpha_F I_{ES} = \alpha_R I_{CS} = I_S$.

Typical values: $\alpha_F \approx 0.99$ (thin base, efficient emitter), $\alpha_R \approx 0.1$–0.5 (the collector is a poor emitter).
Use: forward active ($V_{BE} > 0$, $V_{BC} < 0$): $I_R \approx -I_{CS}$, tiny, so $I_C \approx \alpha_F I_E$. Saturation (both forward): both diodes conduct and $V_{CE}(\text{sat}) = V_t \ln\{[I_C(1-\alpha_R) + I_B]\alpha_F / ([\alpha_F I_B - (1-\alpha_F)I_C]\alpha_R)\}$ (Eq. 12.70). Cut-off (both reverse): only leakage currents.

3. Keyword Explanation

1. Forward and reverse common-base gains

↩ back

α_F and α_R: the fractions of the current injected by the emitter (or by the collector, in inverse mode) that reach the other junction.

2. Reciprocity relation

↩ back

$\alpha_F I_{ES} = \alpha_R I_{CS}$: a consequence of the symmetry of the transport across the base.

4. Notes and Remarks

- The same circuit, with the derivation of $V_{CE}(\text{sat})$, is in 6(a) of the 2020–21 paper (extra reading).
- The Gummel–Poon model of 5(c) extends this circuit.

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6.(d) Show that maximum cut off frequency of a MESFET is related to doping concentration. Marks: 4

1. Theory

1. The MESFET (Core Idea)

A **MESFET** (metal–semiconductor FET) is an n-type channel of thickness a and doping N_d on a semi-insulating substrate, with a Schottky gate of length L and width W between ohmic source and drain contacts. The gate's depletion layer (thickness h) narrows the channel and so controls the current.

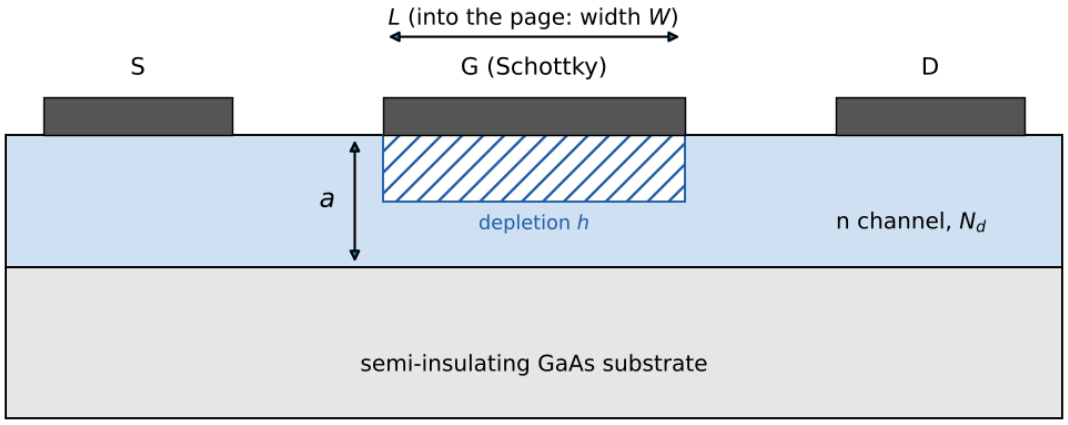


Figure 25: MESFET cross-section: channel thickness a, gate length L, depletion depth h

Reading the figure. The blue layer is the n channel of thickness a on the grey semi-insulating substrate. Under the Schottky gate (length L; width W into the page) the hatched depletion region of depth h reduces the conducting thickness to $a - h$.

2. Cut-off frequency

At high frequency the gate capacitance C_G takes the input current $I_i = j\omega C_G V_{gs}$. The **cut-off frequency** f_T is the frequency at which this input current equals the output current $g_m V_{gs}$ of the intrinsic transistor (Neamen, Section 13.4.2):

$$2\pi f_T C_G V_{gs} = g_m V_{gs} \quad \Rightarrow \quad f_T = \frac{g_m}{2\pi C_G}$$

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2. Answer

Step 1: Definition of f_T

As above (Neamen, Eq. 13.64):

$$f_T = \frac{g_m}{2\pi C_G}$$

Step 2: Maximum transconductance

The largest possible transconductance is that of the fully open channel, the channel conductance (Eq. 13.65):

$$g_m(\max) = G_{01} = \frac{e\mu_n N_d W a}{L}$$

(a channel of thickness a, width W, length L, conductivity $e\mu_n N_d$).

Step 3: Minimum gate capacitance

The gate capacitance is smallest when the depletion region is widest, $h = a$ (Eq. 13.66):

$$C_G(\min) = \frac{\epsilon_s W L}{a}$$

Step 4: Combine

$$f_T(\max) = \frac{g_m(\max)}{2\pi C_G(\min)} = \frac{e\mu_n N_d W a/L}{2\pi\epsilon_s W L/a}$$

$$\boxed{f_T(\max) = \frac{e\mu_n N_d a^2}{2\pi\epsilon_s L^2}} \qquad (13.67)$$

Step 5: Relation to doping

$f_T(\max)$ is **directly proportional to the channel doping** N_d (and to μ_n and a^2, and inversely to L^2). Using the internal pinch-off voltage $V_{p0} = ea^2 N_d/2\epsilon_s$ it can also be written

$$f_T(\max) = \frac{\mu_n V_{p0}}{\pi L^2}$$

so a higher doping gives a higher f_T, provided V_{p0} (hence $a^2 N_d$) and the breakdown voltage stay acceptable; in practice N_d is raised while a is reduced, and the mobility falls slowly with doping.

Example

GaAs MESFET, $\mu_n = 7000$ cm^2/V s, $N_d = 10^{17}$ cm^{-3}, $a = 0.15$ µm, $L = 1$ µm: $f_T(\max) = 346$ GHz (an upper bound; velocity saturation and parasitics give tens of GHz in practice). Neamen's Si example ($\mu_n = 1000$, $N_d = 10^{16}$, $a = 0.6$ µm, $L = 5$ µm) gives 3.54 GHz.

Steps to remember

$f_T = g_m/2\pi C_G \to g_m(\max) = e\mu_n N_d W a/L \to C_G(\min) = \epsilon_s W L/a \to f_T = e\mu_n N_d a^2/(2\pi\epsilon_s L^2) \propto N_d$.

3. Keyword Explanation

1. Cut-off frequency

↩ back

f_T: the frequency at which the short-circuit current gain of the transistor falls to one.

2. Internal pinch-off voltage

↩ back

$V_{p0} = ea^2 N_d/2\epsilon_s$: the potential needed across the gate junction to deplete the whole channel thickness.

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4. Notes and Remarks

- The channel transit time L/v_{sat} (10 ps for 1 µm) is the other limit; for short gates $f_T \approx v_{sat}/(2\pi L)$, independent of doping.
- GaAs MESFETs beat Si ones because μ_n and v_{sat} are higher (Neamen, Section 13.4).

6.(e) Determine the channel thickness of a GaAs MESFET to achieve threshold voltage $V_T = 0.1$ V. Consider an n-channel GaAs MESFET at T = 300 K with a gold Schottky barrier contact. Assume the barrier height is $\varphi_{Bn} = 0.89$ V. The n-channel doping is $N_d = 2 \times 10^{15}$ cm^{-3}. Marks: 5

1. Theory

1. Threshold voltage of an n-channel MESFET (Core Idea)

The gate's Schottky junction has a built-in potential $V_{bi} = \phi_{Bn} - \phi_n$, with $\phi_n = V_t \ln(N_c/N_d)$. A depletion depth h needs the junction potential $V_{bi} - V_{GS} = eN_d h^2/2\epsilon_s$. The channel is fully pinched off when $h = a$, that is when $V_{bi} - V_{GS} = V_{p0} = ea^2 N_d/2\epsilon_s$. So the **threshold voltage** is (Neamen, Eq. 13.42)

$$V_T = V_{bi} - V_{p0}$$

If $V_T > 0$ the device is **enhancement mode**: the built-in potential alone already pinches off the thin channel, and a positive gate voltage opens it.

2. Answer

Given: n-channel GaAs, $T = 300$ K, $\phi_{Bn} = 0.89$ V (gold), $N_d = 2 \times 10^{15}$ cm^{-3}, $V_T = 0.1$ V, $N_c = 4.7 \times 10^{17}$ cm^{-3}, $\epsilon_s = 13.1\epsilon_0 = 1.159 \times 10^{-12}$ F/cm (Neamen, GaAs).

Step 1: Fermi level below E_c

$$\phi_n = V_t \ln \frac{N_c}{N_d} = 0.0259 \ln\left(\frac{4.7 \times 10^{17}}{2 \times 10^{15}}\right) = 0.0259 \times 5.460 = 0.1414 \text{ V}$$

Step 2: Built-in potential

$$V_{bi} = \phi_{Bn} - \phi_n = 0.89 - 0.1414 = 0.7486 \text{ V}$$

Step 3: Required internal pinch-off voltage

$$V_{p0} = V_{bi} - V_T = 0.7486 - 0.1 = 0.6486 \text{ V}$$

Step 4: Channel thickness

$$V_{p0} = \frac{ea^2 N_d}{2\epsilon_s} \quad \Rightarrow \quad a = \left[\frac{2\epsilon_s V_{p0}}{eN_d}\right]^{1/2} = \left[\frac{2(1.159 \times 10^{-12})(0.6486)}{(1.6 \times 10^{-19})(2 \times 10^{15})}\right]^{1/2}$$

$$a = 6.86 \times 10^{-5} \text{ cm}$$

$$\boxed{a = 0.686\ \mu\text{m}}$$

Final Answer: a channel thickness of $a = 0.686$ µm gives $V_T = +0.1$ V (an enhancement-mode MESFET). A thinner channel would give a larger V_T (for $V_T = 0.25$ V, $a = 0.601$ µm, Neamen's Example 13.5).

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3. Keyword Explanation

1. Enhancement-mode MESFET

↩ back

A MESFET whose channel is pinched off at $V_{GS} = 0$ ($V_T > 0$); a positive gate voltage (less than the turn-on of the Schottky diode, about 0.7 V) opens it.

2. Built-in potential of the gate

↩ back

$V_{bi} = \phi_{Bn} - \phi_n$: the band bending under the gate at zero bias.

4. Notes and Remarks

- The precise control of a and N_d needed for small positive V_T makes enhancement-mode MESFETs hard to make (Neamen).
- Check: with $a = 0.686$ µm, $V_{p0} = (1.6\times10^{-19})(6.86\times10^{-5})^2(2\times10^{15})/(2\times1.159\times10^{-12}) = 0.649$ V and $V_T = 0.749 - 0.649 = 0.10$ V.

5. Final Concise Answer

(a) HEMT advantages: very high mobility (2DEG in undoped GaAs, away from the donors; about 8500 cm^2/V s at 300 K, far more at 77 K), high speed (f_T above 100 GHz), high transconductance (the 2DEG close to the gate, about 250 mS/mm), lowest noise, high sheet density (10^{12} cm^{-2}), low-voltage operation, depletion or enhancement mode by design.

(b) N-AlGaAs / spacer / undoped GaAs: electrons from the AlGaAs donors drop by ΔE_c into a triangular well in the GaAs at the interface, forming a 2DEG separated from its donors (high mobility). The Schottky gate controls $n_s = \epsilon_N(V_g - V_{off})/[e(d + \Delta d)]$ with $V_{off} = \phi_B - \Delta E_c/e - eN_d d_d^2/2\epsilon_N$: negative V_g empties the well (cut-off at V_{off}), positive V_g fills it until the AlGaAs conducts; I_D follows JFET-like equations, saturating at v_{sat}.

(c) Two back-to-back diodes $I_F = I_{ES}(e^{V_{BE}/V_t} - 1)$ and $I_R = I_{CS}(e^{V_{BC}/V_t} - 1)$ with current sources $\alpha_F I_F$ (at the collector) and $\alpha_R I_R$ (at the emitter); $I_C = \alpha_F I_F - I_R$, $I_E = I_F - \alpha_R I_R$, $I_B = I_E - I_C$; $\alpha_F I_{ES} = \alpha_R I_{CS}$.

(d) $f_T = g_m/(2\pi C_G)$ with $g_m(\text{max}) = e\mu_n N_d W a/L$ and $C_G(\text{min}) = \epsilon_s WL/a$ gives $f_T(\text{max}) = e\mu_n N_d a^2/(2\pi\epsilon_s L^2) = \mu_n V_{p0}/(\pi L^2)$: proportional to the channel doping N_d.

(e) $\phi_n = 0.1414$ V, $V_{bi} = 0.7486$ V, $V_{p0} = 0.6486$ V, $a = (2\epsilon_s V_{p0}/eN_d)^{1/2} = 0.686$ µm.

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7. (a) What is ohmic contact? Briefly explain the types of ohmic contacts. Marks: 4

(b) Determine the theoretical barrier height, built-in potential barrier, and maximum electric field in a metal–semiconductor diode for zero applied bias. Consider a contact between gold (5.1 V) and n-type germanium (4.13 V) doped to $N_d = 10^{16}$ cm^{-3} at T = 300 K. $N_C = 1.04 \times 10^{19}$ and $\varepsilon_r = 16$. Marks: 6

7.(a) What is ohmic contact? Briefly explain the types of ohmic contacts. Marks: 4

1. Theory

1. What a contact must do (Core Idea)

Every semiconductor device is joined to metal wires. The joint should let current pass in *both* directions with a **small, linear** voltage drop, so that it does not change the device's behaviour. Such a joint is an **ohmic contact**. Its opposite is a **rectifying** (Schottky) contact, which passes current easily in one direction only.

2. Where a barrier comes from

The **work function** ϕ is the energy (in volts) needed to take an electron from the Fermi level to the vacuum level. When a metal (ϕ_m) touches a semiconductor (ϕ_s), electrons flow from the side with the higher Fermi level (smaller work function) to the other until the Fermi levels line up.

- **n-type, $\phi_m > \phi_s$:** electrons leave the semiconductor; positive donors are left behind; the bands bend up into a **barrier** $\phi_{B0} = \phi_m - \chi$: rectifying.
- **n-type, $\phi_m < \phi_s$:** electrons enter the semiconductor and pile up at the surface (**accumulation**); the bands bend down; **no barrier** for electrons.
- **p-type** is the mirror image: $\phi_m > \phi_s$ gives hole accumulation and no barrier.

3. Tunnelling through a thin barrier

The width of a barrier is the depletion width $x_n = (2\epsilon_s V_{bi}/eN_d)^{1/2}$. Heavy doping makes it very thin; at $N_d \gtrsim 10^{19}$ cm^{-3} it is a few nanometres, and electrons pass **through** it by quantum tunnelling, whatever its height.

2. Answer

Ohmic contact: a metal–semiconductor contact with a linear, symmetric current–voltage characteristic and a negligible resistance compared with the bulk of the device; its quality is measured by the specific contact resistance $R_c = (\partial J/\partial V)^{-1}_{V=0}$ (Ω cm^2), which should be as small as possible (Neamen, Section 9.2).

Types:

1. **Ideal non-rectifying (barrier-free) contact:** $\phi_m < \phi_s$ on n-type, or $\phi_m > \phi_s$ on p-type. Majority carriers accumulate at the surface, the bands bend towards them, and there is no barrier: carriers flow freely in both directions. (Surface states can spoil this by pinning the Fermi level.)
2. **Tunnelling (heavily doped) contact:** the semiconductor surface is doped very heavily (n^+ or p^+, $\gtrsim 10^{19}$ cm^{-3}) by diffusion, implantation, alloying or epitaxy. The barrier still exists but is only a few nanometres thick, so carriers tunnel through it; $R_c \propto \exp\left[2\sqrt{\epsilon_s m^*}\,\phi_{Bn}/(\hbar\sqrt{N_d})\right]$ falls steeply with doping. This is the practical type (Au–Ge–Ni alloyed on n-GaAs, silicides on n^+ Si, Al on p^+ Si).

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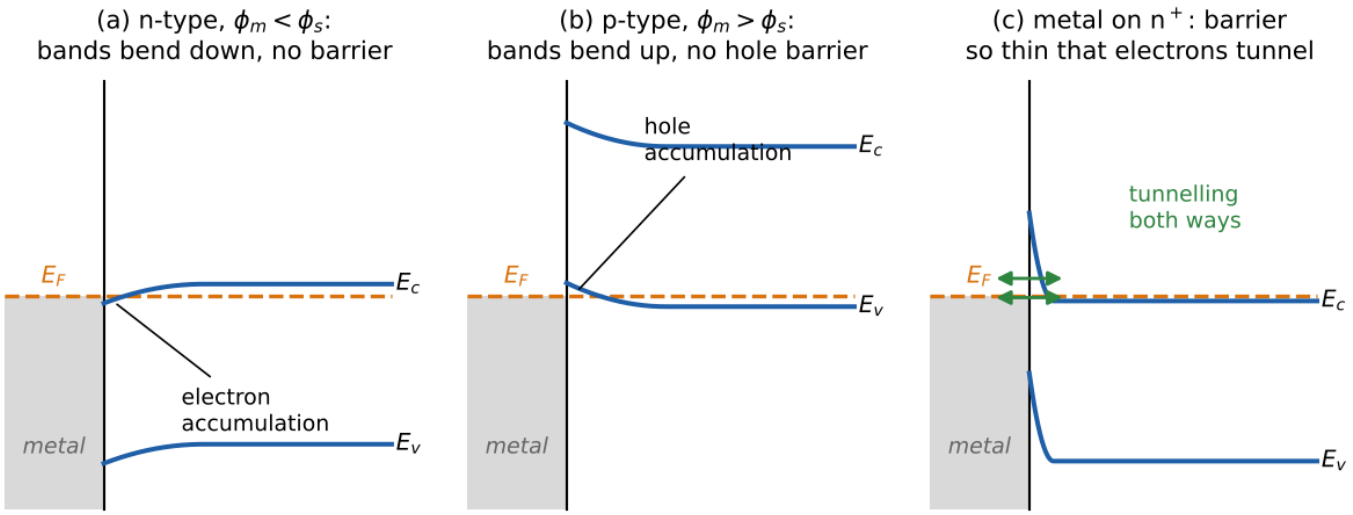


Figure 26: Ohmic contacts: (a) n-type with $\phi_m < \phi_s$, (b) p-type with $\phi_m > \phi_s$, (c) metal on n^+ with a thin tunnelling barrier

Reading the figure. The grey block is the metal, the dashed orange line the common Fermi level, the solid blue lines E_c and E_v. (a) The n-type bands bend down at the surface: electron accumulation, no barrier. (b) The p-type bands bend up: hole accumulation, no barrier for holes. (c) On heavily doped n^+ the barrier is very thin, and electrons tunnel through it both ways (green arrows).

3. Keyword Explanation

1. Specific contact resistance

↩ back

R_c: the contact resistance times the contact area (Ω cm^2).

2. Accumulation layer

↩ back

A surface layer with more majority carriers than the bulk, formed when the bands bend towards the majority-carrier band.

4. Notes and Remarks

- The same question is 3(c) of the final; the specific contact resistance is worked out in 3(a).

7.(b) Determine the theoretical barrier height, built-in potential barrier, and maximum electric field in a metal–semiconductor diode for zero applied bias. Consider a contact between gold (5.1 V) and n-type germanium (4.13 V) doped to $N_d = 10^{16}$ cm^{-3} at T = 300 K. $N_C = 1.04 \times 10^{19}$ and $\varepsilon_r = 16$. Marks: 6

1. Theory

1. The ideal Schottky barrier (Core Idea)

When a metal of work function ϕ_m contacts an n-type semiconductor of electron affinity χ with $\phi_m > \phi_s$, electrons flow from the semiconductor into the metal until the Fermi levels are equal. The positive donors left behind form a depletion layer, and the bands bend up. In the ideal model (no interface states):

- the barrier seen from the metal is $\phi_{B0} = \phi_m - \chi$;
- the Fermi level in the n-type bulk is $\phi_n = (kT/e)\ln(N_c/N_d)$ below E_c (from $n_0 = N_c e^{-(E_c - E_F)/kT}$ with $n_0 = N_d$);
- the band bending, or built-in potential, is $V_{bi} = \phi_{B0} - \phi_n$.

2. Field from Poisson's equation

In the depletion layer ($0 < x < x_n$) the charge is $+eN_d$, so $dE/dx = eN_d/\epsilon_s$. With $E(x_n) = 0$: $E(x) = -(eN_d/\epsilon_s)(x_n - x)$. The potential rises by $V_{bi} = eN_d x_n^2/2\epsilon_s$ across the layer, so

$$x_n = \left[\frac{2\epsilon_s V_{bi}}{eN_d}\right]^{1/2}, \qquad |E_{max}| = \frac{eN_d x_n}{\epsilon_s}$$

at the metal surface (Neamen, Eqs. 9.5–9.7).

2. Answer

Given: $\phi_m = 5.1$ V (gold), $\chi = 4.13$ V (germanium), $N_d = 10^{16}$ cm^{-3}, $N_c = 1.04 \times 10^{19}$ cm^{-3}, $\epsilon_r = 16$ ($\epsilon_s = 1.416 \times 10^{-12}$ F/cm), $T = 300$ K, $kT/e = 0.0259$ V.

Step 1: Barrier height

$$\phi_{B0} = \phi_m - \chi = 5.1 - 4.13 = 0.97 \text{ V}$$

Step 2: Fermi level below E_c

$$\phi_n = 0.0259 \ln\left(\frac{1.04 \times 10^{19}}{10^{16}}\right) = 0.0259 \times 6.947 = 0.180 \text{ V}$$

Step 3: Built-in potential

$$V_{bi} = 0.97 - 0.180 = 0.790 \text{ V}$$

Step 4: Depletion width

$$x_n = \left[\frac{2(1.416 \times 10^{-12})(0.790)}{(1.6 \times 10^{-19})(10^{16})}\right]^{1/2} = 3.74 \times 10^{-5} \text{ cm} = 0.374\ \mu\text{m}$$

Step 5: Maximum field

$$|E_{max}| = \frac{(1.6 \times 10^{-19})(10^{16})(3.74 \times 10^{-5})}{1.416 \times 10^{-12}} = 4.23 \times 10^4 \text{ V/cm}$$

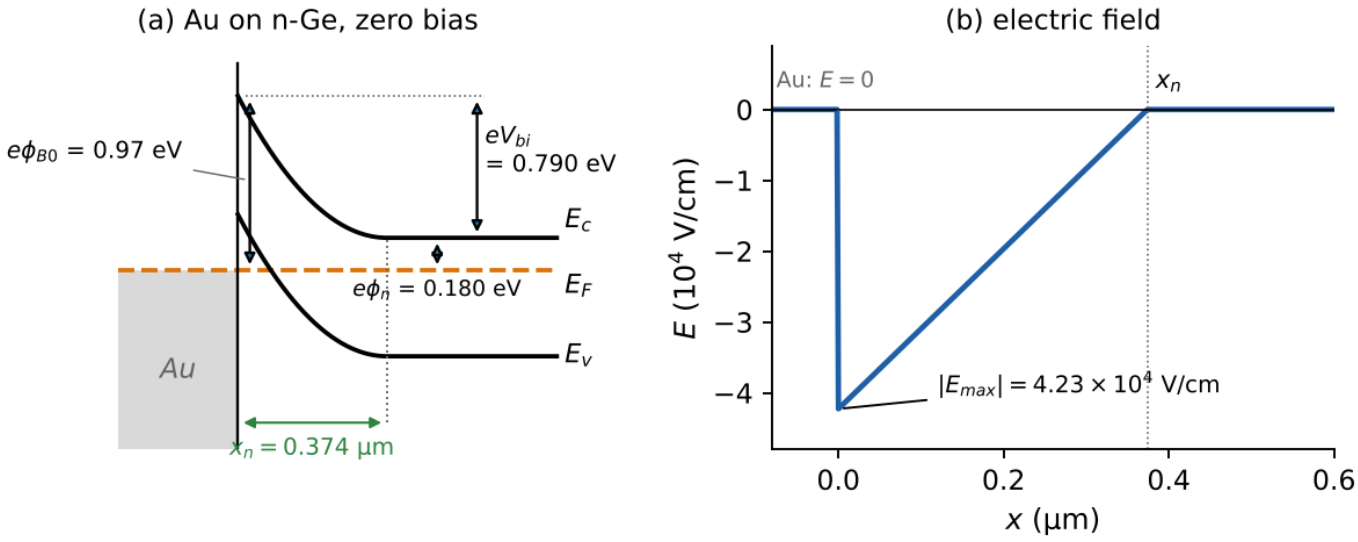


Figure 27: Au on n-Ge at zero bias: band diagram and electric field (same data as 1(c))

Reading the figure. Panel (a): the Ge bands (solid black) bend up towards the gold over $x_n = 0.374$ µm; the barrier from the gold is 0.97 eV and from the Ge 0.790 eV; E_F (dashed orange) is flat. Panel (b): the field (solid blue) is -4.23×10^4 V/cm at the surface and rises linearly to zero at x_n.

Final Answer: $\phi_{B0} = 0.97$ V, $V_{bi} = 0.790$ V, $|E_{max}| = 4.23 \times 10^4$ V/cm.

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3. Keyword Explanation

1. Electron affinity

↩ back

χ: the energy from the conduction-band edge to the vacuum level, a material constant (Ge: 4.13 eV).

2. Built-in potential barrier

↩ back

V_{bi}: the barrier that electrons in the semiconductor must climb to reach the metal at zero bias.

4. Notes and Remarks

- The same problem is 1(c) of the final.
- The test gives the work function and affinity in volts; they equal 5.1 eV and 4.13 eV as energies.

5. Final Concise Answer

(a) Ohmic contact: linear, symmetric I–V, negligible resistance (R_c small). Types: (1) barrier-free, $\phi_m < \phi_s$ on n-type or $\phi_m > \phi_s$ on p-type (accumulation, no barrier); (2) tunnelling, heavily doped surface ($\gtrsim 10^{19}$ cm^{-3}) makes the barrier thin enough for carriers to tunnel through.

(b) $\phi_{B0} = 5.1 - 4.13 = 0.97$ V; $\phi_n = 0.180$ V; $V_{bi} = 0.790$ V; $x_n = 0.374$ µm; $|E_{max}| = 4.23 \times 10^4$ V/cm.

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8. (a) Sketch the energy band diagram of the MOS capacitor under various bias conditions for both a p-type and s-type [*sic*: should be n-type] substrate. Marks: 5

(b) A p-type Si substrate at T = 300 K was doped at $N_a = 10^{14}\ cm^{-3}$ It has $Q'_{ss} = 10^{10}\ cm^{-2}$ and $t_{ox} = 30\ nm$ Determine the threshold voltage of an MOS system with a gold gate. Consider at T = 300 K, p-type Si electron affinity 4.01, $N_v = 1.04 \times 10^{19}$ cm^{-3}, $N_c = 2.8 \times 10^{19}$ cm^{-3}, $n_i = 1.5 \times 10^{10}$ cm^{-3} and $\varepsilon_r = 11.7$ and gold p-Si work function $\varphi_{ms} = 0.07$ V. Marks: 5

8.(a) Sketch the energy band diagram of the MOS capacitor under various bias conditions for both a p-type and s-type [*sic*: should be n-type] **substrate. Marks: 5**

1. Theory

1. The MOS capacitor (Core Idea)

A **MOS capacitor** is a metal gate, a thin oxide (an insulator with a very wide gap, about 9 eV for SiO_2) and a semiconductor substrate. No current flows through the oxide, so the semiconductor stays in equilibrium: its Fermi level E_F is **flat**. The gate voltage V_G shifts the metal's Fermi level by $-eV_G$ relative to E_F and bends the semiconductor's bands near the surface.

Take an **ideal** MOS: $\phi_{ms} = 0$ and no oxide charge, so the bands are flat at $V_G = 0$.

2. Reading the bands

- E_{Fi} (intrinsic level) is in the middle of the gap. Where E_F is below E_{Fi} the material is p-type; above it, n-type.
- **Bands bending up** at the surface (E_{Fi} moves away from E_F for p-type): more holes at the surface.
- **Bands bending down** (for p-type): fewer holes; if E_{Fi} crosses below E_F, electrons become the majority at the surface: **inversion**.

3. The three conditions

For **p-type** substrate ($e\phi_{fp} = E_{Fi} - E_F$ in the bulk):

- **Accumulation ($V_G < 0$):** negative gate charge attracts holes; bands bend *up*; a thin hole layer at the surface.
- **Depletion ($0 < V_G < V_T$):** positive gate repels holes; bands bend *down*; negative acceptor ions are left in a depletion layer.
- **Inversion ($V_G > V_T$):** bands bend down so far that E_{Fi} at the surface is below E_F (by $e\phi_{fp}$ at threshold): an electron inversion layer.

For **n-type** everything reverses: accumulation for $V_G > 0$ (bands down), depletion for small $V_G < 0$ (bands up), inversion (holes) for $V_G < V_T$.

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2. Answer

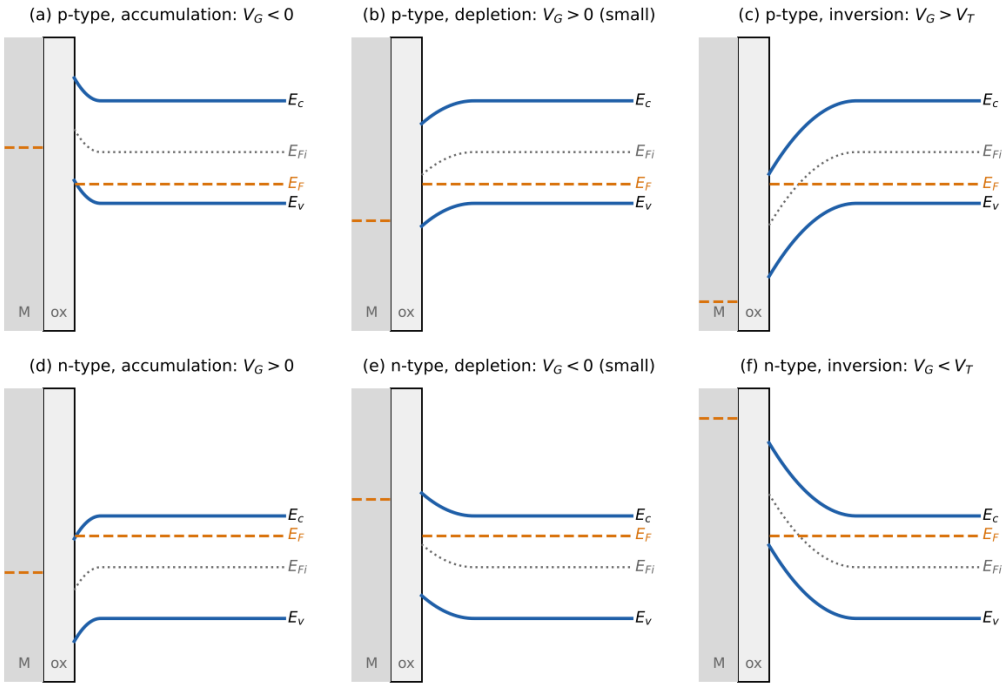


Figure 28: MOS energy-band diagrams (ideal, $\phi_{ms} = 0$): (a)-(c) p-type substrate in accumulation, depletion and inversion; (d)-(f) n-type substrate in accumulation, depletion and inversion

Reading the figure. In each panel the grey block is the metal (M) with its Fermi level (dashed orange), the light block is the oxide (ox), and on the right the semiconductor with E_c and E_v (solid blue), E_{Fi} (dotted grey) and the flat E_F (dashed orange). The metal Fermi level lies eV_G below the semiconductor's E_F for positive V_G and above it for negative V_G.

p-type substrate:

- **(a) Accumulation, $V_G < 0$:** metal E_F raised; bands bend up at the surface; E_v comes close to E_F: many holes.
- **(b) Depletion, $V_G > 0$ small:** metal E_F lowered; bands bend down; E_v moves away from E_F: holes depleted, ionised acceptors remain (width x_d).
- **(c) Inversion, $V_G > V_T$:** bands bend down by more than $2e\phi_{fp}$; at the surface E_{Fi} is below E_F and E_c is close to it: an electron inversion layer; the depletion width stays at x_{dT}.

n-type substrate (mirror image):

- **(d) Accumulation, $V_G > 0$:** bands bend down; E_c close to E_F: electron accumulation.
- **(e) Depletion, $V_G < 0$ small:** bands bend up; electrons depleted, ionised donors remain.
- **(f) Inversion, $V_G < V_T$ (negative):** bands bend up by more than $2e\phi_{fn}$; E_{Fi} at the surface rises above E_F: a hole inversion layer.

3. Keyword Explanation

1. Surface potential

↩ back

ϕ_s: the total band bending from the bulk to the surface, in volts.

2. Threshold inversion

↩ back

$\phi_s = 2\phi_{fp}$ (p-type): the surface is as strongly n-type as the bulk is p-type.

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4. Notes and Remarks

- **Note on the paper:** "s-type substrate" is read as **n-type**: a MOS capacitor has only p-type and n-type substrates, so "both a p-type and s-type" can only mean the two of them.
- With a real gate ($\phi_{ms} \neq 0$) and oxide charge, the same pictures hold with V_G measured from V_{FB} instead of 0 (see 4(d)).

8.(b) A p-type Si substrate at T = 300 K was doped at $N_a = 10^{14}$ cm^{-3} It has $Q'_{ss} = 10^{10}$ cm^{-2} and $t_{ox} = 30$ nm Determine the threshold voltage of an MOS system with a gold gate. Consider at T = 300 K, p-type Si electron affinity 4.01, $N_v = 1.04 \times 10^{19}$ cm^{-3}, $N_c = 2.8 \times 10^{19}$ cm^{-3}, $n_i = 1.5 \times 10^{10}$ cm^{-3} and $\varepsilon_r = 11.7$ and gold p-Si work function $\varphi_{ms} = 0.07$ V. Marks: 5

1. Theory

1. Threshold voltage (Core Idea)

The **threshold voltage** V_{TN} of a MOS on p-type silicon is the gate voltage that makes the surface just inverted, $\phi_s = 2\phi_{fp}$ (Neamen, Section 10.1.6). At that point the depletion layer has its maximum width and charge:

$$\phi_{fp} = V_t \ln \frac{N_a}{n_i}, \qquad x_{dT} = \left(\frac{4\epsilon_s \phi_{fp}}{eN_a}\right)^{1/2}, \qquad |Q'_{SD}(\max)| = eN_a x_{dT}$$

2. The equation

Charge neutrality and the voltage loop (derived step by step in 4(d)) give

$$V_{TN} = (|Q'_{SD}(\max)| - Q'_{ss})\frac{t_{ox}}{\epsilon_{ox}} + \phi_{ms} + 2\phi_{fp} \qquad (10.31b)$$

where Q'_{ss} is the fixed oxide charge per area (C/cm^2; given in cm^{-2} it is multiplied by e), t_{ox} the oxide thickness, $\epsilon_{ox} = 3.9\epsilon_0$ for SiO_2, and ϕ_{ms} the gate–semiconductor work-function difference.

2. Answer

Given: p-Si, $T = 300$ K, $N_a = 10^{14}$ cm^{-3}, $Q'_{ss} = 10^{10}$ cm^{-2}, $t_{ox} = 30$ nm $= 3\times10^{-6}$ cm, $n_i = 1.5\times10^{10}$ cm^{-3}, $\epsilon_r = 11.7$, $\phi_{ms} = 0.07$ V (gold gate, given); oxide SiO_2, $\epsilon_{ox} = 3.9 \times 8.85 \times 10^{-14} = 3.452 \times 10^{-13}$ F/cm.

Step 1: Fermi potential

$$\phi_{fp} = 0.0259 \ln\left(\frac{10^{14}}{1.5 \times 10^{10}}\right) = 0.0259 \times 8.805 = 0.2280 \text{ V}$$

Step 2: Maximum depletion width

$$x_{dT} = \left[\frac{4(11.7)(8.85 \times 10^{-14})(0.2280)}{(1.6 \times 10^{-19})(10^{14})}\right]^{1/2} = 2.43 \times 10^{-4} \text{ cm} = 2.43\ \mu\text{m}$$

Step 3: Maximum depletion charge

$$|Q'_{SD}(\max)| = eN_a x_{dT} = (1.6 \times 10^{-19})(10^{14})(2.43 \times 10^{-4}) = 3.888 \times 10^{-9} \text{ C/cm}^2$$

Step 4: Oxide charge and oxide capacitance

$$Q'_{ss} = (10^{10})(1.6 \times 10^{-19}) = 1.6 \times 10^{-9} \text{ C/cm}^2, \qquad C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.452 \times 10^{-13}}{3 \times 10^{-6}} = 1.151 \times 10^{-7} \text{ F/cm}^2$$

Step 5: Threshold voltage

$$V_{TN} = \frac{3.888 \times 10^{-9} - 1.6 \times 10^{-9}}{1.151 \times 10^{-7}} + 0.07 + 2(0.2280) = 0.0199 + 0.07 + 0.456$$

$$\boxed{V_{TN} = 0.546 \text{ V}}$$

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Final Answer: $V_{TN} \approx +0.55$ V: an enhancement-mode n-channel device (a positive gate voltage is needed to form the inversion layer). The flat-band voltage is $V_{FB} = 0.07 - 1.6\times10^{-9}/1.151\times10^{-7} = 0.056$ V.

3. Keyword Explanation

1. Fermi potential

↩ back

$\phi_{fp} = (E_{Fi} - E_F)/e$ in the p-type bulk.

2. Oxide capacitance

↩ back

$C_{ox} = \epsilon_{ox}/t_{ox}$: the capacitance per area of the gate oxide.

4. Notes and Remarks

- **About ϕ_{ms}:** the paper gives $\phi_{ms} = 0.07$ V and it is used as given. From vacuum work functions ($\phi_m = 5.1$ V for gold, $\chi = 4.01$ V) one would get $\phi_{ms} = 5.1 - (4.01 + 0.56 + 0.228) = 0.302$ V and $V_{TN} = 0.778$ V; MOS work-function differences are better taken from measured oxide barriers (Neamen, Figure 10.15), which is presumably where the given value comes from.
- The very light doping makes the depletion term small (0.034 V); the threshold is set mostly by $2\phi_{fp}$.
- The extra data in the paper (N_v, N_c, χ) are not needed once ϕ_{ms} is given.

5. Final Concise Answer

(a) Ideal MOS ($\phi_{ms} = 0$), E_F flat in the semiconductor, metal E_F shifted by $-eV_G$. p-type: $V_G < 0$ accumulation (bands bend up, holes at the surface); $0 < V_G < V_T$ depletion (bands bend down, acceptor ions); $V_G > V_T$ inversion (E_{Fi} below E_F at the surface, electrons). n-type ("s-type" in the paper): the mirror image, accumulation for $V_G > 0$, depletion for small $V_G < 0$, hole inversion for $V_G < V_T$.
(b) $\phi_{fp} = 0.228$ V, $x_{dT} = 2.43$ µm, $|Q'_{SD}(\text{max})| = 3.89 \times 10^{-9}$ C/cm^2, $Q'_{ss} = 1.6 \times 10^{-9}$ C/cm^2, $C_{ox} = 1.15 \times 10^{-7}$ F/cm^2; $V_{TN} = 0.0199 + 0.07 + 0.456 = 0.546$ V.

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FORMULA SHEET

EEE459

Compound Semiconductor and Hetero-Junction Devices

Formula Sheet

Session 2018–19 paper

Every formula behind the 4th Year 1st Semester Final Examination, 2022, and Term Tests 1 and 2

TOPICS

1 Crystals and growth — 6 formulas

2 Metal–semiconductor contacts — 11 formulas

3 Heterojunctions — 6 formulas

4 Bipolar transistors — 9 formulas

5 MOS capacitor — 7 formulas

6 HEMT and MESFET — 5 formulas

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

Before you use the sheet

How to read a row

★ **Atoms per cell**	corner $\frac{1}{8}$, face $\frac{1}{2}$, edge $\frac{1}{4}$, inside 1	bcc 2, zinc blende 8, hcp prism 6	1(a)
the name you search for	the formula, exactly as derived	every new symbol with its unit	link: opens the derivation

In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

Contents

1 Crystals and growth **4**

2 Metal–semiconductor contacts **4**

3 Heterojunctions **5**

4 Bipolar transistors **7**

5 MOS capacitor **8**

6 HEMT and MESFET **10**

7 Key formulas **11**

8 Symbol index **12**

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Conventions (Neamen): e is the magnitude of the electron charge; potentials (ϕ, V) in volts, energies in eV; $V_t = kT/e = 0.0259$ V at 300 K; $\epsilon_s = \epsilon_r\epsilon_0$, $\epsilon_0 = 8.85 \times 10^{-14}$ F/cm; $n_i = 1.5 \times 10^{10}$ cm^{-3} (Si). References point to the solutions of the 2018–19 paper (Q7 and Q8 are the term tests).

Columns: Title · Equation · Notation · Ref. (reference).

1. Crystals and growth:

Title	Equation	Notation	Ref.
★ **Atoms per cell**	corner $\frac{1}{8}$, face $\frac{1}{2}$, edge $\frac{1}{4}$, inside 1	bcc 2, zinc blende 8, hcp prism 6	1(a)
bcc packing	$4r = \sqrt{3}a$, fraction $\sqrt{3}\pi/8 = 0.68$	r = atom radius (Å) a = lattice constant (Å) touching along the body diagonal	1(a)
hcp ideal ratio	$c/a = \sqrt{8/3} = 1.633$	c = height of the hcp cell (Å) 12 neighbours, packing fraction 0.74	1(a)
Zinc blende bond	$d = \sqrt{3}a/4$	d = nearest-neighbour (bond) distance (Å) 4 tetrahedral neighbours	1(a)
Si CVD	$SiCl_4 + 2H_2 \rightarrow Si + 4HCl$	at about 1200 °C	1(d)
GaAs chloride VPE	$6GaCl + As_4 + 3H_2 \rightarrow 6GaAs + 6HCl$	Ga source 850 °C, deposition 750 °C	1(d)

Table 1: Crystals and growth

2. Metal–semiconductor contacts:

Title	Equation	Notation	Ref.
★ **Barrier, n-type**	$\phi_{B0} = \phi_m - \chi$	ϕ_{B0} = Schottky barrier height for electrons (V) ϕ_m = metal work function (V) χ = electron affinity of the semiconductor (V)	1(c), 3(d), 7(b)
★ **Barrier, p-type**	$\phi_{Bp} = \frac{E_g}{e} + \chi - \phi_m$	ϕ_{Bp} = Schottky barrier height for holes (V) E_g = band gap energy (eV) e = magnitude of the electron charge (C) Schottky if positive	3(d)
$E_c - E_F$ in n-type	$\phi_n = V_t \ln \frac{N_c}{N_d}$	$\phi_n = (E_c - E_F)/e$ in the n-type semiconductor (V) V_t = thermal voltage, kT/e (V) N_c = effective density of states in the conduction band (cm^{-3}) N_d = donor concentration (cm^{-3})	1(c), 7(b)
★ **Built-in potential**	$V_{bi} = \phi_{B0} - \phi_n$	V_{bi} = built-in potential, the band bending at zero bias (V)	1(c), 7(b)
★ **Depletion width**	$x_n = \left[\frac{2\epsilon_s V_{bi}}{eN_d}\right]^{1/2}$	x_n = depletion width at zero bias (cm) ϵ_s = permittivity of the semiconductor (F/cm)	1(c)

Title	Equation	Notation	Ref.
★ **Maximum field**	$\lvert E_{max}\rvert = \dfrac{eN_d x_n}{\epsilon_s}$	E_{max} = peak electric field, at the metal (V/cm)	1(c)
Interface states	$\phi_{B0} = \gamma(\phi_m - \chi) + (1-\gamma)\left(\dfrac{E_g}{e} - \phi_0\right)$, $\gamma = \dfrac{\epsilon_i}{\epsilon_i + e^2\delta D_{it}}$	γ = weight of the metal work function in the barrier ϕ_0 = neutral level of the interface states, above E_v (V) ϵ_i = permittivity of the interfacial layer (F/cm) δ = thickness of the interfacial layer (cm) D_{it} = density of interface states ($\mathrm{cm^{-2}eV^{-1}}$)	1(b)
★ **Specific contact resistance**	$R_c = \left(\dfrac{\partial J}{\partial V}\right)^{-1}_{V=0}$	R_c = specific contact resistance (Ω $\mathrm{cm^2}$) J = current density ($\mathrm{A/cm^2}$) V = applied voltage (V)	3(a)
Thermionic R_c	$R_c = \dfrac{kT/e}{A^*T^2}\exp\left(\dfrac{e\phi_{Bn}}{kT}\right)$	k = Boltzmann constant (J/K) T = temperature (K) A^* = effective Richardson constant ($\mathrm{A/cm^2K^2}$) ϕ_{Bn} = barrier height for electrons (V)	3(a)
Tunnelling R_c	$R_c \propto \exp\left(\dfrac{2\sqrt{\epsilon_s m_n^*}}{\hbar}\cdot\dfrac{\phi_{Bn}}{\sqrt{N_d}}\right)$	m_n^* = electron effective mass (kg) $\hbar$ = reduced Planck constant, $h/2\pi$ (J s) heavy doping, $N_d \gtrsim 10^{19}$ $\mathrm{cm^{-3}}$	3(a), 7(a)
RTD levels	$E_n \approx \dfrac{\hbar^2}{2m^*}\left(\dfrac{n\pi}{L_w}\right)^2$	E_n = energy of level n in the well (J) m^* = effective mass in the well (kg) n = level number (1, 2, ...) L_w = well width (m) one current peak per level	3(b)

Table 2: Metal–semiconductor contacts

3. Heterojunctions:

Title	Equation	Notation	Ref.
★ **Band offsets**	$\Delta E_c = e(\chi_n - \chi_P)$, $\Delta E_v = \Delta E_g - \Delta E_c$	ΔE_c = conduction-band offset (eV) e = magnitude of the electron charge (C) χ_n = electron affinity of the narrow-gap n side (V) χ_P = electron affinity of the wide-gap P side (V) ΔE_v = valence-band offset (eV) ΔE_g = difference of the two band gaps (eV) Anderson's rule	2(a), 2(d)

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Title	Equation	Notation	Ref.
★ **Built-in potential (nP)**	$eV_{bi} = \Delta E_v + kT \ln\left(\frac{p_{P0}}{p_{n0}} \cdot \frac{N_{vn}}{N_{vP}}\right)$	V_{bi} = built-in potential (V) k = Boltzmann constant (eV/K) T = temperature (K) p_{P0}, p_{n0} = equilibrium hole concentrations in the P and n sides (cm^{-3}) N_{vn}, N_{vP} = effective densities of states in the valence band of the n and P sides (cm^{-3}) $p_{n0} = n_i^2/N_d$ n_i = intrinsic carrier concentration of the n side (cm^{-3}) N_d = donor concentration of the n side (cm^{-3})	2(d)
Charge balance	$N_{dn}x_n = N_{aP}x_P$	N_{dn} = donor concentration in the n side (cm^{-3}) N_{aP} = acceptor concentration in the P side (cm^{-3}) x_n, x_P = depletion widths in the n and P sides (cm) from $\epsilon_n E_n(0) = \epsilon_P E_P(0)$ ϵ_n, ϵ_P = permittivities of the n and P sides (F/cm) $E_n(0)$, $E_P(0)$ = fields at the interface (V/cm)	2(b)
Potential drops	$V_{bin} = \frac{eN_{dn}x_n^2}{2\epsilon_n}$, $V_{biP} = \frac{eN_{aP}x_P^2}{2\epsilon_P}$	V_{bin}, V_{biP} = parts of V_{bi} across the n and P sides (V) they add to V_{bi}	2(b)
★ **Depletion widths**	$x_n = \left[\frac{2\epsilon_n\epsilon_P N_{aP}V_{bi}}{eN_{dn}(\epsilon_n N_{dn} + \epsilon_P N_{aP})}\right]^{1/2}$, $x_P = \left[\frac{2\epsilon_n\epsilon_P N_{dn}V_{bi}}{eN_{aP}(\epsilon_n N_{dn} + \epsilon_P N_{aP})}\right]^{1/2}$	V_R = reverse bias (V) replace V_{bi} by $V_{bi} + V_R$ under reverse bias	2(b), 2(d)
Total width	$W = \left[\frac{2\epsilon_n\epsilon_P(N_{dn} + N_{aP})^2 V_{bi}}{eN_{dn}N_{aP}(\epsilon_n N_{dn} + \epsilon_P N_{aP})}\right]^{1/2}$	W = total depletion width, $x_n + x_P$ (cm)	2(b)

Table 3: Heterojunctions

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4. Bipolar transistors:

Title	Equation	Notation	Ref.
★ **Base minority electrons**	$\delta n_B = \dfrac{n_{B0}(e^{eV_{BE}/kT}-1)\sinh\frac{x_B-x}{L_B} - n_{B0}\sinh\frac{x}{L_B}}{\sinh(x_B/L_B)}$	δn_B = excess electron concentration in the base (cm^{-3}) n_{B0} = equilibrium electron concentration in the base (cm^{-3}) e = magnitude of the electron charge (C) V_{BE} = base–emitter voltage (V) k = Boltzmann constant (J/K) T = temperature (K) x = distance from the emitter edge of the base (cm) x_B = neutral base width (cm) L_B = electron diffusion length in the base, $\sqrt{D_B\tau_{B0}}$ (cm) D_B = electron diffusion coefficient in the base (cm^2/s) τ_{B0} = electron lifetime in the base (s) narrow base: $n_{B0}e^{eV_{BE}/kT}(1-x/x_B)$	2(c)
★ **Injection efficiency**	$\gamma = \left[1 + \dfrac{p_{E0}D_E L_B}{n_{B0}D_B L_E}\cdot\dfrac{\tanh(x_B/L_B)}{\tanh(x_E/L_E)}\right]^{-1}$	γ = emitter injection efficiency p_{E0} = equilibrium hole concentration in the emitter (cm^{-3}) D_E = hole diffusion coefficient in the emitter (cm^2/s) L_E = hole diffusion length in the emitter (cm) x_E = neutral emitter width (cm) short regions: $[1 + N_B D_E x_B/(N_E D_B x_E)]^{-1}$ N_B, N_E = base and emitter dopings (cm^{-3})	5(a), 5(d)
Base transport factor	$\alpha_T = \dfrac{1}{\cosh(x_B/L_B)}$	α_T = base transport factor	5(d)
★ **Gain**	$\alpha = \gamma\alpha_T\delta,\ \beta = \dfrac{\alpha}{1-\alpha}$	α = common-base current gain δ = recombination factor β = common-emitter current gain	5(d)
★ **HBT injection ratio**	$\dfrac{J_{nE}}{J_{pE}} = \dfrac{N_{DE}}{N_{AB}}\cdot\dfrac{W_E}{W_B}\cdot\dfrac{D_{nB}}{D_{pE}}\exp\left(\dfrac{\Delta E_g}{kT}\right)$	J_{nE} = electron current density injected into the base ($\mathrm{A/cm}^2$) J_{pE} = hole current density injected into the emitter ($\mathrm{A/cm}^2$) N_{DE} = emitter donor concentration (cm^{-3}) N_{AB} = base acceptor concentration (cm^{-3}) W_E, W_B = emitter and base widths (cm) D_{nB} = electron diffusion coefficient in the base (cm^2/s) D_{pE} = hole diffusion coefficient in the emitter (cm^2/s) ΔE_g = band gap of the emitter minus that of the base (eV) n_{iB}, n_{iE} = intrinsic concentrations of the base and emitter (cm^{-3}) $n_{iB}^2/n_{iE}^2 = e^{\Delta E_g/kT}$	4(b)

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Title	Equation	Notation	Ref.
Early effect	$\frac{dI_C}{dV_{CE}} = \frac{I_C}{V_{CE}+V_A}$, $r_o = \frac{V_A}{I_C}$	I_C = collector current (A) V_{CE} = collector–emitter voltage (V) V_A = Early voltage (V) r_o = output resistance (Ω)	5(b)
Gummel–Poon current	$J_n = \frac{-eD_n n_i^2 e^{eV_{BE}/kT}}{\int_0^{x_B} p\,dx}$	J_n = electron current density through the base (A/cm^2) D_n = electron diffusion coefficient in the base (cm^2/s) n_i = intrinsic carrier concentration (cm^{-3}) p = hole concentration in the base (cm^{-3}) denominator: base Gummel number	5(c)
Normalised base charge	$q_b = \frac{q_1}{2}(1+\sqrt{1+4q_2})$	q_b = base charge over its zero-bias value q_1 = the Early-effect term q_2 = the high-injection term	5(c)
★ **Ebers–Moll**	$I_C = \alpha_F I_F - I_R$, $I_E = I_F - \alpha_R I_R$, $\alpha_F I_{ES} = \alpha_R I_{CS}$	I_E = emitter current (A) α_F, α_R = forward and reverse common-base current gains I_F = forward diode current, $I_{ES}(e^{V_{BE}/V_t}-1)$ (A) I_R = reverse diode current, $I_{CS}(e^{V_{BC}/V_t}-1)$ (A) I_{ES}, I_{CS} = emitter and collector saturation currents (A) V_{BC} = base–collector voltage (V) V_t = thermal voltage, kT/e (V)	6(c)

Table 4: Bipolar transistors

5. MOS capacitor:

Title	Equation	Notation	Ref.
Fermi potential	$\phi_{fp} = V_t \ln \frac{N_a}{n_i}$	$\phi_{fp} = (E_{Fi}-E_F)/e$ in the p-type substrate (V) V_t = thermal voltage, kT/e (V) N_a = substrate acceptor concentration (cm^{-3}) n_i = intrinsic carrier concentration (cm^{-3})	4(c), 8(b)
★ **Maximum depletion width**	$x_{dT} = \left(\frac{4\epsilon_s \phi_{fp}}{eN_a}\right)^{1/2}$	x_{dT} = maximum depletion width, at threshold (cm) ϵ_s = permittivity of the semiconductor (F/cm) e = magnitude of the electron charge (C) at threshold, $\phi_s = 2\phi_{fp}$ ϕ_s = surface potential (V)	4(c), 8(b)
Oxide capacitance	$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$	C_{ox} = oxide capacitance per unit area (F/cm^2) ϵ_{ox} = permittivity of the oxide, $3.9\epsilon_0$ (F/cm) t_{ox} = oxide thickness (cm)	4(c)
Depletion C	$C' = \frac{\epsilon_{ox}}{t_{ox} + (\epsilon_{ox}/\epsilon_s)x_d}$	C' = capacitance per unit area in depletion (F/cm^2) x_d = depletion width (cm) C'_{min} at $x_d = x_{dT}$	4(c)

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Title	Equation	Notation	Ref.
Flat-band C	$C'_{FB} = \dfrac{\epsilon_{ox}}{t_{ox} + (\epsilon_{ox}/\epsilon_s)L_D}$	C'_{FB} = flat-band capacitance per unit area (F/cm^2) L_D = extrinsic Debye length, $\sqrt{\epsilon_s V_t/(eN_a)}$ (cm)	4(c)
★ **Flat-band voltage**	$V_{FB} = \phi_{ms} - \dfrac{Q'_{ss}}{C_{ox}}$	V_{FB} = flat-band voltage (V) ϕ_{ms} = metal–semiconductor work-function difference (V) Q'_{ss} = fixed oxide charge per unit area (C/cm^2)	4(d)
★ **Threshold voltage**	$V_{TN} = V_{FB} + \dfrac{\lvert Q'_{SD}(\max)\rvert}{C_{ox}} + 2\phi_{fp}$	V_{TN} = threshold voltage of the n-channel device (V) $Q'_{SD}(\max)$ = maximum depletion charge per unit area (C/cm^2) $\lvert Q'_{SD}(\max)\rvert = eN_a x_{dT}$	4(d), 8(b)

Table 5: MOS capacitor

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6. HEMT and MESFET:

Title	Equation	Notation	Ref.
★ **2DEG density**	$n_s = \frac{\epsilon_N}{e(d+\Delta d)}(V_g - V_{off})$	n_s = sheet density of the 2DEG (cm^{-2}) ϵ_N = permittivity of the AlGaAs (F/cm) e = magnitude of the electron charge (C) d = AlGaAs thickness (cm) Δd = distance of the 2DEG from the interface (about 80 Å) V_g = gate voltage (V) V_{off} = off voltage (V)	6(b)
HEMT off voltage	$V_{off} = \phi_B - \frac{\Delta E_c}{e} - \frac{eN_d d_d^2}{2\epsilon_N}$	ϕ_B = Schottky barrier height of the gate (V) ΔE_c = conduction-band offset (eV) N_d = donor concentration in the AlGaAs (cm^{-3}) d_d = doped AlGaAs thickness (cm)	6(b)
★ **Cut-off frequency**	$f_T = \frac{g_m}{2\pi C_G}$; $f_T(\text{max}) = \frac{e\mu_n N_d a^2}{2\pi\epsilon_s L^2}$	f_T = cut-off frequency (Hz) g_m = transconductance (S) C_G = gate capacitance (F) μ_n = electron mobility (cm^2/V s) N_d = donor concentration in the channel (cm^{-3}) a = channel thickness (cm) ϵ_s = permittivity of the semiconductor (F/cm) L = gate length (cm)	6(d)
Internal pinch-off	$V_{p0} = \frac{ea^2 N_d}{2\epsilon_s}$	V_{p0} = internal pinch-off voltage (V)	6(d), 6(e)
★ **MESFET threshold**	$V_T = V_{bi} - V_{p0}$, $V_{bi} = \phi_{Bn} - \phi_n$	V_T = threshold voltage (V) V_{bi} = built-in potential of the Schottky gate (V) ϕ_{Bn} = Schottky barrier height (V) $\phi_n = (E_c - E_F)/e$ in the channel (V) enhancement mode if $V_T > 0$	6(e)

Table 6: HEMT and MESFET

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7. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
Atoms per cell	corner $\frac{1}{8}$, face $\frac{1}{2}$, edge $\frac{1}{4}$, inside 1
Barrier, n-type	$\phi_{B0} = \phi_m - \chi$
Barrier, p-type	$\phi_{Bp} = \dfrac{E_g}{e} + \chi - \phi_m$
Built-in potential	$V_{bi} = \phi_{B0} - \phi_n$
Depletion width	$x_n = \left[\dfrac{2\epsilon_s V_{bi}}{eN_d}\right]^{1/2}$
Maximum field	$\lvert E_{max}\rvert = \dfrac{eN_d x_n}{\epsilon_s}$
Specific contact resistance	$R_c = \left(\dfrac{\partial J}{\partial V}\right)^{-1}_{V=0}$
Band offsets	$\Delta E_c = e(\chi_n - \chi_P)$, $\Delta E_v = \Delta E_g - \Delta E_c$
Built-in potential (nP)	$eV_{bi} = \Delta E_v + kT \ln\left(\dfrac{p_{P0}}{p_{n0}} \cdot \dfrac{N_{vn}}{N_{vP}}\right)$
Depletion widths	$x_n = \left[\dfrac{2\epsilon_n\epsilon_P N_{aP} V_{bi}}{eN_{dn}(\epsilon_n N_{dn} + \epsilon_P N_{aP})}\right]^{1/2}$, $x_P = \left[\dfrac{2\epsilon_n\epsilon_P N_{dn} V_{bi}}{eN_{aP}(\epsilon_n N_{dn} + \epsilon_P N_{aP})}\right]^{1/2}$
Base minority electrons	$\delta n_B = \dfrac{n_{B0}(e^{eV_{BE}/kT} - 1)\sinh\frac{x_B - x}{L_B} - n_{B0}\sinh\frac{x}{L_B}}{\sinh(x_B/L_B)}$
Injection efficiency	$\gamma = \left[1 + \dfrac{p_{E0} D_E L_B}{n_{B0} D_B L_E} \cdot \dfrac{\tanh(x_B/L_B)}{\tanh(x_E/L_E)}\right]^{-1}$
Gain	$\alpha = \gamma\alpha_T\delta$, $\beta = \dfrac{\alpha}{1-\alpha}$
HBT injection ratio	$\dfrac{J_{nE}}{J_{pE}} = \dfrac{N_{DE}}{N_{AB}} \cdot \dfrac{W_E}{W_B} \cdot \dfrac{D_{nB}}{D_{pE}} \exp\left(\dfrac{\Delta E_g}{kT}\right)$
Ebers–Moll	$I_C = \alpha_F I_F - I_R$, $I_E = I_F - \alpha_R I_R$, $\alpha_F I_{ES} = \alpha_R I_{CS}$
Maximum depletion width	$x_{dT} = \left(\dfrac{4\epsilon_s\phi_{fp}}{eN_a}\right)^{1/2}$
Flat-band voltage	$V_{FB} = \phi_{ms} - \dfrac{Q'_{ss}}{C_{ox}}$
Threshold voltage	$V_{TN} = V_{FB} + \dfrac{\lvert Q'_{SD}(\max)\rvert}{C_{ox}} + 2\phi_{fp}$
2DEG density	$n_s = \dfrac{\epsilon_N}{e(d + \Delta d)}(V_g - V_{off})$
Cut-off frequency	$f_T = \dfrac{g_m}{2\pi C_G}$; $f_T(\max) = \dfrac{e\mu_n N_d a^2}{2\pi\epsilon_s L^2}$
MESFET threshold	$V_T = V_{bi} - V_{p0}$, $V_{bi} = \phi_{Bn} - \phi_n$

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8. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
r	atom radius (Å)	bcc packing
a	lattice constant (Å)	bcc packing
c	height of the hcp cell (Å)	hcp ideal ratio
d	nearest-neighbour (bond) distance (Å)	Zinc blende bond
ϕ_{B0}	Schottky barrier height for electrons (V)	Barrier, n-type
ϕ_m	metal work function (V)	Barrier, n-type
χ	electron affinity of the semiconductor (V)	Barrier, n-type
ϕ_{Bp}	Schottky barrier height for holes (V)	Barrier, p-type
E_g	band gap energy (eV)	Barrier, p-type
e	magnitude of the electron charge (C)	Barrier, p-type
ϕ_n	$(E_c - E_F)/e$ in the n-type semiconductor (V)	$E_c - E_F$ in n-type
V_t	thermal voltage, kT/e (V)	$E_c - E_F$ in n-type
N_c	effective density of states in the conduction band (cm^{-3})	$E_c - E_F$ in n-type
N_d	donor concentration (cm^{-3})	$E_c - E_F$ in n-type
V_{bi}	built-in potential, the band bending at zero bias (V)	Built-in potential
x_n	depletion width at zero bias (cm)	Depletion width
ϵ_s	permittivity of the semiconductor (F/cm)	Depletion width
E_{max}	peak electric field, at the metal (V/cm)	Maximum field
γ	weight of the metal work function in the barrier	Interface states
ϕ_0	neutral level of the interface states, above E_v (V)	Interface states
ϵ_i	permittivity of the interfacial layer (F/cm)	Interface states
δ	thickness of the interfacial layer (cm)	Interface states
D_{it}	density of interface states ($\mathrm{cm}^{-2}\mathrm{eV}^{-1}$)	Interface states
R_c	specific contact resistance (Ω cm^2)	Specific contact resistance
J	current density ($\mathrm{A/cm}^2$)	Specific contact resistance
V	applied voltage (V)	Specific contact resistance
k	Boltzmann constant (J/K)	Thermionic R_c
T	temperature (K)	Thermionic R_c
A^*	effective Richardson constant ($\mathrm{A/cm}^2\mathrm{K}^2$)	Thermionic R_c
ϕ_{Bn}	barrier height for electrons (V)	Thermionic R_c
m_n^*	electron effective mass (kg)	Tunnelling R_c
$\hbar$	reduced Planck constant, $h/2\pi$ (J s)	Tunnelling R_c
E_n	energy of level n in the well (J)	RTD levels
m^*	effective mass in the well (kg)	RTD levels
n	level number (1, 2, …)	RTD levels
L_w	well width (m)	RTD levels

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Symbol	Meaning	First used in
ΔE_c	conduction-band offset (eV)	Band offsets
χ_n	electron affinity of the narrow-gap n side (V)	Band offsets
χ_P	electron affinity of the wide-gap P side (V)	Band offsets
ΔE_v	valence-band offset (eV)	Band offsets
ΔE_g	difference of the two band gaps (eV)	Band offsets
p_{P0}, p_{n0}	equilibrium hole concentrations in the P and n sides (cm^{-3})	Built-in potential (nP)
N_{vn}, N_{vP}	effective densities of states in the valence band of the n and P sides (cm^{-3})	Built-in potential (nP)
n_i	intrinsic carrier concentration of the n side (cm^{-3})	Built-in potential (nP)
N_{dn}	donor concentration in the n side (cm^{-3})	Charge balance
N_{aP}	acceptor concentration in the P side (cm^{-3})	Charge balance
x_n, x_P	depletion widths in the n and P sides (cm)	Charge balance
ϵ_n, ϵ_P	permittivities of the n and P sides (F/cm)	Charge balance
$E_n(0)$, $E_P(0)$	fields at the interface (V/cm)	Charge balance
V_{bin}, V_{biP}	parts of V_{bi} across the n and P sides (V)	Potential drops
V_R	reverse bias (V)	Depletion widths
W	total depletion width, $x_n + x_P$ (cm)	Total width
δn_B	excess electron concentration in the base (cm^{-3})	Base minority electrons
n_{B0}	equilibrium electron concentration in the base (cm^{-3})	Base minority electrons
V_{BE}	base–emitter voltage (V)	Base minority electrons
x	distance from the emitter edge of the base (cm)	Base minority electrons
x_B	neutral base width (cm)	Base minority electrons
L_B	electron diffusion length in the base, $\sqrt{D_B \tau_{B0}}$ (cm)	Base minority electrons
D_B	electron diffusion coefficient in the base (cm^2/s)	Base minority electrons
τ_{B0}	electron lifetime in the base (s)	Base minority electrons
γ	emitter injection efficiency	Injection efficiency
p_{E0}	equilibrium hole concentration in the emitter (cm^{-3})	Injection efficiency
D_E	hole diffusion coefficient in the emitter (cm^2/s)	Injection efficiency
L_E	hole diffusion length in the emitter (cm)	Injection efficiency
x_E	neutral emitter width (cm)	Injection efficiency
N_B, N_E	base and emitter dopings (cm^{-3})	Injection efficiency
α_T	base transport factor	Base transport factor
α	common-base current gain	Gain
δ	recombination factor	Gain
β	common-emitter current gain	Gain
J_{nE}	electron current density injected into the base (A/cm^2)	HBT injection ratio
J_{pE}	hole current density injected into the emitter (A/cm^2)	HBT injection ratio
N_{DE}	emitter donor concentration (cm^{-3})	HBT injection ratio

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Symbol	Meaning	First used in
N_{AB}	base acceptor concentration (cm^{-3})	HBT injection ratio
W_E, W_B	emitter and base widths (cm)	HBT injection ratio
D_{nB}	electron diffusion coefficient in the base (cm^2/s)	HBT injection ratio
D_{pE}	hole diffusion coefficient in the emitter (cm^2/s)	HBT injection ratio
ΔE_g	band gap of the emitter minus that of the base (eV)	HBT injection ratio
n_{iB}, n_{iE}	intrinsic concentrations of the base and emitter (cm^{-3})	HBT injection ratio
I_C	collector current (A)	Early effect
V_{CE}	collector–emitter voltage (V)	Early effect
V_A	Early voltage (V)	Early effect
r_o	output resistance (Ω)	Early effect
J_n	electron current density through the base (A/cm^2)	Gummel–Poon current
D_n	electron diffusion coefficient in the base (cm^2/s)	Gummel–Poon current
p	hole concentration in the base (cm^{-3})	Gummel–Poon current
q_b	base charge over its zero-bias value	Normalised base charge
q_1	the Early-effect term	Normalised base charge
q_2	the high-injection term	Normalised base charge
I_E	emitter current (A)	Ebers–Moll
α_F, α_R	forward and reverse common-base current gains	Ebers–Moll
I_F	forward diode current, $I_{ES}(e^{V_{BE}/V_t}-1)$ (A)	Ebers–Moll
I_R	reverse diode current, $I_{CS}(e^{V_{BC}/V_t}-1)$ (A)	Ebers–Moll
I_{ES}, I_{CS}	emitter and collector saturation currents (A)	Ebers–Moll
V_{BC}	base–collector voltage (V)	Ebers–Moll
ϕ_{fp}	$(E_{Fi}-E_F)/e$ in the p-type substrate (V)	Fermi potential
N_a	substrate acceptor concentration (cm^{-3})	Fermi potential
x_{dT}	maximum depletion width, at threshold (cm)	Maximum depletion width
ϕ_s	surface potential (V)	Maximum depletion width
C_{ox}	oxide capacitance per unit area (F/cm^2)	Oxide capacitance
ϵ_{ox}	permittivity of the oxide, $3.9\epsilon_0$ (F/cm)	Oxide capacitance
t_{ox}	oxide thickness (cm)	Oxide capacitance
C'	capacitance per unit area in depletion (F/cm^2)	Depletion C
x_d	depletion width (cm)	Depletion C
C'_{FB}	flat-band capacitance per unit area (F/cm^2)	Flat-band C
L_D	extrinsic Debye length, $\sqrt{\epsilon_s V_t/(eN_a)}$ (cm)	Flat-band C
V_{FB}	flat-band voltage (V)	Flat-band voltage
ϕ_{ms}	metal–semiconductor work-function difference (V)	Flat-band voltage
Q'_{ss}	fixed oxide charge per unit area (C/cm^2)	Flat-band voltage
V_{TN}	threshold voltage of the n-channel device (V)	Threshold voltage
$Q'_{SD}(\mathrm{max})$	maximum depletion charge per unit area (C/cm^2)	Threshold voltage
n_s	sheet density of the 2DEG (cm^{-2})	2DEG density

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Symbol	Meaning	First used in
ϵ_N	permittivity of the AlGaAs (F/cm)	2DEG density
d	AlGaAs thickness (cm)	2DEG density
Δd	distance of the 2DEG from the interface (about 80 Å)	2DEG density
V_g	gate voltage (V)	2DEG density
V_{off}	off voltage (V)	2DEG density
ϕ_B	Schottky barrier height of the gate (V)	HEMT off voltage
d_d	doped AlGaAs thickness (cm)	HEMT off voltage
f_T	cut-off frequency (Hz)	Cut-off frequency
g_m	transconductance (S)	Cut-off frequency
C_G	gate capacitance (F)	Cut-off frequency
μ_n	electron mobility (cm^2/V s)	Cut-off frequency
a	channel thickness (cm)	Cut-off frequency
L	gate length (cm)	Cut-off frequency
V_{p0}	internal pinch-off voltage (V)	Internal pinch-off
V_T	threshold voltage (V)	MESFET threshold
V_{bi}	built-in potential of the Schottky gate (V)	MESFET threshold
ϕ_{Bn}	Schottky barrier height (V)	MESFET threshold
ϕ_n	$(E_c - E_F)/e$ in the channel (V)	MESFET threshold

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